NN Builder

User manual: using the application, designing a neural network,
and the demo architectures explained

The NN Builder project
https://github.com/varunjha089/NN-Builder

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Abstract

NN Builder is a desktop application for designing neural networks on a canvas, the way one draws a block diagram. The design is checked while it is drawn: every connection shows the shape of the tensor it carries, every block shows its output size and parameter count, and mistakes are explained in plain words on the block itself. A finished design becomes a complete, readable PyTorch training script that also loads the data (a built-in dataset, Hugging Face or Kaggle), and the application runs it on a GPU somewhere else: a free Kaggle notebook, a college GPU server over SSH, or an Amazon EC2 instance (including spot instances, which may be interrupted). The computer the application runs on only designs: it never trains, never downloads datasets and needs no PyTorch. This manual explains the application screen by screen, walks through designing a network from scratch, explains how to decide what to build (the parts of every network, the kinds of data between blocks, the common motifs) with four worked examples, gives the arithmetic behind every number the application shows, explains the fifteen demo architectures (from LeNet-5 to a one-billion-parameter Vision Transformer, plus one original design), and ends with troubleshooting. References point to the original papers, on arXiv wherever they are published there.

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1 What NN Builder is

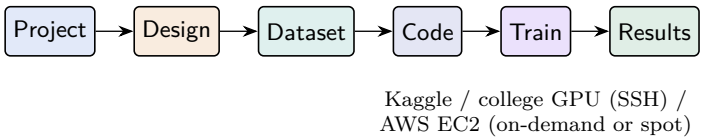
1.1 The idea

Building a neural network usually means writing code, running it, and finding out from an error message (or worse, from silently bad results) that a layer received a tensor of the wrong shape. NN Builder moves that discovery to the moment of design. It is inspired by visual programming tools such as MIT App Inventor: blocks are placed and connected, and the tool refuses or explains combinations that cannot work.

Three principles shape the application:

1. **The design computer only designs.** Shapes, parameter counts and the generated code are computed by an engine written in Rust inside the application, instantly and offline. No deep-learning library is installed locally, and no dataset is downloaded to the design computer.
2. **Training happens on a GPU elsewhere, automatically.** One button generates the script, uploads it, prepares the machine if needed, starts the run, follows it, and brings the results back.
3. **A run does not depend on the connection.** On a server or an EC2 instance the run is started detached; it continues when the laptop sleeps, the network drops or the application is closed, and the application reconnects to it later.

1.2 The workflow at a glance



The tabs in the top bar follow this order. **Connection**, set apart on the right, holds the tokens and machines that training needs, and is usually visited once.

1.3 What it can build

- **Tasks:** classification, multi-label (Section 10.8), regression (10.9), segmentation, object detection (10.11), and several of them at once (extra outputs, 10.10).
- **Data:** pictures, satellite images of any number of bands (Sentinel-2, Sentinel-1, 11), two sensors together (fusion, also at their own sizes, 10.14), before/after pairs (10.13), time series of dated images (10.12); from Hugging Face, Kaggle, your own files or a cloud folder; a data check before training (14.4).
- **Networks:** CNNs, U-Nets, Vision Transformers and hybrids from the block list; Siamese networks with shared weights; Conv3D, temporal attention, LSTM / GRU and temporal transformers for time series; transformer options (causal, rotary, RMSNorm, SwiGLU, cross-attention, 8.2); mixture of experts and early exits, frozen layers and LoRA (8.3); and anything else as your own PyTorch class (Custom block, 8.1).
- **Training:** the usual recipe, unbalanced classes and their losses (14.3), ImageNet weights, several seeds, several GPUs (8.4), on Kaggle, an SSH machine or EC2 (spot included).

1.4 What it is not

NN Builder trains on images (pictures, satellite bands, time series of them). It does not train on text, audio or tabular data, it does not generate images (no diffusion models or GANs), and it does not run training on the design computer, even for a tiny network. The generated script, however, is ordinary PyTorch [31]: it can be copied, read, edited and run anywhere by hand.

2 Installing, updating and the window

2.1 Installing

Installers are published with every release on GitHub (`https://github.com/varunjha089/NN-Builder/releases/latest`):

- **macOS** (Apple silicon): the file ending in `_aarch64.app.zip`. Unzip it and move `NN Builder.app` to the Applications folder. The application is not signed by Apple, so the first start may need a right-click on the application and **Open**.
- **Windows**: `NN-Builder_X_x64-setup.exe`.
- **Linux**: `nn-builder_X_amd64.deb` or the portable `.AppImage`.

2.2 Automatic updates

Once a day (if enabled), the application asks GitHub for the latest release. When a newer version exists, a banner appears under the top bar with **Install and restart**. The download is checked against the SHA-256 checksum published with the release before anything is replaced. The switch is in **Settings** (and on the Projects page); **Help** → **Check for Updates...** checks at once.

2.3 The top bar

From left to right: the name, the **Project** button (the current project, or *No project*; it opens the Projects page), a **not saved** badge and a **Save...** button when the open project is an unsaved demo, the tabs **Design**, **Dataset**, **Code**, **Train**, **Results**, then on the right a summary ("*N params · ready*" in green, or "*design has problems*" / "*dataset incomplete*" in red), the day/night button (a moon in day mode, a sun in night mode), the **Settings** button, and **Connection** (a red dot means something is still missing there).

2.4 Menus and keyboard shortcuts

Menu	Item (shortcut)	What it does
NN Builder	Settings... (Cmd+,)	opens the Settings page
File	New Project... (Cmd+N)	Projects page, cursor in the name field
	Open Project... (Cmd+O)	choose a project folder
	Save Project... (Cmd+S)	keeps an unsaved demo as a real project
	Close Project	back to the Projects page with no project open (an unsaved demo is discarded)

Menu	Item (shortcut)	What it does
	Delete (Projects page)	asks in the row, then moves the project's folder (design, runs, results) to the Trash, from where it can be put back; a missing project is only taken off the list
	New from Demo ▷	opens one of the fifteen demo architectures
	Import / Export Design…	reads or writes a design (`model.json`)
	Export train.py… (Cmd+E)	saves the generated training script
	Show Project in Finder (Shift+Cmd+R)	reveals the project folder
Edit	Undo (Cmd+Z), Redo (Shift+Cmd+Z)	undo/redo changes to the design (in a text field: the typing)
	Arrange Left to Right (Cmd+L)	lays the canvas out in columns by depth
View	Night Mode (Shift+Cmd+N)	switches the neon night theme
Connection	Open Connection (Shift+Cmd+K)	the Connection page
	Install Tools…	installs the `kaggle` and `boto3` helpers
Help	Documentation (Cmd+?)	opens this manual (a PDF inside the application)
	Check for Updates…	asks GitHub for a newer release

2.5 Day and night themes, Settings

The night theme draws panels dark and boxes and text as neon glows; every layer glows in the colour of its kind (Conv2D orange, pooling teal, normalisation violet, Dense blue, joins red, outputs green, transformer blocks teal and violet). In night mode the Design screen is split into three separate panels (blocks on the left, canvas in the middle, inspector on the right). Their background colours are chosen in **Settings** → **Night panel colours**: six presets (Midnight, Deep purple, Ocean, Carbon, Forest, Crimson) or a colour picker per panel, with a live preview; a warning appears for colours so light that neon text becomes hard to read. The theme, the panel colours and the automatic-update switch are all on the Settings page (Figure 1); Figure 2 shows the day theme.

3 Projects

3.1 Creating, opening, saving

A *project* is a folder. The Projects page has two cards side by side:

- **New project**: type a name and press **Create** (or Enter). The folder is made in the projects folder shown underneath (**change** picks another one).
- **Open**: a field to open any project folder: type or paste its path and press **Open folder**, or press **Choose…** and pick the folder in the Finder dialog (as **File** → **Open Project…**). A pasted path may keep its quotes, start with `~/`, or name a file inside the project (e.g. `model.json`); the project folder is the one that holds `project.json`.

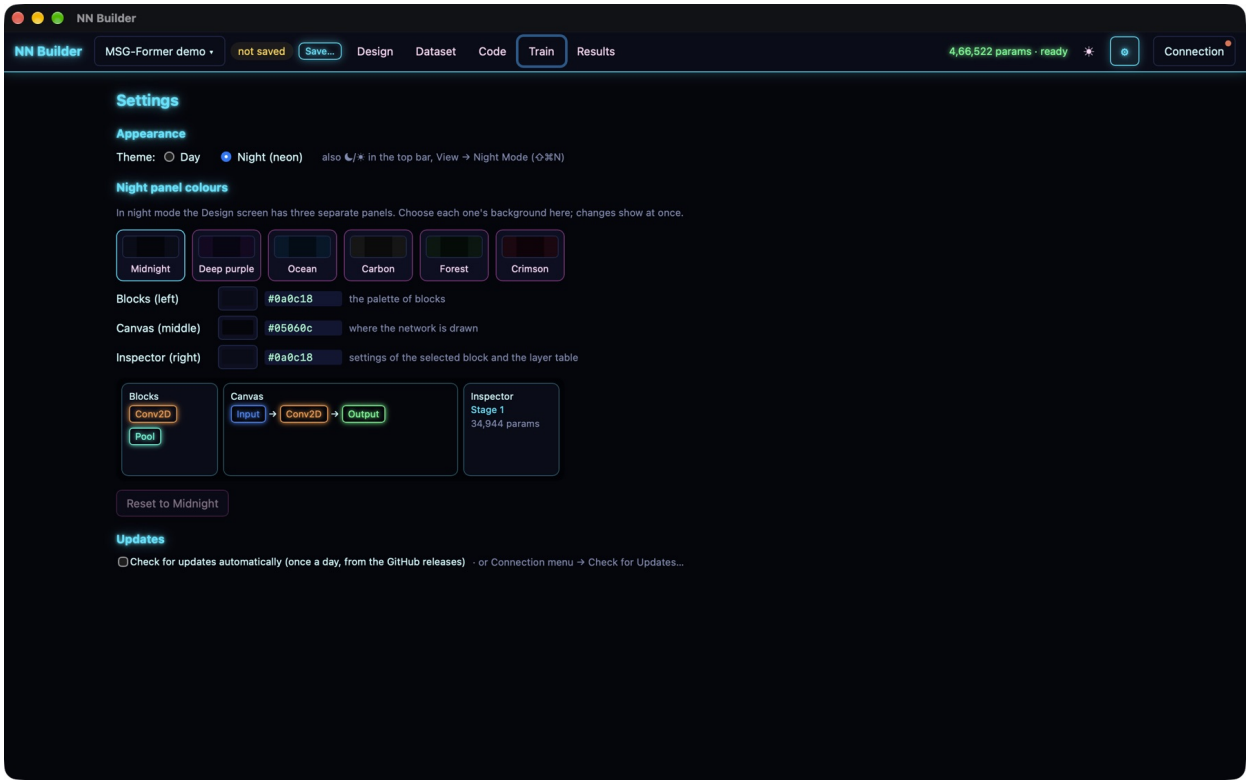


Figure 1: Settings: the theme, the six presets and a colour picker per panel for the night Design screen, a live preview, and the automatic-update switch.

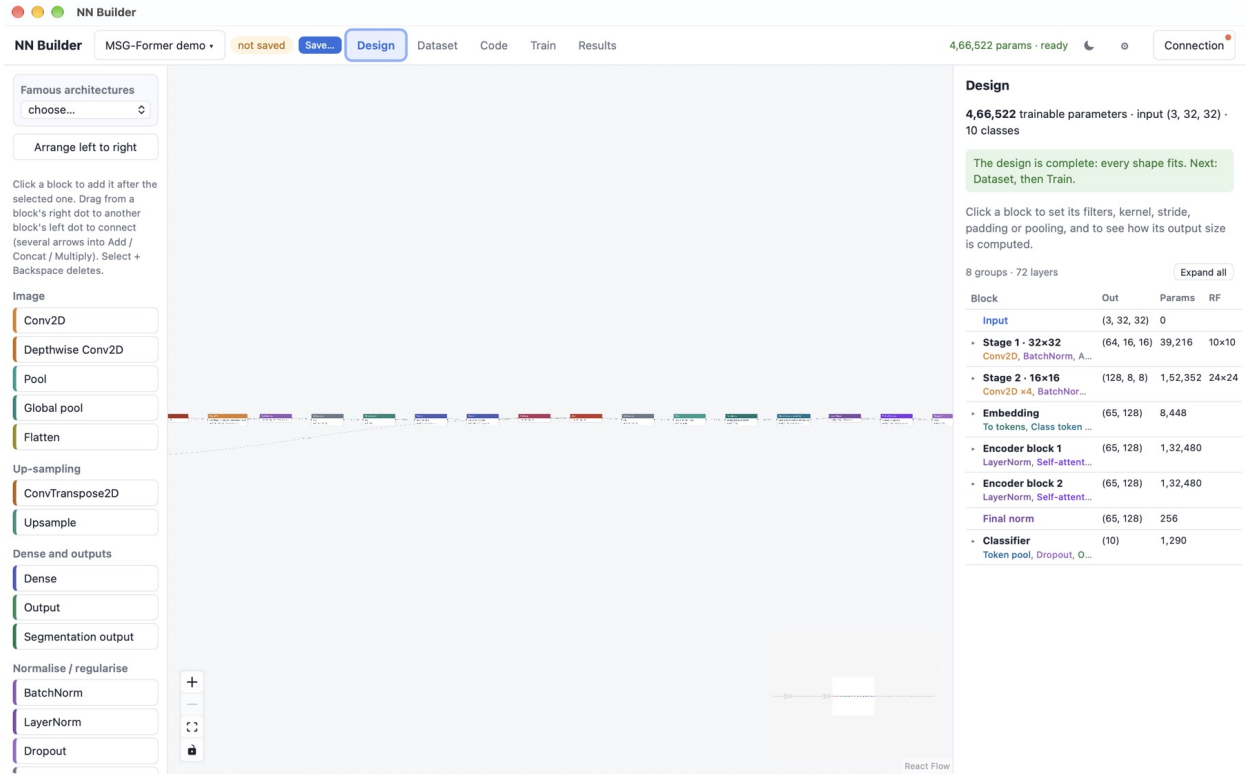


Figure 2: The Design screen in the day theme (MSG-Former, fitted to the canvas; the layer table on the right).

Under the two cards come these sections:

- **Running Task** (its heading in pink): every run still starting, running or queued, in any project, as a tile with the project, the machine, the dataset and the epochs done. The numbers are as the app last saw them (it follows a run while its project is open); click a tile to open that project at that run on the **Train** tab, where it is followed again.
- **Overnight report** (when there is something to report): every run, in any project, that ended since you last pressed **Mark as seen** (the first time: in the last 24 hours), with its result (test accuracy, or R^2), the epochs done, how long it took, the machine and when it ended; done in green, failed or crashed in red, stopped or interrupted in amber. Click one to open it. While this page is shown, the application looks at every unfinished run on its machine every two minutes (machines out of reach, e.g. the lab computers from outside the lab network, are skipped until they answer); when it starts, it looks once and says how many runs ended while it was closed. So: start runs on the lab machines in the evening, and next morning on the lab Wi-Fi open the application and read the report.
- **Opened projects**: the projects used before as tiles, the most recently opened first, each with its description, its folder, **Delete** and **Open**. **Add a description** (or **Edit**) writes a few words about the project; **Save** (or **Cmd**-Return) keeps them in the project's `project.json`, and an empty description removes it.
- The demo architectures (section 16).

Click the heading of **Opened projects** or of the demos to fold it away; the app remembers which are folded. Every change is saved as it is made; there is no "save" step for a normal project.

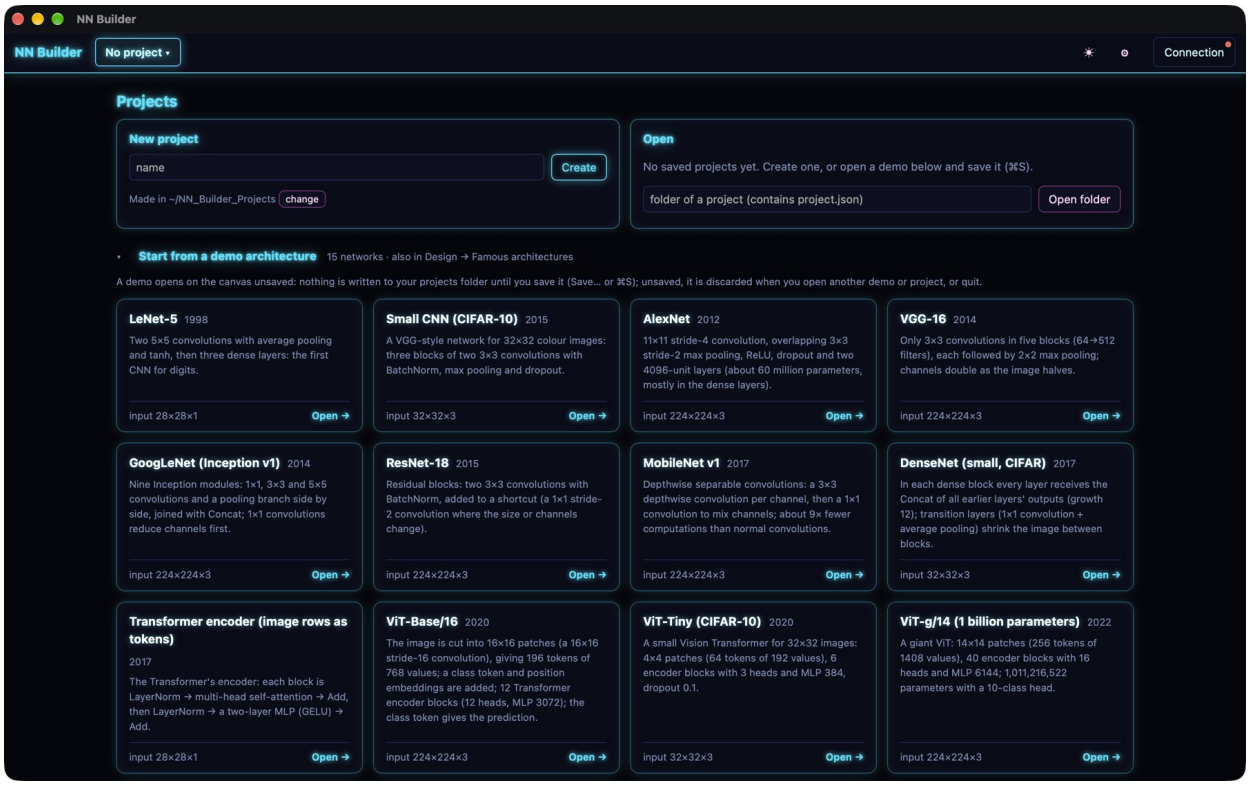


Figure 3: The Projects page (night theme): New project and Open side by side, then the demo architectures as tiles.

3.2 Demo architectures are unsaved until you save them

Below the two cards, **Start from a demo architecture** shows the sixteen demos as tiles (the section folds with its arrow). Clicking a tile opens that network on the canvas as an *unsaved* project: nothing is written to the projects folder. The top bar then shows **not saved** and **Save...**. Saving (**Save...** or Cmd+S) asks for a name and moves it into the projects folder as a normal project. An unsaved demo is discarded, without asking, when another demo or project is opened or created, and when the application quits. This keeps the projects folder free of "AlexNet demo 2, 3, 4...". A demo cannot be trained before it is saved, so that runs always belong to a kept project.

3.3 What a project folder contains

`project.json`	name, creation date
`model.json`	the design: blocks (with their settings and positions) and arrows
`dataset.json`	the dataset choice
`train.json`	the training settings
`runs/<id>/`	one folder per training run (code, metrics, log, results, model)

A design can also be exported on its own (**File** → **Export Design...**) and imported into another project.

4 The Design screen

4.1 Layout

Blocks (left). The palette, grouped as *Image* (Conv2D, Depthwise Conv2D, Pool, Global pool, Flatten), *Up-sampling* (ConvTranspose2D, Upsample), *Dense and outputs* (Dense, Output, Segmentation output), *Normalise / regularise* (BatchNorm, LayerNorm, Dropout, Activation), *Transformer* (To tokens, Class token + position, Self-attention, Token pool) and *Branches* (Add, Concat, Multiply). Above them: the **Famous architectures** picker and **Arrange left to right**.

Canvas (middle). The network, read from left to right. Each block shows its title (with × for an error and a warning sign for a warning), a one-line summary of its settings, and its output shape with its parameter count. Each arrow shows the shape of the tensor it carries. Zoom and pan with the trackpad or the controls; a minimap sits in the corner.

Inspector (right). With a block selected: its settings, its shape formula with its own numbers, its receptive field, the sliding-window animation (convolution and pooling), and the skip-connection tool. With nothing selected: the whole design's parameter count, its problems, and the layer table.

4.2 Working with blocks

- **Add:** click a block in the palette; it is added after the selected block (and connected to it).
- **Connect:** drag from a block's right dot to another block's left dot. Add, Concat and Multiply accept several arrows; all other blocks take one.
- **Delete:** select a block or an arrow and press Backspace. The Input block cannot be deleted.

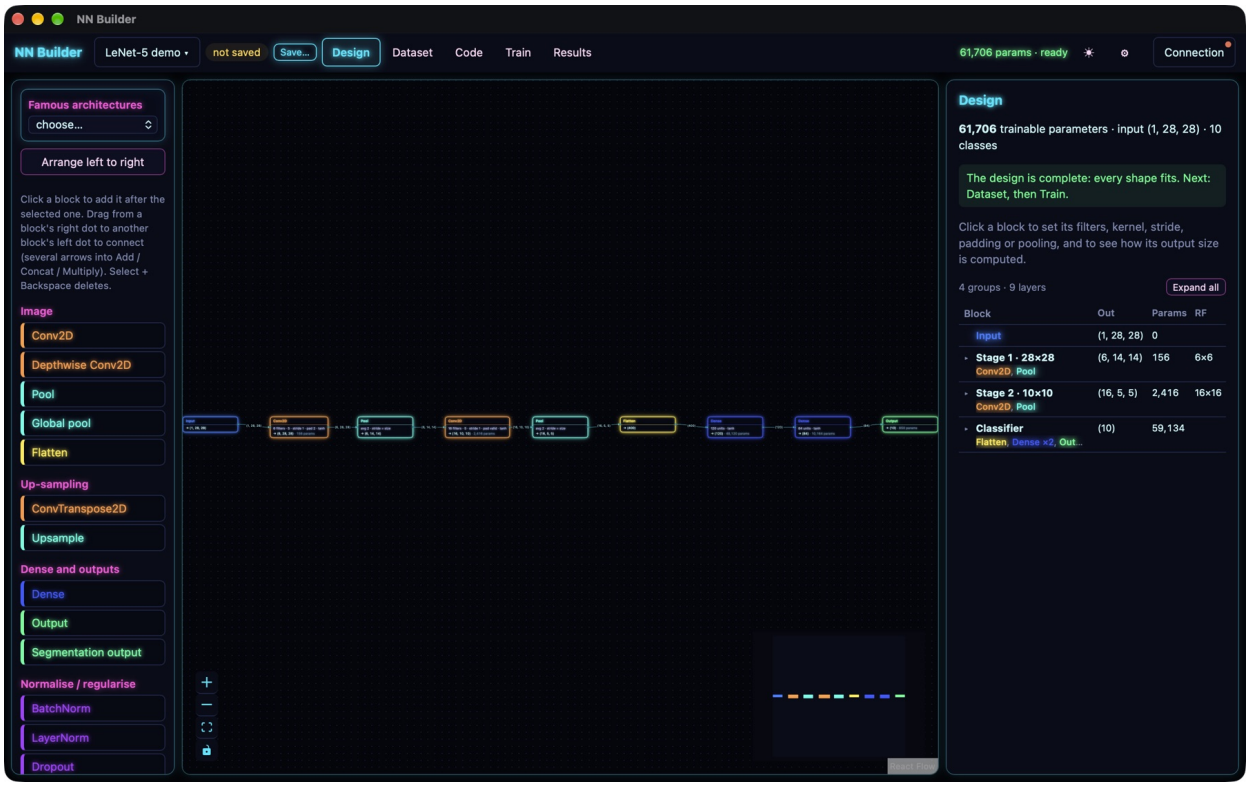


Figure 4: The Design screen with LeNet-5: blocks on the left, the network on the canvas (every arrow labelled with its tensor shape, every block with its output and parameters), and on the right the whole design's summary and the grouped layer table.

- **Move:** drag blocks; **Arrange left to right** (Cmd+L) places them in columns by depth (the longest path from the Input), stacking parallel branches in their column.
- **Undo / Redo:** Cmd+Z / Shift+Cmd+Z; changes made within 0.6 seconds of each other count as one step.
- **Copy / paste:** select blocks (Shift-click or drag a box with Shift) and use Cmd+C / Cmd+V: the copies, with the arrows between them, appear below the originals, ready to be connected.
- **Residual block after selected:** inserts Conv 3×3 → BatchNorm → ReLU → Conv 3×3 → BatchNorm → Add (with a skip from the selected block) → ReLU, keeping its size and channels.
- **Collapse repeated blocks (×N):** draws identical consecutive groups as one node, for example the 40 encoder blocks of ViT-g/14 as "Encoder block ×40" (287 blocks become 8). It is a read-only view; double-click the node or press **Show every block** to edit again.
- **Save as PNG:** draws the whole design into one picture, whatever part of it the window shows and however far it is zoomed: every block at twice its normal size (the longest side at most 16 000 pixels), with the tensor shapes on the arrows, in the current theme (day or night), without the minimap and the zoom buttons. It is saved as `export/<project> architecture.png` in the project folder and shown in the Finder (also **File** → **Export Design as PNG**). Collapsed repeated blocks stay collapsed in the picture. Section 17 has one for each test architecture.

4.3 The layer table

With nothing selected, the inspector lists the layers grouped into stages, each with its output shape, its parameter subtotal and the receptive field at its end:

- **Input**.
- **Image stages** by resolution ("Stage 2 · 27×27"); a pooling layer, a Dropout after it, and small vector layers inside a stage (a squeeze-and-excitation gate) stay with their stage; decoder stages that grow the image are marked with an up arrow.
- **Transformers**: *Embedding* (with a patch convolution before it), *Encoder block 1, 2, …*, *Final norm*.
- **Classifier**: the vector layers at the end.

Each group shows its contents in block colours ("Conv2D ×3, Pool") and opens with a click (**Expand all** opens all).

4.4 Famous architectures and the skip-connection tool

Famous architectures replaces the current design with a demo (the previous design is kept as `model.previous.json`) and adjusts the dataset's image size to the one the architecture was made for. The skip-connection tool (inspector, with a block selected) joins an earlier block's output to the selected block's output with *Add* (a residual connection, as in ResNet [10]) or *Concat* (as in DenseNet [17] and U-Net [34]). When the shapes differ, it inserts a projection: a 1×1 convolution with the right stride (images), a pooling layer, or a Dense layer (vectors).

5 Designing a network, step by step

This section builds a small convolutional network for CIFAR-10 [20] (32×32 colour photos, 10 classes).

1. A project and the data first. Create a project "CIFAR CNN". Open **Dataset**, choose **Built-in (torchvision)** and **CIFAR-10**. The Input block now says $(3, 32, 32)$: three colour channels of 32×32 pixels. (Choosing the data first means every shape shown while designing is the real one.)

2. A convolution. In **Design**, select the Input block and click **Conv2D**. In the inspector set **Filters** to 32 and choose the preset "3×3, stride 1 (keep size)". The block shows $\rightarrow (32, 32, 32)$ and 896 parameters ($3 \cdot 3 \cdot 3 \cdot 32$ weights + 32 biases). The arrow from Input shows $(3, 32, 32)$.

3. Downsampling. With the convolution selected, click **Pool**: max pooling 2×2, stride "same as pool size". Output $(32, 16, 16)$, no parameters. The inspector animates the 2×2 window sliding with stride 2: every window position gives one output value.

4. Repeat, wider. Add another Conv2D (64 filters, 3×3, same) and Pool: $(64, 8, 8)$. A third pair with 128 filters gives $(128, 4, 4)$. This is the common pattern: *the image gets smaller and deeper*.

5. From image to vector. A classifier needs one vector per image. Two choices:

- **Flatten**: $(128, 4, 4) \to (2048)$. Keeps the spatial layout, but the next Dense layer then has many weights.
- **Global pool** (average): $(128, 4, 4) \to (128)$, one number per channel [25]. Far fewer parameters, and it works for any image size.

Choose Global pool here.

6. The head. Add **Dense** (128 units, ReLU), **Dropout** (rate 0.3) [14, 38], then **Output**. The Output block reads the number of classes from the dataset (10) and has $128 \cdot 10 + 10 = 1290$ parameters. The top bar now says "*... params · ready*".

7. Make it better.

- Insert **BatchNorm** [18] after each convolution (set the convolution's activation to *none*, and add an **Activation** ReLU after the BatchNorm): faster, more stable training.
- Select the second convolution and use the skip-connection tool to add its input to its output (*Add*): a residual block. Where the channel counts differ, the tool adds the 1×1 projection for you.
- Look at the layer table: it now shows Stage 1 · 32×32, Stage 2 · 16×16, Stage 3 · 8×8, Classifier, each with its parameter subtotal.

8. Check, then train. The **Code** tab shows the generated script; **Train** starts it on Kaggle, a server or EC2 (Section 14).

5.1 When something is wrong

A block turns **red** when the design cannot work (training is blocked) and **yellow** when it works but probably not as intended. Blocks after a red block are greyed out ("fix the layer above"); blocks not connected to the Input are faded and ignored. The reasons are written on the block and in the inspector. The checks include:

Level	Check (as the application words it, shortened)
error	exactly one Input; exactly one Output (or Segmentation output), and it must be last; no loops
error	the image is too small for the window ("use a smaller kernel, padding, or fewer pooling layers before this")
error	Dense / Output on an image: "add Flatten or Global pool before it"
error	an image block (convolution, pooling) after Flatten or after To tokens
error	Add with different shapes ("use a 1×1 Conv2D to match channels, or a stride to match sizes")
error	Concat with different heights/widths; Multiply with incompatible shapes
error	attention: the token dimension must divide evenly by the number of heads
error	BatchNorm on tokens (use LayerNorm); Output on tokens (add Token pool)
error	segmentation: the output must return to the mask size ("Upsample / ConvTranspose2D")
error	pooling padding more than half the window; output padding not smaller than the stride (PyTorch rules)
error	the dataset has no classes yet

Level	Check (as the application words it, shortened)
warning	the stride does not divide the input evenly: the last rows/columns are ignored
warning	stride larger than the window: pixels between windows are skipped
warning	"same" padding with an even kernel cannot keep the size exactly
warning	a very large Dense layer after Flatten ("try Global pool instead")
warning	a block whose result is not used; Flatten on an input that is already flat

6 How to build a network: the ideas, then four worked examples

The step-by-step section above builds one network click by click. This section is about *deciding what to build*: the few ideas every design follows, the handful of small groups of blocks ("motifs") that almost every network is made of, and four complete examples for different jobs, each built from those motifs with the reasoning written down. Each example is given as text (the NNB language, Section 9): paste it into the **Text** tab, or open the file from `doc/examples/` with **File** → **Import Design**, and the canvas draws it. The shapes and parameter counts printed here are the application's own.

6.1 Every network has four parts

Whatever its name, a network for images is four parts in a row:

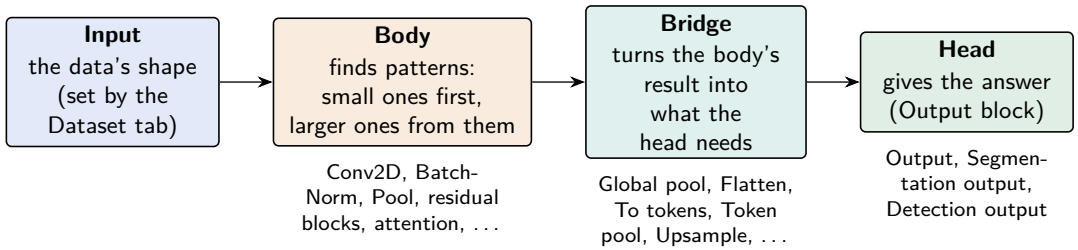


Figure 5: The four parts of every design. Most of the thinking (and most of the parameters) goes into the body; the bridge and the head follow from the task.

Input. Fixed by the data: $(3, 32, 32)$ for CIFAR-10, $(6, 128, 128)$ for six Sentinel-2 bands in 128-pixel patches. Choose the dataset first, so that every shape shown while building is the real one.

Body. Where the network learns. Its job is to turn pixels into *features*: numbers that say "there is an edge here", "a texture like water here", "something shaped like a car here". Early layers see a few pixels; every convolution and every halving of the size lets the next layer see a larger part of the image (its *receptive field*, Section 7).

Bridge. The body's output has the wrong shape for the answer, so one block changes it: Global pool or Flatten from an image to one vector, To tokens from an image to a sequence, Token pool from a sequence to one vector, Upsample / ConvTranspose2D back to the image's size for a mask.

Head. The answer, matched to the task of the Dataset tab: **Output** for classes (also multi-label and numbers), **Segmentation output** for a class per pixel, **Detection output** for boxes. A design has exactly one head (several only for multi-task data).

6.2 What travels along the arrows

Every arrow carries one of four kinds of data, and each block accepts some kinds and gives one kind. Knowing which kind you hold, and which block turns it into the next kind, is most of what

building takes (Figure 6).

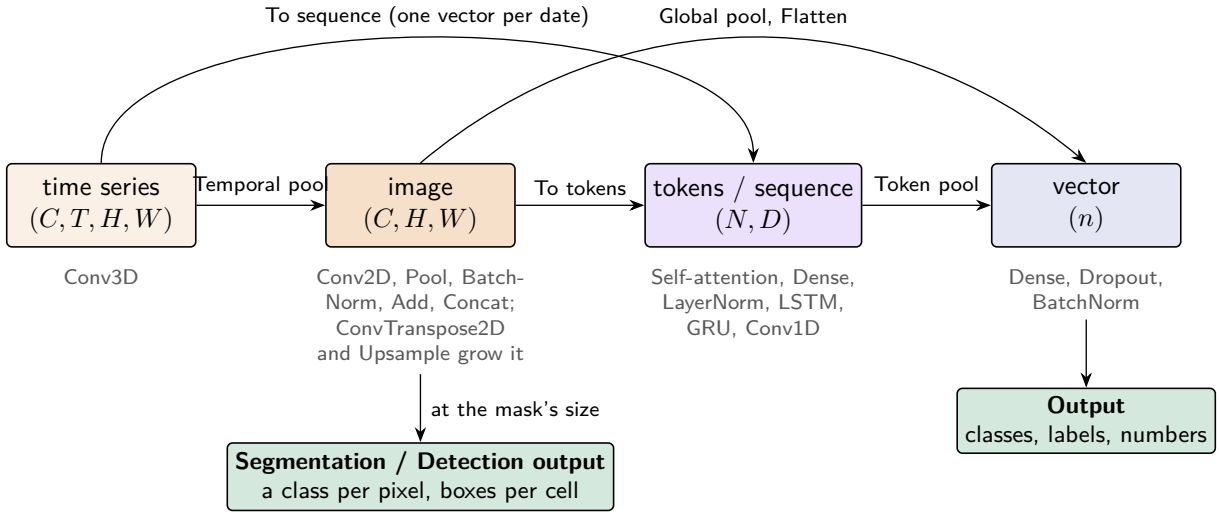


Figure 6: The four kinds of data inside a network, the blocks that keep each kind (written under it), and the blocks that turn one kind into another (on the arrows). There is no way back from a vector to an image or from tokens to an image: plan the order so you never need one.

Two consequences explain most red blocks. A block must receive the kind it works on: Dense and Output need a vector ("add Flatten or Global pool before it"), Conv2D needs an image, Self-attention needs tokens, and Output after To tokens needs a Token pool first. And the head must match the task: a segmentation dataset wants an image at the mask's size at the end, not a vector.

6.3 Seven questions to answer before the first block

1. **What is the answer?** One class per image, several (multi-label), a few numbers, a class per pixel, or boxes. This fixes the head, and with it the bridge (Figure 6).
2. **How large are the images, and how small must they get?** A classifier usually shrinks the image until it is about 4–8 pixels across before Global pool, halving it each time: 32 pixels need 2–3 halvings, 64 pixels 3–4, 224 pixels 5. Each halving is a Pool or a convolution with stride 2. A segmentation network shrinks the same way and then grows back, one step for each halving.
3. **How many channels?** Start with 16–64 and double the channels whenever the size halves, so every stage costs about the same: $(32, 32, 32) \to (64, 16, 16) \to (128, 8, 8)$. The variable `let w = 32` in the text makes the width one number to change.
4. **How much data is there?** The number of parameters should grow with the number of training images. As a rough guide from practice, not a law: a few thousand images suit up to a few hundred thousand parameters (or a large network started from ImageNet weights, Section 14); tens of thousands of images suit 0.3–5 million; more needs more. A network much larger than its data learns the training images by heart: training accuracy far above validation accuracy.
5. **Convolutions or a Transformer?** Convolutions assume that nearby pixels belong together, which is true of photos and satellite images, and they learn well from little data. Self-attention relates every position to every other, which helps with long-range context, but it needs much more data to learn what convolutions know from the start. On small data, use convolutions, or convolutions first and a few attention blocks after them (Example 2).
6. **How deep?** More than about ten convolutions in a row train badly without help. The help is

BatchNorm (or GroupNorm for small batches) after each convolution, and residual connections: an **Add** that adds a block's input to its output, so each block only has to learn a correction.

7. **How will it be checked?** Plan a validation part (Train tab, 10–20 %) and early stopping from the first run. Compare a small design with a larger one before making it larger still: if training accuracy is low, the network is too small or too shallow (TEST 25 in Section 17: 25 % on CIFAR-10 with 1 898 parameters); if training accuracy is far above validation accuracy, it is too large for the data or needs more augmentation and dropout.

6.4 The motifs: small groups that networks are made of

Famous networks look complicated, but they are a few small groups of blocks repeated with different numbers. Learn these and almost every design in this manual reads as a short sentence. Each motif below is a function (`def`, Section 9): written once, then used like a block, with its numbers as arguments.

1. The convolution unit (image → image). Convolution, BatchNorm, ReLU. The convolution has no bias and no activation of its own: BatchNorm's shift replaces the bias, and the activation comes after the normalisation. With `s = 2` it halves the size (a 4×4 kernel with stride 2 and padding 1 halves it exactly).

```
def unit(c, s = 1) {
  a conv c kernel=s+2 stride s pad 1 none nobias
  b bn
  r relu
}
2 unit(32)            # (3, 32, 32) -> (32, 32, 32)
3 unit(64, 2)         # -> (64, 16, 16)
```

2. The stage (image → smaller image). One halving and a few units at the new size. A classifier's body is two to five stages, each with twice the channels of the one before. The simplest stage is a Pool and two units:

```
def unit(c) {
  a conv c 3x3 none nobias
  b bn
  r relu
}
def stage(c) {
  p maxpool 2x2
  x unit(c)
  y unit(c)
}
2 unit(32)
3 for c in 64, 128 { s stage(c) }     # (32, 32, 32) -> (64, 16, 16) -> (128, 8, 8)
```

3. The residual block (image → image). Two convolutions whose result is *added* to the block's input, so the block learns a correction to what it receives. When the block changes the size or the channels, the input is first brought to the new shape by a *shortcut* (here an average pool and a 1×1 convolution). On the canvas, **Residual block after selected** inserts one; the skip-connection tool adds a shortcut to blocks you already have.

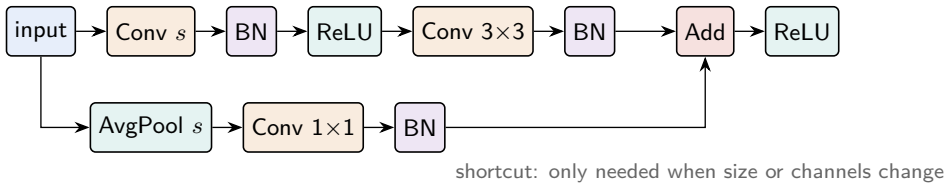


Figure 7: The residual block of Example 1 (without its gate). With $s = 2$ the main path halves the size, and the shortcut does too.

4. The channel gate (image → image, same shape). Squeeze-and-excitation [16]: Global pool summarises each channel in one number, two small Dense layers decide how much each channel matters for this image (a sigmoid gives 0 to 1), and Multiply scales the channels by it. Cheap (a few thousand parameters) and often worth one to two points of accuracy.

```
def gate(c) {
  g global_pool
  d dense c // 4
  e dense c sigmoid
  m mul <- in, e          # (c, H, W) times (c): each channel scaled
}
2 conv 64 3x3
3 gate(64)
```

5. The Transformer encoder block (tokens → tokens). LayerNorm, Self-attention, Add; LayerNorm, a two-layer MLP, Add (the figure in "Transformers in NN Builder", Section 8.5). Attention lets every token look at every other; the MLP works on each token alone. Before the first block, To tokens turns the image into tokens and Class token + position adds the class token and the positions; after the last, a LayerNorm and Token pool give the vector. The width D must divide by the number of heads.

```
let d = 64
2 conv d 4x4 stride 4 valid none   # patches
3 to_tokens patches
4 cls_pos
5 repeat 2 {
    a ln
    b attention heads 4
    c add <- in, b
    e ln
    f dense d * 2 gelu
    g dense d none
    h add <- c, g
  }
6 ln
7 token_pool cls
8 output
```

6. The decoder step (smaller image + skip → larger image). For segmentation: double the size (Upsample or ConvTranspose2D), join the encoder's map of that size with Concat (the *skip connection*, which brings back the fine detail that shrinking lost), then units. U-Net [34] is an encoder of stages and a decoder of these steps. A function with two inputs calls the second one `in2`:

```
def up(c) {
  u upsample x2 bilinear
  j concat <- u, in2
  k conv c 3x3
}
2 conv 16 3x3
3 maxpool 2x2
4 conv 32 3x3
5 up(16) <- 4, 2       # (32, H/2, W/2) and (16, H, W) -> (16, H, W)
6 seg_output
```

7. Joining two inputs (two images → one). Two band groups (Sentinel-2 and Sentinel-1, or the same place at two dates) start from two Input blocks. *Different* data gets its own branch, joined by Concat (fusion, Section 10.14). *The same* data at two dates should be described in the same terms: the second branch reuses the first branch's layers with **Shared weights**, and **Subtract** (absolute) compares them (Example 4).

6.5 From the questions to a design

If you want ...	build ...
one class per photo, little data	3–4 stages of units, Global pool, Dropout, Output; or a demo started from ImageNet weights
one class per photo, more accuracy	the same with residual blocks (Example 1); add channel gates; widen with `let w`
to use attention on images	a convolution stem down to 8–16 positions across, then To tokens and 2–6 encoder blocks (Example 2)
a class for every pixel	an encoder of stages and a decoder of steps with skips, ending at the mask's size (Example 3)
to compare two dates	a twin branch with shared weights and Subtract (Example 4)
two sensors together	one Input per band group, a branch each, Concat (Section 10.14)
several labels per image, or numbers	the same body as for one class; only the Dataset tab's task changes (the Output adapts)
images over time	Conv3D or Temporal pool first (Section 10.12), then any image body
a layer that is not in the list	a Custom block in PyTorch (Section 8.1)

6.6 Example 1: a classifier for satellite photos

The job. EuroSAT RGB [12] (Hugging Face `timm/eurosat-rgb`): 27 000 Sentinel-2 photos of 64×64 pixels in 10 land-cover classes (forest, river, highway, ...). One class per photo.

The decisions.

- Head and bridge: one class per image → Output, after Global pool.
- Size: 64 pixels to 8 needs three halvings. The first is done at once by the stem (a unit with stride 2): the finest details of a 10-metre pixel matter less than the texture of a field or a town, and every later layer costs a quarter as much.

- Body: three stages of one residual block each (32, 64, 128 channels at 32×32, 16×16, 8×8), each with a channel gate, since land cover is very much about *which* textures are present.
- Size of the whole: 27 000 images suit a few hundred thousand parameters: about 0.4 million here. Dropout 0.2 before the Output.

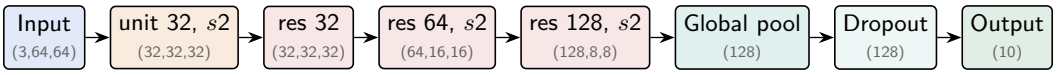


Figure 8: Example 1 at the level of motifs ("res" is the residual block of Figure 7 with a channel gate); the shapes are what each part gives. On the canvas it is 49 blocks.

build_classifier.nnb

```
net "Satellite photo classifier"
# EuroSAT RGB: 3×64×64 photos, 10 land-cover classes.
# Three motifs: a unit, a gate, a residual block.
let w = 32

def unit(c, s = 1) {        # convolution -> BatchNorm -> ReLU; s = 2 halves the size
  a conv c kernel=s+2 stride s pad 1 none nobias
  b bn
  r relu
}

def gate(c) {               # squeeze-and-excitation: weigh each channel by the image
  g global_pool
  d dense c // 4
  e dense c sigmoid
  m mul <- in, e
}

def res(c, s = 1) {         # residual block: two convolutions, the gate, a shortcut
  u unit(c, s)
  v conv c 3x3 none nobias
  n bn
  q gate(c)
  p avgpool s <- in                # the shortcut starts from the block's input
  k conv c 1x1 none nobias
  kn bn
  j add <- q, kn
  o relu
}

1 input
2 unit(w, 2)                       # stem: (32, 32, 32)
3 for c, s in (w, 1), (w * 2, 2), (w * 4, 2) {   # stages at 32×32, 16×16, 8×8
    x res(c, s)
  }
4 global_pool                      # (128)
5 dropout 0.2
6 output
```

Reading it. The three functions are the motifs 1, 4 and 3. The line labelled 3 is the body: a `for` over (channels, stride) that calls `res` three times, each pass taking the previous pass's output. A residual block with stride 1 and unchanged channels would not need its shortcut convolution; it is kept so that one function serves every stage (it costs 1 088 parameters).

The numbers. 393 122 parameters, 58 M multiply–accumulates per image. Where they are:

Part	Output	Parameters
stem: unit 32, stride 2	$(32, 32, 32)$	1 600
stage 1: residual block 32 + gate	$(32, 32, 32)$	20 200
stage 2: residual block 64, stride 2 + gate	$(64, 16, 16)$	74 192
stage 3: residual block 128, stride 2 + gate	$(128, 8, 8)$	295 840
Global pool, Dropout, Output	(10)	1 290
total		393 122

Three quarters of the parameters sit in the last stage, as usual: its convolutions have the most channels. The first stage has few parameters but the most computation (it works on 32×32 positions).

Training it. **Use recommended** (AdamW, cosine schedule with warm-up, label smoothing), 30–50 epochs, batch 128, validation 10 % with early stopping; augmentation: horizontal *and* vertical flips (a satellite photo has no "up"). Kaggle, one GPU: it is small.

Changing it. Wider: `let w = 48`. Deeper: a second call in the loop, `y res(c)` after `x res(c, s)` (this is the Small ResNet of Section 9). More bands (EuroSAT-MS, 13 bands): nothing changes in the text; only the Input's shape and the stem's parameters grow. Larger images (e.g. 128 pixels): one more stage, `(w * 8, 2)`.

6.7 Example 2: convolutions, then a Transformer

The job. CIFAR-10: 50 000 photos of 32×32 pixels, 10 classes. This example grows TEST 25 (Section 17: one patch convolution, To tokens, Token pool, 25 % accuracy) into a network that uses its tokens.

The decisions.

- Why TEST 25 stayed at 25 %: To tokens only reshapes, and Token pool (mean) averages the 64 patch tokens straight away. Nothing ever relates one patch to another, and a single 4×4 convolution sees only its own patch. The network is too *simple*, not too small: its training accuracy was as low as its test accuracy.
- The fix is two things. Convolutions first (a stem), so that each position already describes its neighbourhood: two units at 32×32, a Pool, one more unit, then a 2×2 convolution with stride 2 that makes an 8×8 grid of 128-value positions. Then encoder blocks, so that positions exchange information: four, width 128, 4 heads.
- Bridge: To tokens (64 tokens), Class token + position (65 tokens: attention ignores order unless positions are added); at the end LayerNorm and Token pool (class token).
- Size: 0.72 million parameters for 50 000 images.

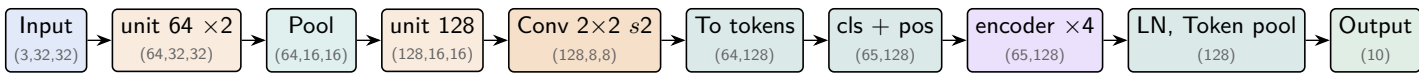


Figure 9: Example 2: image kind on the left, tokens in the middle, a vector at the end (Figure 6).

build_hybrid.nnb

```
net "Convolutions, then a Transformer"
# CIFAR-10, 3×32×32. Convolutions find local patterns cheaply; the Transformer then
# relates the 64 positions to each other.
let d = 128

```

```
def unit(c) {                       # convolution -> BatchNorm -> ReLU, size kept
  a conv c 3x3 none nobias
  b bn
  r relu
}

1 input
2 unit(64)                          # (64, 32, 32)
3 unit(64)
4 maxpool 2x2                       # (64, 16, 16)
5 unit(d)                           # (128, 16, 16)
6 conv d 2x2 stride 2 valid none    # (128, 8, 8): each 2×2 patch one position
7 to_tokens patches                 # (64, 128): 64 tokens of 128 values
8 cls_pos                           # (65, 128): class token first, positions added
9 repeat 4 {                        # four pre-norm encoder blocks
    a ln
    b attention heads 4 drop 0.1
    c add <- in, b
    e ln
    f dense d * 2 gelu
    g dense d none
    h add <- c, g
  }
10 ln
11 token_pool cls                   # (128): the class token's vector
12 output
```

The numbers. 718 410 parameters, 101 M multiply–accumulates per image: the stem 178 496 (with the 2×2 convolution), Class token + position 8 448, each encoder block 132 480 (attention 66 048, MLP 65 920, two LayerNorms 512), the end 1 546. Each line's shape can be checked against Figure 6: the block labelled 6 is the last of image kind, 7 the first of token kind, 11 the first vector. (The labels, not the listing's grey line numbers.)

Training it. **Use recommended** (Transformers need the warm-up and weight decay more than CNNs do), 50–100 epochs, batch 128, horizontal flip and random crop, validation 10 %. On one T4 an epoch should take seconds, not minutes; one GPU is enough.

Changing it. To see what the stem is worth, delete the lines labelled 2–5 and make 6 `conv d 4x4 stride 4 valid none`: a pure small ViT, which on CIFAR-10 from scratch usually does clearly worse. More encoder blocks: `repeat 6`. Wider: `let d = 192` with `heads 6` or `heads 3` (192 must divide by the heads). A mean of all tokens instead of the class token: `token_pool mean` (and `cls_pos nocls`).

6.8 Example 3: a water map from Sentinel-2

The job. Six Sentinel-2 bands (B2, B3, B4, B8, B11, B12) in 128×128 patches, each with a mask of the same size: water or not water for every pixel. Dataset tab: **My own files**, task **Segmentation**, images and masks folders (Section 11).

The decisions.

- Head: a class per pixel → Segmentation output, which must receive an image at the mask's size, 128×128.

- Body: an encoder that halves three times (128 → 16), so that one value at the narrowest level sees 68 × 68 input pixels (a river's banks, a lake's shape; the inspector shows this receptive field), and a decoder that doubles three times back to 128. A skip connection at every size brings back the sharp edges.
- Motifs: `block` (two units) at every level, `down` (Pool + block) in the encoder, `up` (Upsample + Concat with the skip + block) in the decoder: motifs 1, 2 and 6.
- Bilinear Upsample instead of ConvTranspose2D: no parameters and no checkerboard patterns; the block after it learns what is needed. Dropout 0.3 at the narrowest point, where the network holds the most channels.

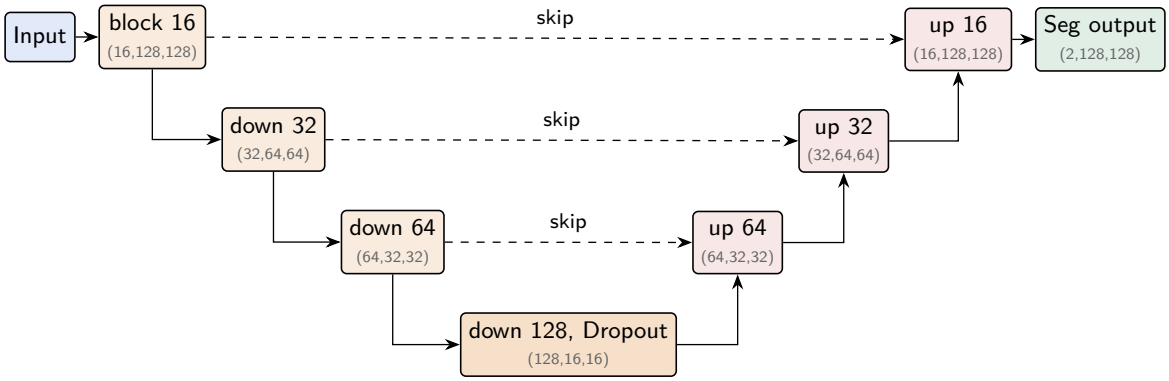


Figure 10: Example 3, a U-Net: down the left, up the right, a skip (Concat) across at every size.

build_segment.nnb

```
net "Water from Sentinel-2"
# 6 Sentinel-2 bands, 128×128 patches with a mask each (water / not water).
# An encoder that shrinks, a decoder that grows back.
def unit(c) {
  a conv c 3x3 none nobias
  b bn
  r relu
}
def block(c) {                      # two units: the step every level makes
  a unit(c)
  b unit(c)
}
def down(c) {                       # half the size, more channels
  p maxpool 2x2
  b block(c)
}
def up(c) {                   # twice the size, joined with the encoder's map (in2)
  u upsample x2 bilinear
  j concat <- u, in2
  b block(c)
}

1 input                             # (6, 128, 128)
2 block(16)                         # (16, 128, 128)
3 down(32)                          # (32, 64, 64)
4 down(64)                          # (64, 32, 32)
5 down(128)                         # (128, 16, 16): the bottleneck
6 dropout 0.3
7 up(64) <- 6, 4                    # (64, 32, 32)
8 up(32) <- 7, 3                    # (32, 64, 64)
9 up(16) <- 8, 2                    # (16, 128, 128)
10 seg_output                       # (2, 128, 128): a score per class per pixel
```

Reading it. `up(64) <- 6, 4` gives the call two inputs: the line above's output (`in`, the dropout's $(128, 16, 16)$) and that of the line labelled 4 (`in2`, the encoder's $(64, 32, 32)$). After Upsample the two have the same height and width, so Concat can join them: $(128 + 64, 32, 32) = (192, 32, 32)$, which the block brings down to 64 channels.

The numbers. 488 450 parameters, 675 M multiply–accumulates per patch (the full-size levels are the expensive ones). The encoder holds 294 432 parameters, the decoder 193 984, the Segmentation output 34 (a 1×1 convolution from 16 channels to 2 classes).

Training it. **Use recommended**, batch 16–32, horizontal and vertical flips (applied to image and mask alike), validation 10 % and early stopping on mIoU. Water is often a small part of each patch: look at **Unbalanced classes** (Section 14.3) and at the per-class IoU in Results, not only at pixel accuracy.

Changing it. Smaller objects or larger patches: one more level (`down(256)` and `up(128)`). Fewer parameters: start at 8 channels. More classes (a land-cover map): only the Dataset tab changes; Segmentation output follows the classes. Sentinel-1 radar as well: a second Input and a branch, joined at the bottleneck with Concat (Section 10.14).

6.9 Example 4: what changed between two dates

The job. Pairs of Sentinel-2 patches of the same place (4 bands, 64×64) before and after: has the land changed (cleared, built on, flooded) or not? The Dataset tab gives the "before" folder as band group 1 and the "after" folder as band group 2 with the same bands (Section 10.13).

The decisions.

- Two Inputs (one per band group). Both images should be described *in the same terms*, otherwise their difference measures the two branches, not the land. So the second branch uses the first branch's layers: **Shared weights**. Build the "before" branch, select its last block, press **Twin branch (shared weights)**, and the "after" branch appears.
- The comparison: **Subtract** with *absolute difference*: large where the two descriptions differ, near zero where nothing changed.
- After it, one more convolution to judge the differences in context, then Global pool and Output (two classes).

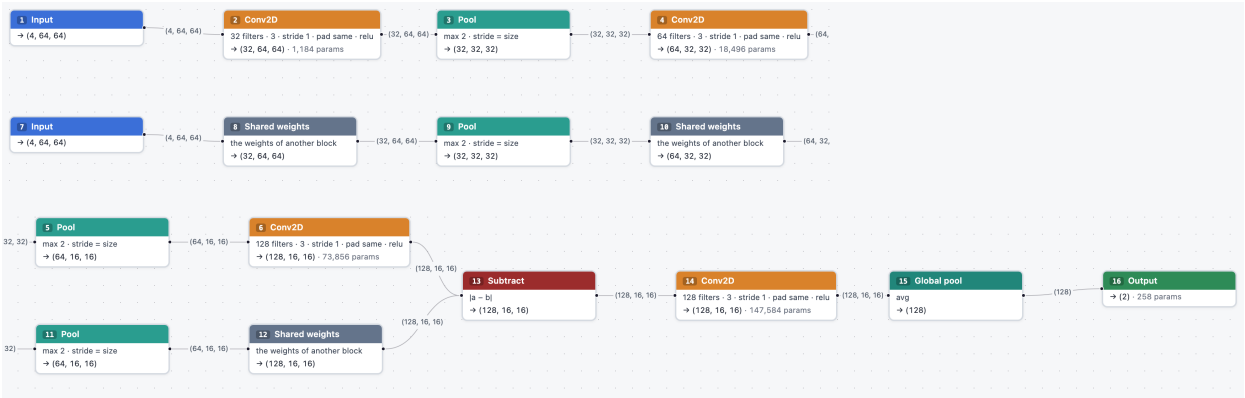


Figure 11: Example 4 on the canvas (drawn by the application; the right half is shown under the left). The upper branch's convolutions (2, 4, 6) are used again by the lower branch's Shared-weights blocks (8, 10, 12); Subtract (13) compares the two at 16×16.

build_change.nnb

```
net "What changed between two dates"
# Two Sentinel-2 images of one place (4 bands, 64×64): band group 1 before,
# band group 2 after. Classes: changed / unchanged.
1 input 1                          # before: (4, 64, 64)
2 conv 32 3x3
3 maxpool 2x2                      # (32, 32, 32)
4 conv 64 3x3
5 maxpool 2x2                      # (64, 16, 16)
6 conv 128 3x3                     # (128, 16, 16)
7 input 2                          # after: the same bands
8 shared 2                         # block 2's convolution with block 2's weights
9 maxpool 2x2
10 shared 4
11 maxpool 2x2
12 shared 6
13 subtract abs <- 6, 12           # |before - after|: large where something changed
14 conv 128 3x3
15 global_pool                     # (128)
16 output                          # changed / unchanged
```

The numbers. 241 378 parameters. The twin branch adds none: its three Shared-weights blocks show "the weights of another block" and 0 parameters, and the convolutions 2, 4 and 6 (93 536 together) are trained by both images. Convolution 14 (147 584) and the Output (258) are the rest.

Training it. **Use recommended**, flips and turns (applied to both images alike), validation 10 %. Changed pairs are usually rare: see Section 14.3.

Changing it. A change *map* instead of one answer per pair: masks in the Dataset tab (segmentation), and after Subtract a decoder of `up` steps to the image's size and a Segmentation output, as in Example 3 (the skips can come from either branch). Keep both descriptions instead of their difference: `concat <- 6, 12` instead of the Subtract.

6.10 When a design does not learn

What the Results show	What to change
training and validation accuracy both low, and close	the network is too simple or too small: add stages or blocks, widen it, bring attention or a stem (Example 2); check the receptive field covers the objects
training accuracy high, validation much lower	too large for the data: more augmentation, Dropout, weight decay, fewer channels; or ImageNet weights
the loss does not fall at all	learning rate too high or too low (try 0.001 with AdamW), missing normalisation in a deep network, or the data: run **Check data** (Section 14)
the loss is NaN	lower the learning rate, switch on gradient clipping (part of **Use recommended**)
"CUDA out of memory"	smaller batch, **Save memory**, fewer channels at the full-size levels
segmentation edges blurred	skip connections missing or joined at the wrong size; fewer halvings

7 The arithmetic behind the numbers

The engine computes everything below exactly as PyTorch does; on every change to the code, an automatic test on GitHub builds hundreds of random designs (and all demos) in real PyTorch and compares every shape and parameter count with the application's (most recently: 296 designs, 0 differences). A good visual guide to this arithmetic is Dumoulin and Visin [8].

7.1 Output size

For a convolution or pooling layer with input size n (height or width), kernel k, stride s, padding p and dilation d:

$$n_{\text{out}} = \left\lfloor \frac{n + 2p - d\,(k-1) - 1}{s} \right\rfloor + 1. \tag{1}$$

- **same**: padding chosen so that, with stride 1, $n_{\text{out}} = n$ (for an odd kernel, $p = d(k-1)/2$). With stride $s > 1$ the application computes the explicit padding PyTorch needs (it refuses `padding="same"` with a stride), giving $n_{\text{out}} = \lceil n/s \rceil$.
- **valid**: $p = 0$.
- **a number**: that many pixels of zeros on each side.

Dilation [46] spreads the kernel: a 3×3 kernel with dilation 2 covers 5×5 pixels with 9 weights. For a transposed convolution (up-sampling):

$$n_{\text{out}} = (n-1)\,s - 2p + d\,(k-1) + p_{\text{out}} + 1, \tag{2}$$

so a 2×2 kernel with stride 2 exactly doubles the size. Upsample multiplies the size by its scale.

7.2 Computation

The Design screen's summary also gives the **multiply–accumulate operations (MACs) per image**: convolutions (weights × output positions), dense layers (per token on a sequence) and attention ($4Nd^2 + 2N^2d$); normalisation, activations, pooling and joins are counted as zero. ResNet-18 at 224 × 224 comes to the usual 1.8 G.

7.3 Parameters

Block	Parameters	Example
Conv2D	$k_h k_w C_{\text{in}} C_{\text{out}} + C_{\text{out}}$ (bias)	3×3, 64→128: 73 856
Depthwise Conv2D	$k_h k_w C_{\text{in}} m + C_{\text{in}} m$	3×3, 512 channels: 5 120
ConvTranspose2D	$k_h k_w C_{\text{in}} C_{\text{out}} + C_{\text{out}}$	2×2, 512→256: 524 544
Dense / Output	$n_{\text{in}} n_{\text{out}} + n_{\text{out}}$	4096→4096: 16 781 312
BatchNorm	$2C$ (scale and shift)	64 channels: 128
LayerNorm	$2D$	768 values: 1 536
Self-attention	$4d^2 + 4d$ (query, key, value, output)	$d = 768$: 2 362 368
Class token + position	$d + (N+1)\,d$	196 tokens, $d = 768$: 152 064
Pool, Global pool, Flatten, Dropout, Activation, joins	0	

Convolutions followed by BatchNorm are created without a bias in the demos (the BatchNorm shift makes it redundant), as torchvision does.

7.4 Receptive field

The receptive field of a block is the size of the input region that one of its output values can see. With $r_0 = 1$ and $j_0 = 1$ (the "jump", the distance in input pixels between neighbouring outputs), each layer with kernel k, dilation d and stride s gives

$$r_\ell = r_{\ell-1} + d\,(k-1)\,j_{\ell-1}, \qquad j_\ell = j_{\ell-1}\,s. \tag{3}$$

The inspector shows it for every block ("one output value here sees 99×99 input pixels"). A network should usually reach a receptive field about the size of the objects it must recognise; the *effective* receptive field is smaller and concentrated in the centre [27].

7.5 Memory needed for training

A useful estimate for training with Adam [19] in mixed precision [29] is about **16 bytes per parameter** before any images: 4 for the weights, 4 for the gradients and 8 for Adam's two running averages. On top come the activations kept for the backward pass, which grow with the batch size and the image size. Examples: ResNet-18 (11.2 M parameters) needs about 0.2 GB for its state; ViT-Base (86 M) about 1.4 GB; ViT-g/14 (1.01 B) about 16 GB, more than a 16 GB T4 has. If a run fails with "CUDA out of memory", halve the batch size; beyond that, gradient checkpointing [3] and sharding the optimizer state over several GPUs [32] are the standard remedies (not built into NN Builder).

8 Reference of all blocks

Block	Shape in → out	Settings and notes
Input	$\to (C, H, W)$	set by the dataset (channels, height, width)
Conv2D	$(C, H, W) \to (F, H', W')$	filters, kernel (with a "square" link: unlink for height×width), stride, padding (same / valid / number), dilation, activation, bias; presets: 3×3 keep size, 3×3 stride 2 (halve), 1×1 (mix channels), 5×5, 7×7 stride 2 (ResNet stem)
Depthwise Conv2D	$(C, H, W) \to (Cm, H', W')$	one filter per input channel (channel multiplier m); with a following 1×1 Conv2D it forms a depthwise separable convolution [15]
ConvTranspose2D	$(C, H, W) \to (F, H'', W'')$	learnt up-sampling; filters, kernel, stride, padding, output padding
Pool	$(C, H, W) \to (C, H', W')$	max or average; pool size; stride (default: same as size); padding
Global pool	$(C, H, W) \to (C)$	average or max over the whole image
Upsample	$(C, H, W) \to (C, sH, sW)$	scale, nearest or bilinear; no parameters
BatchNorm	same shape	normalises each channel over the batch [18]; images and vectors only
LayerNorm	same shape	normalises each sample [1]; on tokens, over the dimension; on an image, over (channels, height, width)
GroupNorm	same shape	normalises each image over groups of channels; independent of the batch, so good for small batches; the channels must divide by the number of groups (often 32)

Block	Shape in → out	Settings and notes
Dropout	same shape	rate: the fraction of values set to zero during training [38]
Flatten	$(C, H, W) \to (CHW)$	
Dense	$(n) \to (u)$; tokens $(N, D) \to (N, u)$	units, activation; on tokens it acts on each token (the transformer MLP)
Activation	same shape	ReLU, Leaky ReLU, GELU [13], SiLU [33], sigmoid, tanh
Add	equal shapes → same	element-wise sum: residual connections [10]
Concat	joins along channels	images need equal height and width; the arrow order is the channel order
Multiply	see notes	equal shapes; a one-channel map times an image (attention gate [30]); a per-channel vector times an image (squeeze-and-excitation [16])
To tokens	$(C, H, W) \to (N, D)$	*patches*: every pixel of the feature map becomes a token of C values ($N = HW$), used after a patch convolution as in ViT [7]; *rows*: every image row is a token
Class token + position	$(N, D) \to (N{+}1, D)$	a learnt class token in front, and learnt position embeddings added
Self-attention	$(N, D) \to (N, D)$	heads, dropout; multi-head scaled dot-product attention [42]
Token pool	$(N, D) \to (D)$	the class token, the mean of all tokens, or the last one (after an LSTM / GRU)
Cross-attention	$(N, D), (M, D_2) \to (N, D)$	the first input's tokens attend to the second's (queries; keys and values); heads, dropout
RMSNorm	same shape	per token: divided by its root mean square, times a learnt scale (LLaMA)
SwiGLU	same shape	the gated MLP $W_2(\text{silu}(W_1 x) \cdot W_3 x)$; hidden units
Output	$(n) \to (K)$	K = number of classes; softmax/cross-entropy is in the training script
Segmentation output	$(C, H, W) \to (K, H, W)$	a 1×1 convolution giving a class score for every pixel; must match the mask size
Early exit	$(n) \to (K)$	an extra classifier part way through, trained with the same labels (its loss × weight); Results show what stopping confident images there would give
Mixture of experts	same shape	a router sends every token to its top k of E expert MLPs; a load-balancing loss keeps all in use
Detection output	$(C, h, w) \to (K{+}4, h, w)$	anchor-free boxes (CenterNet [51]): per cell a score per class, the box's width and height and the centre's offset; the image must be a whole number of cells (Section 10.11)
Conv3D	$(C, T, H, W) \to (F, T', H', W')$	a time series: filters, kernel (pixels), dates per window, stride (pixels), padding same / valid (Section 10.12)
Temporal pool	$(C, T, H, W) \to (C, H, W)$	the dates of every pixel combined: mean, maximum, or attention (weights learnt per pixel, as in U-TAE [48])
To sequence	$(C, T, H, W) \to (T, C)$	the mean of every date's pixels: one vector per date
Conv1D	$(T, D) \to (T', F)$	a convolution along a sequence: filters, kernel, stride, padding

Block	Shape in → out	Settings and notes
LSTM, GRU	$(T, D) \to (T, u)$	recurrent layers along a sequence [50]; units, both directions ($2u$ out); Token pool *last* keeps the last step
Subtract	equal shapes → same	the first input minus the second, or $\lvert a - b \rvert$: the difference of two branches (Siamese change detection)
Shared weights	as the shared block	runs another block's layer, with the same weights, on its own input of the same shape; the weights count once (Siamese / twin networks, Section 10.13)
Custom (PyTorch)	declared by the class	runs a class from the design's own PyTorch code (an `nn.Module` given the input shape; a `# out:` line declares the output shape); the inspector edits the code; checked on the training machine (Section 8.1)

8.1 Your own layers in PyTorch (Custom block)

When a layer is not in the block list (a new attention variant, a gate from a paper, a pixel shuffle), write it in PyTorch. Add a **Custom (PyTorch)** block (palette: *Your own*) and select it. The inspector shows the design's PyTorch code, shared by all its Custom blocks, and a **Class** list with the classes in it; **Insert an example** adds a squeeze-and-excitation class to start from. Each class is an `nn.Module` at the top level of the code:

- `__init__(self, c, h, w)` receives the shape of the block's input (`t, d` for tokens, `n` for a vector), so the class can size its layers;
- a line `# out: c * 2, h // 2, w // 2` in the class declares the output shape in those names (without it, the output has the input's shape). NN Builder uses it to check and size the rest of the design;
- `forward(self, x)` returns the output. `torch` and `nn` are imported; import anything else at the top.

The Mac never runs this code. It is copied into `train.py` above the network, and on the training machine the one-batch check compares each Custom block's real output shape with the declared one before any training, stopping with the block's name if they differ. The parameters of Custom blocks are counted there too (the Design tab's total leaves them out). In the Text tab the code is the `python """ ... """` section and the block is `custom CLASS` (Section 9.11).

The loss can be your own as well: **Loss: your own (Python)** in the Train tab opens a box with `def custom_loss(out, y):` (an example for the task is filled in). `out` is the network's output for a batch and `y` the labels as the data gives them (class numbers; 0/1 per class for multi-label; a class per pixel with −100 for unlabelled pixels; the standardised targets for regression); it returns one number. The validation loss uses it too.

8.2 Transformer options: masks, positions, LLaMA-style blocks, cross-attention

Self-attention has two switches: **causal** (each token sees only itself and the tokens before it, as in a decoder or for a time series that must not look ahead) and **rotary positions** (RoPE: the queries and keys are rotated by an angle that grows with the position, so attention depends on how far apart two tokens are; use it with **Positions: none** in Class token + position). **Positions** can also be fixed sines and cosines (no parameters). **RMSNorm** and **SwiGLU** give LLaMA-style blocks (RMSNorm

→ attention → Add, RMSNorm → SwiGLU → Add). **Cross-attention** joins two token streams: the first arrow's tokens ask, the second's are looked at, e.g. Sentinel-2 patch tokens attending to Sentinel-1 tokens (mid fusion with attention instead of Concat), or one date's tokens attending to the others. Concat joins token sequences one after the other. In a time series, dates with no data at all (every pixel empty) are filled from the dates before and after, so attention needs no mask for them; a token embedding for text is not part of NN Builder (it trains on images).

8.3 Frozen layers, LoRA, experts and early exits

Frozen. Every block with weights has a **frozen** switch (text: the word `frozen`): its weights stay as they are, e.g. the first layers of a network started from ImageNet weights. The Design tab shows how many weights are trained and how many frozen.

LoRA. Dense and Self-attention have a **LoRA rank**. The layer's own weights are frozen and a low-rank change $B \cdot A$ (rank r, B starting at zero) is learnt instead [52]: with ImageNet weights and **train only the new layers**, the loaded weights stay and only the adapters and the new head learn, a small fraction of the weights. Text: `dense 64 lora 8`, `attention heads 8 lora 4`.

Mixture of experts. The **Mixture of experts** block (tokens or vectors) holds E small expert MLPs and a router; every token goes to its top k experts, whose outputs are mixed by the router's weights (Switch, Mixtral). Many more weights at the cost of k experts per token; a load-balancing loss (Switch Transformer, $\times 0.01$) keeps every expert in use. Text: `moe 8 top 2 hidden 256`.

Two options from GShard [6]. **Capacity** (0 = no limit) caps the tokens each expert takes in a batch at capacity × its fair share, $\lceil \text{capacity} \cdot Nk/E \rceil$ for N tokens; first choices are served before second choices, and a token over the limit skips that expert. A token that gets no expert at all leaves the block as zeros, so put the block inside a residual **Add** (as in a Transformer layer) and the token goes on unchanged. Values from 1 to 2 are usual (GShard used 2); below 1 drops tokens even when the router is balanced, and the block says so. **Random second expert**: while training, a token goes to its second expert only with probability $2\times$ that expert's weight, which spreads the load and saves work; evaluation always uses both. Text: `moe 8 top 2 hidden 256 cap 1.25 random`. GShard also places the experts on different GPUs (expert parallelism, all-to-all exchange of tokens); NN Builder's several-GPU training instead gives every GPU all the experts, which gives the same results while the experts fit in one GPU's memory.

Early exits. An **Early exit** after a Global pool part way through a classifier is an extra Output trained with the same labels (its loss times its **share**). Results list each exit's accuracy and what stopping every image at the first exit that is at least 90 % sure would give: the accuracy and the share of images leaving at each exit, i.e. how much computation could be saved. Text: `exit 0.3`.

8.4 Several GPUs

On a machine with several GPUs (a college server, a multi-GPU EC2 instance such as g4dn.12xlarge or g5.12xlarge, a Kaggle notebook with two T4s) the Train tab's **GPUs** can be *all the machine has* or a number. `train.py` then starts one worker per GPU itself (nothing else changes in how the run is started) and trains with PyTorch's DistributedDataParallel: every worker holds the whole network and takes its share of every batch, the gradients are averaged after each step, and BatchNorm counts the images of all GPUs. The batch size stays the total, so the training goes as on one GPU, in about $1/N$ of the time when the data loading keeps up. Worker 0 evaluates, writes the events, checkpoints and results; the stop button, the spot interruption notice and the timed checkpoints reach every worker (the run's process forwards them), all stop at the same step, and a resumed run continues at the same batch. With several GPUs a batch that does not fit is

not split automatically (use a smaller batch or **save memory**); a data check uses one GPU. Kaggle may give a notebook two T4s (the accelerator "GPU T4 x2"); with **GPUs: all** a Kaggle run then trains on both (tried: two T4s, R^2 0.998 in 28 seconds). A worker that waits for another more than an hour ends the run, and once worker 0 has written the results the others are ended within a minute, so a run never stays open after it finished. A model too big for one GPU's memory (FSDP, splitting the weights over the GPUs) is not supported yet.

8.5 Transformers in NN Builder

A transformer works on a *sequence of tokens*, shape (N, D): N tokens of D values. To use one on images:

1. cut the image into patches with a convolution whose kernel equals its stride (ViT: 16×16, stride 16, D filters), giving $(D, H/16, W/16)$;
2. **To tokens** (patches) turns that map into $N = (H/16)(W/16)$ tokens;
3. **Class token + position** adds a learnt class token and position information (attention by itself ignores order);
4. repeat the *encoder block*: LayerNorm → Self-attention → Add (with the block's input), then LayerNorm → Dense (GELU, usually $4D$ units) → Dense (D) → Add. Normalising before each part ("pre-norm") trains more stably than after it [45];
5. a final LayerNorm, **Token pool** (class token), **Output**.

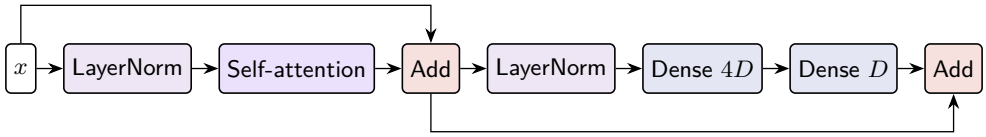


Figure 12: A pre-norm Transformer encoder block as it is built on the canvas.

8.6 Segmentation

For segmentation (Dataset → **Segmentation**) every pixel gets a class. The network must end with **Segmentation output** at the size of the masks, so an encoder that shrinks the image needs a decoder that grows it again (ConvTranspose2D or Upsample), as in U-Net [34]. The training script uses a per-pixel cross-entropy loss and reports pixel accuracy and the mean intersection over union (mIoU) of the classes.

9 Writing designs as text: the NNB language

Every design can also be written as text, in the **Text** tab. The text and the canvas are two views of the same design and are kept in step both ways: typing changes the canvas a moment after the text is valid, and a change on the canvas changes the text. This chapter teaches the language from the first line to functions and loops, then lists every block and every rule.

Why write text at all? A network of fifty blocks is fifty short lines, typed in a minute; every number of every block is visible at once; repeated parts are written once (`repeat 40 { ... }`); and the text can be copied into a paper, an e-mail or a chat and pasted back unchanged (**Copy**, or **File** → **Export Design...** as a `.nnb` file).

9.1 A first design, line by line

The smallest useful design is a chain of blocks. LeNet-5 (Section 16) for the $1 \times 28 \times 28$ MNIST digits:

lenet5.nnb

```
net "LeNet-5"
# MNIST digits, 1×28×28. One block per line: a label, the block, its settings.
1 input
2 conv 6 5x5 pad 2 tanh      # 6 filters, 5×5 kernel, padding 2 keeps 28×28
3 avgpool 2x2                # 28×28 -> 14×14
4 conv 16 5x5 valid tanh     # no padding: 14 -> 10
5 avgpool 2x2                # 10 -> 5
6 flatten                    # 16×5×5 = 400 values
7 dense 120 tanh
8 dense 84 tanh
9 output                     # one score per class (10 for MNIST)
```

- `net "LeNet-5"` gives the design a name (optional; see Section 9.2). It is not a block: it makes nothing on the canvas.
- Each further line is one block: a **label** (`1`, `2`, ...), the **block** (`conv`, `avgpool`, ...) and its **settings** in any order. Left-out settings take the same defaults as the canvas (a `conv` without a kernel is 3×3, without an activation it uses ReLU, ...).
- A block takes its input from **the line above** unless it says otherwise.
- `#` starts a comment, to the end of the line. Blank lines and indentation do not matter.
- The first block is `input`: its shape comes from the Dataset tab, not from the text. `output` gets as many classes as the dataset has.

Typed into the Text tab, this draws the nine blocks on the canvas, each with its label in its header, and the shapes and parameters are those of Section 7: 61 706 parameters.

9.2 The design's name: `net`

```
net "NAME"
```

A line `net` followed by a name in double quotes names the design, for example `net "LeNet-5"` or `net "ResNet-18, width 32"`. It is optional and is usually the first line. What it does:

- it says what the text is when the text travels on its own: in a `.nnb` file, pasted into a paper, an e-mail or a chat;
- the generated `train.py` carries it in its first line, after the project's name (`My project (design LeNet-5): generated by NN Builder ...`), so a script says which design it trains;
- it stays when the text is printed again (**Format**, or after a change on the canvas).

It does not rename the project (the project's name is set when the project is created or saved, and shown in the top bar), and it does not create a block. The name is any text without a double quote; if there are several `net` lines, the last one counts. It may only stand at the top level, not inside braces.

9.3 Settings: words and numbers

Settings are written the way you would say them. For a convolution, the first plain number is the number of filters, `5x5` is the kernel, and words set the rest. In general (capitals are your values, `[ ]` is optional, `|` means one of):

```
LABEL conv FILTERS HxW [stride S] [pad P | same | valid] [dil D] [ACTIVATION] [nobias]
      ACTIVATION: relu | leaky_relu | gelu | silu | sigmoid | tanh | none
```

For example:

```
2 conv 64 3x3                       # 64 filters, 3×3, stride 1, padding "same", ReLU
3 conv 128 3x3 stride 2 pad 1 gelu # stride 2, padding 1, GELU
4 conv 128 1x1 valid none nobias   # no padding, no activation, no bias
5 conv 64 3x3 dil 2                # dilation 2
6 conv filters=64 kernel=7x7 stride=2 padding=3   # any setting as name=value
```

Every setting also has a *name*=*value* form (the names are those of the inspector), useful when a value is computed (`kernel=s+2`, below). A word that a block does not know is an error, never ignored: `conv 64 stide 2` says "`'stide' means nothing for conv (did you mean 'stride'?)`". A value outside its range (a kernel of 40, 100 000 filters) is an error on its block.

9.4 Connections: `<-`

A block with a different input, or with several (Add, Concat, Multiply), names them after `<-`:

```
1 input
2 conv 64 3x3
3 conv 64 3x3 none
4 add <- 2, 3          # residual: the output of block 2 plus that of block 3
5 relu
6 conv 32 1x1 <- 2     # a side branch from block 2 (not from the line above)
7 concat <- 5, 6       # channels joined in this order: 5's first, then 6's
```

A block may only use blocks above it (a network computes each block once, from the blocks before it), so the text can never contain a loop. `<- none` makes a block without input (shown as not connected).

9.5 Labels

Labels are names, not positions: `1, 2, 5, 9` is fine, and so is a word (`stem conv 32 3x3`, then `<- stem`). The label may also be left out: `conv 32 3x3` on its own gets the next free number. Two lines may not have the same label. The labels are what the canvas shows in each block's header and what the layer table uses, so canvas, text and table name blocks the same way.

9.6 Names and arithmetic: `let`

`let` gives a value a name; every number may be an expression of names and numbers with `+ - * /`, `//` (whole-number division), `%` (remainder), `^` (power) and parentheses:

```
let w = 32
let depth = 6
2 conv w 3x3
3 conv w * 2 3x3 stride 2
4 dense 2 ^ 10 gelu        # 1024
```

Changing `w` changes every block that uses it: the quickest way to try a wider or narrower network.

9.7 Repeating a group: `repeat`

Identical parts are written once. A `repeat` line has a label, a count and the group in braces; the passes are chained (each pass takes the previous pass's output), and the line's output is the last pass's last block:

vit_small.nnb

```
net "Small ViT"
# CIFAR-10, 3×32×32: 4×4 patches give 64 tokens of d values; depth encoder blocks.
let d = 192
let depth = 6

1 input
2 conv d 4x4 stride 4 valid none     # patch embedding: (192, 8, 8)
3 to_tokens patches                  # (64, 192)
4 cls_pos                            # class token + positions: (65, 192)
5 repeat depth {                     # one pre-norm Transformer encoder block per pass
    a ln
    b attention heads 3 drop 0.1
    c add <- in, b                   # residual: the pass's input + attention
    d ln
    e dense d * 2 gelu               # MLP ratio 2
    f dense d none
    g add <- c, f
  }
6 ln
7 token_pool cls
8 output
```

Inside the braces:

- blocks have **local labels** (`a`, `b`, ...); on the canvas they become `5.1.a` ... `5.6.g` (line, pass, block);
- `in` is the group's input (for the first pass, the block above the `repeat` line);
- `i` is the pass number, 1, 2, ...: `conv 32 * i 3x3` gets wider in every pass;
- the count may be a name (`repeat depth`); `repeat 12 as enc` names the group for readers.

This ViT has 49 blocks and 1 806 538 parameters at $3 \times 32 \times 32$; ViT-g/14 is the same pattern with `repeat 40` and 16 lines in all.

9.8 A loop over values: `for`

`for` repeats a group once per value, chained like `repeat`. A value can be a tuple, for several names at once:

```
5 for c in 64, 128, 256 { a conv c 3x3 }
6 for c, s in (64, 1), (128, 2), (256, 2) {
    x conv c 3x3 stride s
    y bn
  }
```

`i` is also the pass number in a `for`.

9.9 Functions: `def`

A function is a group of blocks with values, written once and used many times. It is defined with `def`, its parameters may have defaults, and a line calls it like a block:

resnet_small.nnb

```
net "Small ResNet"
# CIFAR-10, 3×32×32. A function for the residual block, a loop over the stages.
let w = 32

# two convolutions and a shortcut. With s = 2 the first convolution is 4×4 / stride 2 / padding 1, wh
# (kernel = s + 2), and the shortcut averages 2×2 patches before its 1×1 convolution (ResNet-D); with
def basic(c, s = 1) {
  a conv c kernel=s+2 stride s pad 1 none nobias
  b bn
  r relu
  d conv c 3x3 none nobias
  e bn
  p avgpool s <- in                  # the shortcut starts from the block's input
  q conv c 1x1 none nobias
  n bn
  j add <- e, n
  o relu
}

1 input
2 conv w 3x3 none nobias
3 bn
4 relu
5 for c, s in (w, 1), (w * 2, 2), (w * 4, 2) {   # three stages: 32×32, 16×16, 8×8
    x basic(c, s)
    y basic(c)                                    # s takes its default, 1
  }
6 global_pool
7 output
```

- `basic(c, s)` passes the values by position; `basic(c)` leaves `s` at its default; `basic(c=64, s=2)` names them.
- Inside the function, `in` is the input of the call (here the line above the call); the last block is the call's output.
- A function may call other functions (up to 16 deep: a function that calls itself is reported).
- Functions may be defined anywhere in the text, also below their first use.

This network has six residual blocks (66 blocks on the canvas, 791 338 parameters): about forty lines of text.

9.10 Several inputs, and reaching into a group

A call (or a `repeat` / `for`) can take several inputs: `<- 3, 2.3.e` gives the call two, called `in` and `in2` inside. A block made inside a group or a call is named by its path: `2.3.e` is line 2, pass 3, block `e` (for a call: line, then block, as in `7.u`). Together they make U-Net's skip connections:

unet_small.nnb

```
net "Small U-Net"
# Segmentation, 3×64×64 with masks. Functions with two inputs (in, in2) and skips into a repeat by pa
def double(c) {
  a conv c 3x3
  b conv c 3x3
}
def up(c) {
  u convT c 2x2 stride 2 none    # doubles the size
  j concat <- u, in2             # in2: the skip from the encoder
  k double(c)
}
```

```
12
13 1 input
14 2 repeat 3 {                      # encoder: 16, 32, 64 filters, each halving the size after
15     e double(16 * 2 ^ (i - 1))
16     p maxpool 2x2
17   }
18 3 double(128)                    # bottleneck at 8×8
19 4 up(64) <- 3, 2.3.e             # 2.3.e = line 2, pass 3, block e (its output: 64 filters at 16×16)
20 5 up(32) <- 4, 2.2.e
21 6 up(16) <- 5, 2.1.e
22 7 seg_output
```

Line 2 builds the encoder as three passes whose width depends on the pass number ($16 \cdot 2^{i-1}$: 16, 32, 64); lines 4–6 each take the block above and the matching encoder output (25 blocks, 482 050 parameters for 64×64 images and two classes).

9.11 Every block and how its values are set

How a value is written

A block's line is its word (`conv`, `dense`, ...) followed by its values in any order. Each value can be written in a short form, and every value also in a long form *name*=*value*:

Form	Example	Sets
a plain number	`conv 64`, `dense 128`, `dropout 0.3`	the block's *main number* (filters, units, rate, ...: see each block; a second plain number is an error)
a size `HxW`	`3x3`, `5x3`, `2x2`	the kernel (convolutions) or the window (pools); height × width
`x`*n*	`x2`	a multiplier (depthwise) or a scale (upsample)
a word and its value	`stride 2`, `pad 1`, `dil 2`, `heads 8`, `drop 0.1`	that setting
a word alone	`gelu`, `none`, `same`, `nobias`, `max`, `bilinear`	a choice (activation, padding mode, bias off, pool mode, ...)
name=*value*	`filters=64`, `kernel=7`, `stride=2x1`, `bias=false`	any setting by its name (the names are those of the inspector); a single number for a size means a square (`kernel=7` is 7×7)

Wherever a number goes, an expression may go (`conv w * 2 3x3`, `kernel=s+2`). Left-out values take the defaults below, which are also the canvas's. A value outside its range, a word the block does not have, or a second plain number is an error on its line (Section 9.13); nothing is ever changed silently. Block words are not case-sensitive (`Conv2D 16 3x3` works).

Each block starts with its general form: capitals are your values, `[ ]` is optional, `|` means one of, and `ACTIVATION` is one of `relu`, `leaky_relu`, `gelu`, `silu`, `sigmoid`, `tanh`, `none`. Every value can also be given as *name*=*value*, the label may be left out, and every block except `input` may end with `<- INPUT` (by default the line above). In the tables: *short* is how the value is written briefly, *name* its long form.

`input` — the images

```
LABEL input [GROUP]
```

`1 input`: its shape (channels, height, width) comes from the Dataset tab; it takes no input. A design has one, except with two band groups (Dataset tab, Section 10.14: e.g. Sentinel-2 and Sentinel-1), where it may have one Input per group, each starting its own branch (mid fusion):

```
1 input 1                  # band group 1 (Sentinel-2)
2 conv 32 3x3
3 input 2                  # band group 2 (Sentinel-1)
4 conv 16 3x3
5 concat <- 2, 4           # the two branches joined
6 global_pool
7 output
```

value	short	name	default	allowed
band group	`1`	`group`	0	0 (all bands), 1, 2

`conv` (also `conv2d`) — 2-D convolution

```
LABEL conv FILTERS HxW [stride S] [pad P | same | valid] [dil D] [ACTIVATION] [nobias]
```

```
2 conv 64 3x3                              # 64 filters, 3×3, stride 1, "same" padding, ReLU
3 conv 128 3x3 stride 2 pad 1 gelu nobias
4 conv 32 5x3 dil 2 valid none
5 conv filters=64 kernel=7 stride=2 padding=3 activation=silu bias=false
```

Value	short	name	default	range	meaning
filters	the plain number: `64`	`filters`	32	1–65 536	output channels
kernel	`3x3`, `5x3`	`kernel`	3×3	1–32	window size
stride	`stride 2`	`stride`	1	1–32	step (`stride=2x1` for different steps)
padding	`same`, `valid`, `pad 1`	`padding`	same	0–10	`same` keeps the size; `valid` = 0
dilation	`dil 2`	`dilation`	1	1–6	spacing inside the window
activation	`relu` `leaky_relu` `gelu` `silu` `sigmoid` `tanh` `none`	`activation`	relu		applied after the convolution
bias	`nobias` (`bias`)	`bias`	true		off before a BatchNorm

`dwconv` (also `dwconv2d`) — depthwise convolution

```
LABEL dwconv [xM] HxW [stride S] [pad P | same | valid] [ACTIVATION] [nobias]
```

One filter per input channel (MobileNet). `dwconv 3x3`, `dwconv x2 3x3 stride 2 pad 1 none nobias`, `dwconv multiplier=3 kernel=5x5`.

Value	short	name	default	range	meaning
multiplier	`x2`	`multiplier`	1	1–8	output channels = input channels × multiplier
kernel, stride, padding, activation, bias	as for `conv`				(no dilation; no plain number)

`convT` (also `convtranspose`) — transposed convolution

```
LABEL convT FILTERS HxW [stride S] [pad P] [outpad O] [ACTIVATION]
```

Makes the image larger (U-Net's decoder). `convT 64 2x2 stride 2 none` doubles height and width; `convT 32 3x3 stride 2 pad 1 outpad 1` too.

Value	short	name	default	range	meaning
filters	the plain number	`filters`	32	1–65 536	output channels
kernel	`2x2`	`kernel`	2×2	1–32	
stride	`stride 2`	`stride`	2	1–32	the enlargement
padding	`pad 1`	`padding`	0	0–10	a number only (no `same`)
output padding	`outpad 1`	`output_padding`	0	0–7	extra rows / columns at the end
activation	as for `conv`	`activation`	relu		(no bias setting)

`maxpool`, `avgpool` (also `pool`) — pooling

```
LABEL maxpool | avgpool HxW [stride S | stride size] [pad P]
```

`maxpool 2x2` (halves the size), `maxpool 3x3 stride 2 pad 1`, `avgpool 3x3 stride 1 pad 1`, `maxpool 3` (3×3), `pool avg size=2 stride=size`.

Value	short	name	default	range	meaning
mode	the word `maxpool` / `avgpool`, or `max`, `avg`	`mode`	max		maximum or average of the window
size	`2x2`, or the plain number `3`	`size`	2×2	1–8	window
stride	`stride 1`, `stride size`	`stride`	size	1–32	`size`: the same as the window
padding	`pad 1`	`padding`	0	0–4	a number only

`global_pool` (also `gap`) — one value per channel

```
LABEL global_pool [avg | max]
```

`global_pool` (average over the whole image), `global_pool max`. Value: mode, `avg` / `max` (`mode=`), default avg. From (C, H, W) to (C).

`upsample` — enlarge without weights

```
LABEL upsample [xN] [nearest | bilinear]
```

`upsample x2`, `upsample 2`, `upsample x4 bilinear`. Values: scale, xn or the plain number (`scale=`), default 2, range 2–8; mode, `nearest` / `bilinear` (`mode=`), default nearest.

`bn`, `ln`, `gn` — normalisation

```
LABEL bn | ln
LABEL gn GROUPS
```

`bn` (also `batchnorm`) and `ln` (also `layernorm`) have no values. `gn 8` (also `groupnorm`, `groups=8`): GroupNorm with 8 groups, default 8, range 1–65 536; the number must divide the channels.

`dropout`

```
LABEL dropout RATE
```

`dropout 0.3`, `dropout rate=0.5`. Value: rate, the plain number (`rate=`), default 0.25, range 0–0.95: the fraction of values set to zero during training.

`flatten`

```
LABEL flatten
```

`flatten`. No values: (C, H, W) becomes one vector of $C \cdot H \cdot W$ values (before `dense`).

`dense` (also `linear`) — fully connected

```
LABEL dense UNITS [ACTIVATION]
```

`dense 128`, `dense 10 none`, `linear units=256 activation=tanh`. Values: units, the plain number (`units=`), default 128, range 1–1 048 576; activation, as for `conv` (`activation=`), default relu. On tokens it acts on every token separately (a Transformer's MLP).

Activation blocks

```
LABEL relu | leaky_relu | gelu | silu | sigmoid | tanh
```

A line with only an activation word is an activation block: `relu`, `leaky_relu`, `gelu`, `silu`, `sigmoid`, `tanh`. Also `activation gelu`, `act fn=silu` (the name of the value is `fn`).

`add`, `concat` (`cat`), `mul` (`multiply`) — joining

```
LABEL add | concat | mul <- INPUT, INPUT [, INPUT ...]
```

No values; the inputs come after `<-`: `add <- 2, 5` (same shapes, summed: residual), `concat <- 3, 7` (channels joined in this order; same height and width), `mul <- 4, 6` (element-wise; a vector of C values may multiply a (C, H, W) image: an attention gate or squeeze-and-excitation). Two or more inputs.

Transformer blocks: `to_tokens`, `cls_pos`, `attention`, `cross_attention`, `rmsnorm`, `swiglu`, `token_pool`

```
LABEL to_tokens [patches | rows]
LABEL cls_pos [cls | nocls] [learned | sinusoidal | nopos]
LABEL attention [heads] HEADS [drop RATE] [causal] [rotary]
LABEL cross_attention [heads] HEADS [drop RATE] <- ASKING, LOOKED_AT       (also xattn)
LABEL rmsnorm                                                            (also rms)
LABEL swiglu HIDDEN
LABEL moe EXPERTS [top K] [hidden UNITS] [cap FACTOR] [random]
LABEL exit [WEIGHT]                 an early exit (classification): needs a vector
... frozen                          any block with weights: not trained;  dense / attention ... lora
LABEL token_pool [cls | mean | last]
```

```
3 to_tokens patches        # (C, H, W) -> H·W tokens of C values;  "rows": H tokens of C·W va
4 cls_pos                  # a learnt class token in front + learnt positions;  "nocls": posi
5 attention heads 8 drop 0.1      # also: attention 8, attn heads=8 dropout=0.1
6 token_pool cls           # the class token's vector;  "mean": the mean of all tokens
```

`to_tokens`: mode `patches` / `rows` (`mode=`), default patches. `cls_pos`: `cls` / `nocls` (`cls=true` / `false`), default cls. `attention` (also `attn`): heads, the plain number or `heads 8` (`heads=`), default 4, range 1–512, must divide the token width; dropout, `drop 0.1` (`dropout=`), default 0, range 0–0.9; `causal`: each token sees only itself and those before it; `rotary` (also `rope`): rotary positions (RoPE) inside the attention, each head's width must be even. `cls_pos`'s positions: `learned` (default), `sinusoidal` (fixed sines and cosines; even width), `nopos` (none; `pos=`). `cross_attention`: the first input's tokens are the queries, the second's the keys and values (their widths may differ); the output has the first input's shape. `rmsnorm`: per token, LayerNorm without centring (LLaMA). `swiglu`: the gated MLP $W_2(\text{silu}(W_1x) \cdot W_3x)$ with HIDDEN units (`units=`, default 128), back to the token width. `token_pool`: `cls` / `mean` / `last` (`mode=`), default cls. Concat joins token sequences one after the other (the same width).

```
1 input
2 conv 64 4x4 stride 4 valid none
3 to_tokens
4 cls_pos nopos
5 rmsnorm
6 attention heads 4 rotary       # a LLaMA-style block: RMSNorm, rotary attention, RMSNorm, S
7 add <- 4, 6
8 rmsnorm
9 swiglu 256
10 add <- 7, 9
11 token_pool cls
12 output
```

`output`, `seg_output`

```
LABEL output [HEAD] | seg_output | detect
```

`output` gives one score per class of the dataset (classification); `seg_output` one score per class for every pixel (segmentation; it must be at the size of the masks); `detect` (`detect_output`) the boxes of an object-detection dataset, per cell of its input map (the image must be a whole number of cells). A network has one of them, at the end of its path, unless the Dataset tab lists *extra outputs* (multi-task): then each has its own `output` with its number, `HEAD` (also `head=`; default 0, the main task):

```
1 input
2 conv 32 3x3
3 global_pool
4 output                    # the main task: its classes
5 dense 32 <- 3
6 output 1                  # extra output 1 (e.g. biomass and height: two numbers)
7 output 2 <- 3             # extra output 2 (e.g. dry / wet)
```

Time series: `conv3d`, `temporal_pool`, `to_sequence`, `conv1d`, `lstm`, `gru`

```
LABEL conv3d FILTERS KxK [time T] [stride S] [same | valid] [ACTIVATION] [nobias]
LABEL temporal_pool [attention | mean | max]          (also tpool)
LABEL to_sequence                                      (also seq)
LABEL conv1d FILTERS [KERNEL] [stride S] [same | valid] [ACTIVATION] [nobias]
LABEL lstm UNITS [bi]  |  gru UNITS [bi]
```

For a time-series dataset, whose Input is (bands, dates, height, width). `conv3d`: filters (default 16), the pixel kernel (3x3), `time` = dates per window (`tkernel=`, default 3), stride over the pixels, padding same (odd kernels) or valid. `temporal_pool`: to an image (`mode=`, default attention). `to_sequence`: one vector per date. `conv1d`: filters and kernel (the first and second number; default 64 and 3), `stride`, padding. `lstm` / `gru`: units (default 64), `bi` for both directions (`bidirectional=true`). `token_pool last` keeps the last step.

```
1 input                     # (4, 6, 32, 32): 4 bands at 6 dates
2 conv3d 16 3x3 time 3
3 temporal_pool attention   # (16, 32, 32): one image, the dates weighted per pixel
4 conv 32 3x3
5 global_pool
6 seq <- 1                  # (6, 4): one vector per date
7 lstm 16 bi
8 token_pool last
9 concat <- 5, 8
10 output
```

`subtract` (`sub`), `shared` — comparing branches, shared weights

```
LABEL subtract [abs] <- A, B
LABEL shared OTHER
```

`subtract`: the first input minus the second (`abs`: $|a-b|$; `abs=true`); equal shapes. `shared OTHER`: the layer of block `OTHER` (its label) again, with the same weights, on this block's input (a Siamese network); the label may be a whole path or one of the same function call or pass. Its input must have the shape that block's input has.

```
1 input 1                   # the "before" images (band group 1)
2 conv 16 3x3
3 conv 16 3x3
4 input 2                   # the "after" images: the same bands (band group 2)
5 shared 2                  # block 2's convolution, with its weights
6 shared 3
7 subtract abs <- 3, 6      # what changed
8 global_pool
9 output
```

`custom` — your own PyTorch class

```
LABEL custom CLASS
python """   ...  """          the design's PyTorch code: one section, anywhere (usually at the top)
```

A layer the block list does not have is written in PyTorch. The code is a section that begins with a line `python """` and ends with a line holding only `"""` (inside it, write docstrings on one line or with `'''`); every class at its top level is an `nn.Module` whose `__init__` receives the input's shape (`c, h, w` for an image, `t, d` for tokens, `n` for a vector), and a comment line `# out:` declares the output's shape in those names (whole-number arithmetic: `+ - * // %`); without it the output has the input's shape. `custom CLASS` uses a class:

```
python """
class Down(nn.Module):
    # out: c * 2, h // 2, w // 2
    def __init__(self, c, h, w):
        super().__init__()
        self.conv = nn.Conv2d(c, c * 2, 3, stride=2, padding=1)

    def forward(self, x):
        return torch.relu(self.conv(x))
"""
1 input
2 conv 16 3x3
3 custom Down              # (16, 32, 32) -> (32, 16, 16), from the # out: line
4 custom Down              # (32, 16, 16) -> (64, 8, 8): one class, used twice (two separate
5 global_pool
6 output
```

The Mac never runs the code: it checks the rest of the design with the declared shapes, and the training machine's one-batch check compares every Custom block's real output with its declaration before training (a wrong `# out:` line stops the run there). The parameters of these classes are counted on the training machine. On the canvas the Custom block's inspector edits the same code.

9.12 The rules in brief

```
net "name"                                    name of the design (optional)
let NAME = EXPR                               a named value
def NAME(P, Q = EXPR) { LINES }               a function (P, Q: parameters; Q has a default)
LABEL BLOCK SETTINGS [<- REF, REF]            one block; without <- it takes the line above
LABEL NAME(EXPR, Q=EXPR) [<- REF, REF]        a function call (inside: in, in2, ...)
LABEL repeat EXPR [as NAME] [<- REF] { LINES }       passes chained; i = pass number
LABEL for A, B in (EXPR, EXPR), ... [<- REF] { LINES }
python """ ... """                           the design's PyTorch classes (for custom CLASS)
REF:  a label | in | in2 | a path like 5.2.b    EXPR: numbers, names, + - * / // % ^ ( )
```

9.13 Mistakes and what the application says

A text with a problem is not applied: the canvas keeps the last valid design, the status says "not applied", the line number in the gutter turns red and the message below the text names the line (click it to jump there). Typical messages:

Written	Message
`conv 64 3x3 stide 2`	'stide' means nothing for conv (did you mean 'stride'?)
`convv 64 3x3`	unknown block 'convv' (did you mean 'conv'?)
`add <- 2, 7` (no line 7 above)	'7': there is no block '7' above this line (a block may only use blocks above it)
two lines labelled `2`	the label '2' is used twice
`dense 10 pad 1`	'pad' means nothing for dense
`conv 8 3x3 filtrs=4`	conv has no setting 'filtrs' (its settings: activation, bias, dilation, filters, kernel, padding, stride)
`conv 8 3x3 99`	what is the number 99? write it as name=value (e.g. filters=99)
a `{` without `}`	a '{' is not closed with '}'
`f(1, 2)` for `def f(c)`	'f' takes 1 value(s), more were given
`repeat k` without `let k`	repeat: 'k' is not a number
`repeat 5000 { ... }` too large	5000 passes × 7 blocks = 35000 blocks, more than the 13000 a design may have (Settings → Design size ...). At most 1856 passes fit here.

Shape problems (an Add of different shapes, a kernel larger than the image) are found by the same checks as on the canvas and listed under the text with the line of their block; click *block 5.2.a* to see it selected on the canvas. Cmd-click on any label in the text also selects its block.

9.14 How the text and the canvas stay in step

- **Typing:** a quarter of a second after the last key, a valid text replaces the design (one undo step). Blocks keep their place on the canvas when their label and kind stay the same; new blocks are placed next to their input.
- **Moving blocks** on the canvas keeps the text exactly as written.
- **Changing a setting** of a block that has its own line rewrites only that line (its label, inputs and comment stay).
- **Adding, removing or connecting** blocks of plain lines changes only those lines: a new block gets the next free number and goes after its input; a line that took "the line above" gets an explicit `<-` when that changed; a block inserted in front of a function call or loop becomes its input. Names, functions, loops and comments stay.
- **Changing a block that a function or loop made** would change every call, so the text is printed again from the blocks: identical consecutive groups become `repeat`, the rest plain lines. Cmd+Z brings the hand-written text back.
- **Format** prints the text again from the blocks on purpose; **Copy** copies it.
- **Auto numbering:** Enter at the end of a numbered line types the next number on the new line (inside braces the next letter; after a line ending in `{` the letter `a`, after `}` the number after the group's line). When that number is taken, the lines below move on by one, and every `<-` and `shared` that names them changes with them; the blocks keep their places on the canvas. Shift+Enter is a plain new line; Enter on a line holding only its number takes the number away again (and closes the gap). **Renumber** numbers the lines 1, 2, 3, ... and the blocks inside each `repeat` / `for` a, b, c, ... in order, references included; named labels and function bodies stay.

In the layer table, the blocks of one `repeat`, `for` or call line form one group named after the line.

9.15 A design that uses everything

`doc/examples/everything.nnb` in the repository is a hybrid CNN and Transformer written only as text: functions calling functions with defaults and named values, a function with two inputs, `for` with tuples, `repeat` with `i`, `let` and every operator, paths into loop passes, a left-out label, comments, and every block kind except the segmentation output and the time-series blocks (those are in `doc/examples/timeseries.nnb`), including a Custom block with its `python` section and a shared-weights twin, cross-attention, RMSNorm, SwiGLU and rotary attention, a mixture of experts, an early exit, a LoRA adapter and a frozen layer (97 blocks, 217 396 parameters of which 16 832 frozen, no warning at $3 \times 32 \times 32$). The test suite builds it and every example of this chapter, and GitHub Actions compares its layers with real PyTorch.

10 The Dataset tab

Only the *description* of a dataset is read on the design computer; the images are downloaded on the GPU machine when training starts (and cached there in `~/.nn_builder/data`).

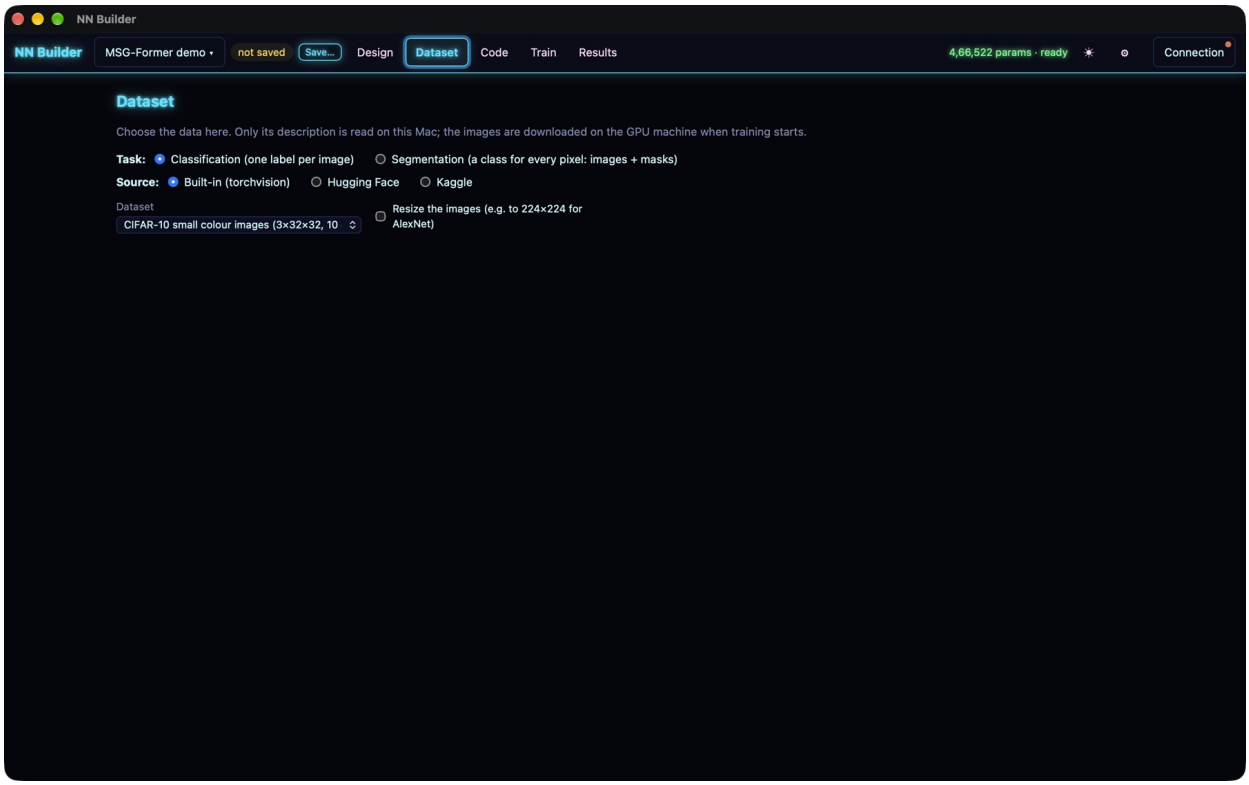


Figure 13: The Dataset tab with built-in CIFAR-10, as MSG-Former opens (the demo picks the built-in set that fits its input, so nothing is resized).

What data NN Builder reads. Every kind, with a picture of such data and the Dataset tab set up for it, is in Section 11.

Data	Sources	Task and files
MNIST, Fashion-MNIST, CIFAR-10	built-in	classification
Pictures (PNG, JPEG, ...; grey or colour)	Hugging Face, Kaggle, own files	classification: a label column, or one folder per class
Pictures + masks	Hugging Face, Kaggle, own files	segmentation: a mask column, or images and masks with the same names
Satellite / multispectral images (Sentinel-2, any bands)	Kaggle, own files	GeoTIFF or `.npy`, 1–32 bands; classification, multi-label, segmentation
Radar (Sentinel-1 VV, VH)	Kaggle, own files	as above; linear or dB
Large exported scenes + label image (Google Earth Engine)	own files, Kaggle	segmentation: cut into tiles
Several classes per image (BigEarthNet)	own files, Kaggle, Hugging Face	multi-label: a CSV (or a list column)
Two sensors for the same places (S1 + S2)	own files, Kaggle	fusion: two band groups, early or mid

10.1 Built-in datasets

MNIST (handwritten digits, $1 \times 28 \times 28$) [23], Fashion-MNIST (clothing, $1 \times 28 \times 28$) [44] and CIFAR-10 (photos, $3 \times 32 \times 32$) [20], ten classes each, downloaded by torchvision on the GPU machine. **Resize the images** enlarges them for networks made for larger inputs (for example 224×224 for AlexNet); the number of channels can be changed as well (grey images repeated into three channels).

10.2 Hugging Face

Type a dataset id (`owner/name`) and press **Read info**: the application reads its configurations, splits, columns and first rows from the public datasets-server API, without downloading [24]. Then choose:

- **Config** and **Training split** (with its number of rows);
- **Test split**: a separate split, or "10% of the training split" (a fixed, seeded split);
- **Image column** and **Label column**: the column that holds the pictures (an *Image* feature) and the one that holds the class (a *ClassLabel* feature, whose class names fill the class list).

Gated or private datasets need a Hugging Face token (Connection tab), which the run can copy to the machine.

Only image datasets work. A text corpus such as `HuggingFaceFW/fineweb-edu` (web pages and their metadata) has no image column and no class column: the image and label columns then point at unrelated text, the class list stays empty, and the design is marked as having problems. Use datasets such as `uoft-cs/cifar10`, `ylecun/mnist` or `food101`. The Dataset tab now says so at once when a dataset has no image column.

After **Read info**, the tab also shows the **class balance** of the training split (counted by Hugging Face's dataset viewer) with a warning when the largest class has more than three times the images of the smallest, and **Show 8 example images** fetches just eight pictures to check that the right columns are chosen (the dataset itself is still downloaded only on the GPU machine).

10.3 Kaggle

Type the dataset slug (`owner/name`); with the Kaggle token stored, the application lists its files and recognises the common layout of *one folder per class*. Set the training folder (and optionally

a test folder; otherwise 10% of the training images are kept for testing) and the class names (the folder names, in order). On Kaggle the dataset is attached to the notebook; on a server or EC2 it is downloaded with the Kaggle tool, which needs the token on that machine (the Train tab offers to copy it there).

10.4 Segmentation data

Hugging Face: an image column and a **Mask column**. Kaggle: an **Images folder** and a **Masks folder** whose files have the same names. The class list maps mask values to classes (value 0 is the first class; with two classes any non-zero pixel is the second).

10.5 Image size and classes

Image height, **Image width** and **Channels** set the shape of the Input block; images are resized to it. The class names, one per line in label order, set the Output block's size.

10.6 Satellite images: Sentinel-2, Sentinel-1 and other bands

Multispectral and radar images have more than three bands and values that are not 0–255 colours, so they are read as arrays, not as pictures. A Kaggle dataset with one folder per class of GeoTIFF (`.tif`, `.tiff`) or `.npy` files works, as in EuroSAT-MS. Choose it as usual (**Read files**, the training folder), then under **Image bands**:

- **Sentinel-2 (13 bands)**: the files' band order is filled in as EuroSAT-MS stores it (B01 … B12, then B8A; correct it if the dataset's description says otherwise). Tick the bands the network should see, or use a preset: all 13, the ten 10 m and 20 m bands, true colour (B04 B03 B02), or colour and near infrared. Each band shows its wavelength and resolution.
- **Sentinel-1 radar (VV, VH)**: the two polarisations; tick **convert to decibels** when the files hold linear backscatter (values around 0.01–0.5), since a network learns radar much better in dB.
- **other bands**: type the bands of the files in their order and tick the ones to use.

The Input block then has as many channels as bands are ticked (up to 32), and the first convolution's parameters follow (13 bands into a 3×3 convolution with 16 filters: $13 \cdot 9 \cdot 16 + 16 = 1\,888$). On the GPU machine `train.py`:

- reads each file (bands first or last), picks the ticked bands by name, converts Sentinel-1 to dB if asked, and resizes to the image size of the Dataset tab;
- measures the mean and spread of every band on up to 256 training images and normalises each band with its own values (kept in `band_stats.json`, so a resumed run normalises the same way);
- augments with horizontal / vertical flips and quarter turns (a satellite image has no "up"), when flips or rotation are ticked in the Train tab;
- shows the result pictures as composites: true colour (B04 B03 B02) for Sentinel-2, VV / VH / VV−VH for Sentinel-1, else the first three bands, each stretched between its 2nd and 98th percentile.

ImageNet weights with bands. **Start from ImageNet weights** (Train tab) also works for satellite images: the first convolution's red, green and blue filters go to B04, B03 and B02 (matched by name), every other band gets the mean of the three, and all are scaled by $3/C$ for C bands so the layer's output keeps its size; the rest of the network loads as usual. The images are still normalised per band (not with ImageNet's colour statistics).

Hugging Face. A Hugging Face dataset whose image column holds band arrays (an array feature, or GeoTIFF / `.npy` bytes) is read the same way: choose the bands under **Image bands** (type the

band order, since it cannot be read on the Mac). Classification, multi-label (a list column) and segmentation (a mask column) work; two band groups need files (own files or Kaggle).

Several classes per image (Section 10.8) and Sentinel-1 with Sentinel-2 (Section 10.14) are below. GitHub Actions trains on generated Sentinel-2 (mixed `.tif` and `.npy`) and Sentinel-1 folders on every change and checks the bands, the statistics and the accuracy; it also trains on every layout of the next section.

10.7 Your own files, e.g. Sentinel-1/2 from Google Earth Engine

Sentinel data is usually exported from Google Earth Engine (GEE) to Google Drive and downloaded, so it is not on Kaggle or Hugging Face. Choose **Source: My own files**, then the folder (**Choose...**) and **Read folder**. The app lists the files, finds class folders, and reads a GeoTIFF's header: its size, number of bands and, when GEE or GDAL wrote them, the band names, which fill **Bands in the files** by themselves (nothing is opened beyond the header, and nothing in the folder is changed). The names come from a file in the **Training folder**, and with fusion the second group's from a file in its own folder, so Sentinel-1 files in `s1` never rename the Sentinel-2 bands of `s2`. Compressed GeoTIFFs (Earth Engine writes LZW) are read on every machine: the training code installs the decoder (`imagecodecs`) when a file needs it.

Where the files are. on this Mac: when a run starts, the folder is copied to the training machine (`~/.nn_builder/data/local/nnb-<folder>-<code>`, with rsync, so only the first time and after a change), or, for Kaggle, uploaded under your account as a *private* Kaggle dataset of the same name and attached to the notebook (again only when it changed; the folders go up as zip files, which Kaggle unpacks). **already on the training machine**: give its path there (e.g. a large export copied to the college server yourself); nothing is copied, and Kaggle cannot be used.

In the cloud. The training machine (or the Kaggle notebook) fetches the files itself, so a large Earth Engine export never passes through the Mac: `gs://bucket/folder` (Earth Engine's Cloud Storage export: in the export script choose **Export to: Cloud Storage**; a private bucket needs a service-account key, Connection tab, which is copied to an SSH or AWS machine with the other tokens; Kaggle notebooks read public buckets only), a Google Drive folder or file shared as "anyone with the link" (fetched with `gdown`: at most 50 files per folder, a zip is unpacked), or `rclone:remote:path` for rclone set up on the machine (any of its 70 storage services, Drive included). The files are kept in the machine's data folder; a later run fetches only what is new.

Classification. One subfolder per class (`forest/`, `water/` ...) of `.tif` or `.npy` patches, or ordinary pictures. An optional test folder; else 10 % of the images.

Segmentation: images + masks. An images folder and a masks folder with the same file names (a mask may add `_mask`); masks may be PNG, GeoTIFF or `.npy` with class numbers 0, 1, 2 ...

Segmentation: a large image and a label image. What GEE exports for a region: one image (possibly split into several files named `scene-0000000000-0000000000.tif` ...) and a label image on the same grid with the same file names (in another folder). The images are cut into tiles of the image size; a tile is used when at least half of its pixels have data and at least one has a class. **Test tiles**: 10 % at random (quick, but neighbouring tiles are alike, so the score is optimistic), the eastern tenth of every image with one column of tiles left out between it and the training tiles (*strip*: no test tile touches a training tile), or a test region exported separately (*files*: the most honest). The arrays are kept as `.npy` files in the data folder and read with memory mapping, so a scene larger than the memory works.

GEE's conventions are handled: band names B1 ... B9 are the same as B01 ... B09 everywhere in the app (**Earth Engine SR (12, B1 ... B12)** sets that band order; the surface-reflectance collection has no B10); empty (masked) pixels written as NaN are left out of the band statistics

and filled with the band's mean; label values that are not a class (no data, 255, negative) are left out of the loss and of the scores; and Sentinel-1 from `COPERNICUS/S1_GRD` is already in dB: such files are recognised by their negative values and not converted again, even with **convert to decibels** ticked.

10.8 Several classes per image (multi-label)

In benchmarks such as BigEarthNet a patch can be forest *and* water *and* farmland at once. Choose **Task: Multi-label**. The labels come from a CSV file (own files or Kaggle): one row per image, a file column (`file`, `image`, `name`, `path`, `patch_id` ...; with or without the extension, relative to the CSV's folder or the images folder) and either a labels column with the classes separated by `;` `,` `|` or spaces, or one 0/1 column per class. **Read labels** reads the class names and how many images have each. From Hugging Face, choose a column that is a list of class labels. Rows that name a missing file are left out (with a warning). A `split` column (`train` / `validation` / `test`) is used as given instead of a random split.

BigEarthNet v2 as it is downloaded. Each patch is a folder of single-band GeoTIFFs (`<patch>/<patch>_B02.tif` ..., the 20 m and 60 m bands at their own sizes) below a tile folder, and the labels are in `metadata.parquet` (a list of class names per patch, the split, and `s1_name`, the matching Sentinel-1 patch). Choose **BigEarthNet (19 classes, metadata.parquet)**, set the **Images folder** to the Sentinel-2 folder, and for fusion add Sentinel-1 as the second band group with its folder: the patch folders are found below them, every band is brought to the largest band's size, and `s1_name` pairs the two sensors. Parquet tables are read on the training machine only (the class names come from the preset or are typed).

The network ends in the ordinary Output block, one score per class, but every class is its own yes/no: the score goes through a sigmoid, a class is predicted when it is at least 50 %, and the loss is binary cross-entropy summed over the classes (label smoothing pulls the targets towards 0.5). The run reports micro F1 (every image–class decision together; shown where accuracy usually is), macro F1 (the classes averaged, so rare classes count as much as common ones), mean average precision (mAP: how well each class's scores rank its images, independent of the threshold) and exact match (every class of an image right). Results lists precision, recall, F1 and AP per class, and the sample pictures mark each class as right, wrong or missed.

10.9 Numbers per image (regression)

To predict a quantity instead of a class (above-ground biomass, canopy height, crop yield, soil moisture of a plot), choose **Task: Regression**. The numbers come from a CSV file next to the images (own files or Kaggle): one row per image, the same file column as for multi-label, and one numeric column per target. **Read labels** lists the columns that hold a number in every row, with their range; tick the targets (one or several). From Hugging Face, tick the numeric columns of the table. Rows whose target is empty or not a number are left out with a warning; a `split` column is used as given.

The Output block gives one number per target (no softmax). Before training the run measures every target's mean and spread on the training images, and the network learns the targets standardised (mean 0, spread 1), so targets of very different sizes train equally well. **Loss** in the Train tab: *mean squared error* (MSE, the default; large errors count most), *L1* (mean absolute error; less swayed by a few wild values) or *Huber* (squared for errors below one standard deviation of the target, linear above). The scores are in the targets' own units: R^2 (the share of the targets' variation the network explains: 1 is perfect, 0 is no better than always predicting the mean; shown where accuracy usually is, and what early stopping on "validation R^2" watches), MAE (mean absolute error) and RMSE (root mean squared error), per target and averaged. Results shows a predicted-against-true scatter plot per target and sample images with both numbers. `model.pt`

outputs the standardised numbers; `target_stats.json` next to it holds each target's mean and standard deviation (value = output × std + mean). Flips and rotations are fine for quantities that do not depend on direction. A number for every pixel (a canopy height map) is not supported yet.

10.10 Several outputs (multi-task)

One network can predict several things about the same image at once: the land-cover class *and* the biomass, or the crop type *and* whether the field is irrigated. The shared layers then learn features useful for all of them, which often helps the smaller tasks. In the Dataset tab, **Extra outputs (multi-task)** → **Add an extra output**, for each one:

- a **name** (shown in the results), what it **Predicts** (one class, several classes, or numbers), and its column(s) of the labels table: the class column; the labels column (several classes; empty when they are 0/1 columns); or the number columns;
- for classes, the class names in label order; its **Loss** (cross-entropy or focal; binary cross-entropy; MSE, L1 or Huber); and its **Weight**: the total loss is the main task's plus each extra output's times its weight.

The values come from the labels table next to the images (own files or Kaggle: the same CSV / TSV / Parquet as for multi-label and regression; with the main task *classification* the image's class is its folder, or the **Main task's class column**), or from columns of a Hugging Face dataset. Rows without a usable value for an output are left out, with a warning.

In the Design tab each output gets its own Output block: select an Output and choose **Output for** (the main task, 1, 2, …; in text `output 1`). Usually they share the body of the network and each has a small branch of its own (a Dense or two) before its Output. The network returns all outputs; the run reports the main task as usual (and its score decides early stopping and the best epoch) and every extra output in a table: accuracy, micro F1 and mAP, or R^2, MAE and RMSE per number, in the numbers' own units. The Train tab's chart draws each output's validation score.

10.11 Object detection (boxes)

To find and count objects (ships, cars, buildings, trees, animals) with a box around each, choose **Task: Object detection**. The data are an **Images folder** and a **Labels folder** with one text file per image of the same name (`im17.png` and `im17.txt`): one line per box, `class cx cy w h`, the class number (0 = the first line of the class list) and the box's centre, width and height as fractions of the image. This is the YOLO format that Roboflow, CVAT and most converted datasets offer; an image without a label file has no object. Pictures and multiband files both work; 10 % of the images are the test data; horizontal and vertical flips move the boxes with the image.

The network ends in a **Detection output** on a feature map, typically a quarter or an eighth of the image size (two or three 2×2 pools or stride-2 convolutions; the image must be a whole number of cells). For every cell it gives a score per class (is an object's centre here?), the box's width and height in cells and the centre's offset inside the cell: the anchor-free CenterNet design [51]. Training draws every box as a Gaussian peak on its class's map and uses CenterNet's loss (a focal loss on the peaks, L1 on size and offset). The boxes found are the peaks of the scores; the run reports mAP@0.5 (a box counts when it overlaps a true one by at least half, IoU ≥ 0.5) and mAP@0.5:0.95 (stricter overlaps, as COCO), with the AP of every class, and Results draws the boxes found (score $\geq 30\,\%$) and the true ones on test images. Not yet: rotated boxes (DOTA's oriented format), COCO JSON labels, very large scenes cut into tiles with their boxes, and non-maximum suppression across classes.

10.12 Time series of satellite images

A field's crop shows in *when* it greens up and is harvested, more than in any single image. A time series gives the network the same place at several dates. In the Dataset tab, after choosing the bands, tick **a time series** and give the **dates per sample**. Each file then holds one place at all its dates: a `.npy` of shape (dates, bands, height, width) (or (dates, height, width, bands)), or a GeoTIFF whose bands go date by date: all bands of the first date, then of the second, ... (as Earth Engine exports a stack of monthly composites, `ee.ImageCollection.toBands()`). Class folders, a labels table (multi-label, regression, extra outputs) and images + masks (a crop-type map) all work; every band is normalised over all its dates, every date is resized like an image, and flips and turns apply to all dates alike. The result pictures show the median of the dates.

The Input is then (C, T, H, W): bands, dates, height, width. Image blocks do not take it directly; the Time series blocks do:

- **Conv3D** convolves over neighbouring dates and pixels at once (*dates per window*, e.g. 3);
- **Temporal pool** combines the dates of every pixel into one image (C, H, W): the mean, the maximum, or *attention*: a score for every date and pixel from the pixel's channels, a softmax over the dates, and the weighted sum (the idea of U-TAE [48]). After it every 2-D block works: convolutions, a U-Net decoder, a Segmentation output for a crop map;
- **To sequence** takes the mean of every date's pixels, one vector per date (T, C): the time series of the whole patch. Then **Conv1D**, **LSTM** and **GRU** work along the dates, and so do the transformer blocks (Class token + position, Self-attention, Token pool), which gives a small temporal transformer (as in TSViT [49]). Token pool's *last* keeps the last step of a recurrent layer.

Both kinds of path can start from the same Input and be joined (`doc/examples/timeseries.nnb`). BatchNorm, Dropout, the activations, Add / Subtract / Multiply, Concat (equal dates and size), Flatten and Global pool work on a time series too. Not yet: a date per image (the dates are taken as evenly spaced), missing dates, a second band group with a time series, Hugging Face time series, the Earth Engine script for a time stack.

10.13 Shared weights: Siamese networks and change detection

A Siamese (twin) network looks at two images with the *same* layers, so that both are described in the same terms, and then compares the two descriptions: has this field been built on, flooded or cleared between two dates? NN Builder does it with:

- **the data as a pair in time:** the second band group with the *same bands* as the first, from the folder of the later images (same subfolders and file names): e.g. `before/changed/17.tif` and `after/changed/17.tif`. The second copy of each band is reported as `B04_2` etc.; flips and turns are applied to both images alike;
- **the Twin branch button** (Design tab): build one branch from the Input (convolutions, pools, ...), select its last block and press **Twin branch (shared weights)**. The same branch appears again from an Input for band group 2, made of **Shared weights** blocks (each runs the original layer with the original weights) and copies of the layers without weights. In text: `5 shared 2` uses block 2's weights (a label inside a function or repeat may name a block of the same call or pass);
- **a join that compares: Subtract** with **absolute difference** gives $|a - b|$, what changed (FC-Siam-diff); Concat keeps both (FC-Siam-conc). Then the head: Global pool + Output for "changed / not changed", or a decoder to a Segmentation output for a change map (images + masks).

The shared layers are trained by both images and counted once in the parameters. A Shared-weights block's input must have the shape of the original block's input, or the weights would not fit; the inspector lists the blocks whose weights can be used.

10.14 Sentinel-1 and Sentinel-2 together (fusion)

Radar sees through clouds and shows water and structure; optical bands show vegetation and colour. Under **Second band group (fusion)** choose the second sensor, its folder and its bands. Its files mirror the first group's: the same subfolders and file names, any extension (`s2/forest/x.tif` and `s1/forest/x.npy`); a different pixel size is resized to the image size. The two groups are stacked into one image (the first group's bands, then the second's), each band normalised with its own statistics, and the second Sentinel-1 group converted to dB when asked. Then, in the Design tab:

Early fusion. One Input takes all bands: nothing else to do.

Mid fusion. Select the Input, set **Bands** to band group 1, and press **Add an Input for band group 2**: each Input starts a branch of its own (e.g. a few convolutions per sensor), and a Concat (or Add) joins them. In text: `input 1` and `input 2`. The generated code takes each group's channels from the stacked image.

Fusion works for classification, multi-label (BigEarthNet: S1 and S2 per patch) and segmentation (images + masks, and scenes: the second group's scene files in its folder with the same names). With two groups the test data are 10 % of the images (no separate test folder).

Each group at its own size. Resizing the second group to the first group's grid invents pixels (Sentinel-1 at 20 m blown up to 10 m) or throws them away. Tick **its own image size** under the second band group and give its size, e.g. 32×32 next to Sentinel-2's 64×64: each group is then resized only to its own size and reaches the network as an input of its own. That needs mid fusion: one Input per group (**Bands** 1 and 2), each with its own branch, joined where their sizes match (here after a 2×2 pool or a stride 2 on the Sentinel-2 branch); an Input for all bands is refused, as the groups cannot be stacked. The second group may not be larger than the first (make the sharper sensor group 1), and scenes keep one grid. Flips and quarter turns are applied to both groups alike, and the band statistics are measured on each group's own pixels. (Inside, the second group fills the top-left corner of its channels in the batch tensor and the generated code cuts it out: the Code tab shows `x[:, 4:, :32, :32]`.)

The export script. Under **Get Sentinel-1/2 data from Google Earth Engine** the Dataset tab writes a script that exports exactly what the tab expects: the ticked bands in their order, from the chosen dates, with clouds masked (Sentinel-2 SR: the scene classification band; with B10: the top-of-atmosphere collection and its QA60 band) or the median of Sentinel-1 IW scenes with VV and VH. For segmentation it is a Code Editor script: draw the region, **Run**, then run the two tasks; it exports the image and the ESA WorldCover 2021 map (11 classes, as class numbers 0–10; replace it by your own labelled image) on one grid into the Drive folders `nnb_image` and `nnb_label`. For classification it is a Python script (`earthengine-api`) that exports one patch per labelled point into a Drive folder per class, starting every export itself. **Set this tab to match the export** fills in the band order, the layout and (WorldCover) the class names. The app does not run the script: it needs your Earth Engine account, so check it there.

11 Kinds of data, step by step

For each kind of data: what such data looks like, the Dataset tab set up for it, and the steps. The data pictures are *generated* examples (a random land-cover map turned into typical Sentinel-2 reflectances and Sentinel-1 backscatter; `doc/screens/data_pictures.py`), not real images; the Dataset tab pictures are the application's own screen (`doc/screens/render.sh`). The folder paths in them are examples. After any of these, the Design tab's Input block shows the resulting shape, and the Output block must have as many classes as the class list.

11.1 Built-in sets (MNIST, Fashion-MNIST, CIFAR-10)

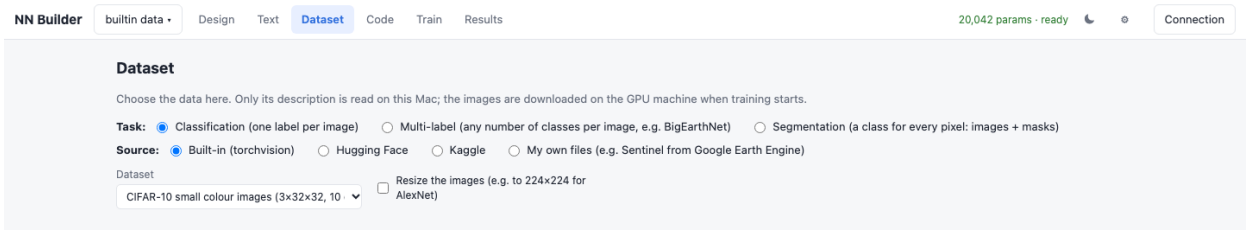


Figure 14: Dataset tab: built-in CIFAR-10.

1. **Task: Classification**, **Source: Built-in (torchvision)**.
2. Choose the set. The Input becomes its own size ($1 \times 28 \times 28$ or $3 \times 32 \times 32$); tick **Resize the images** for a network made for larger pictures and set the size and channels.

Nothing else: torchvision downloads the set on the GPU machine.

11.2 Pictures from Hugging Face

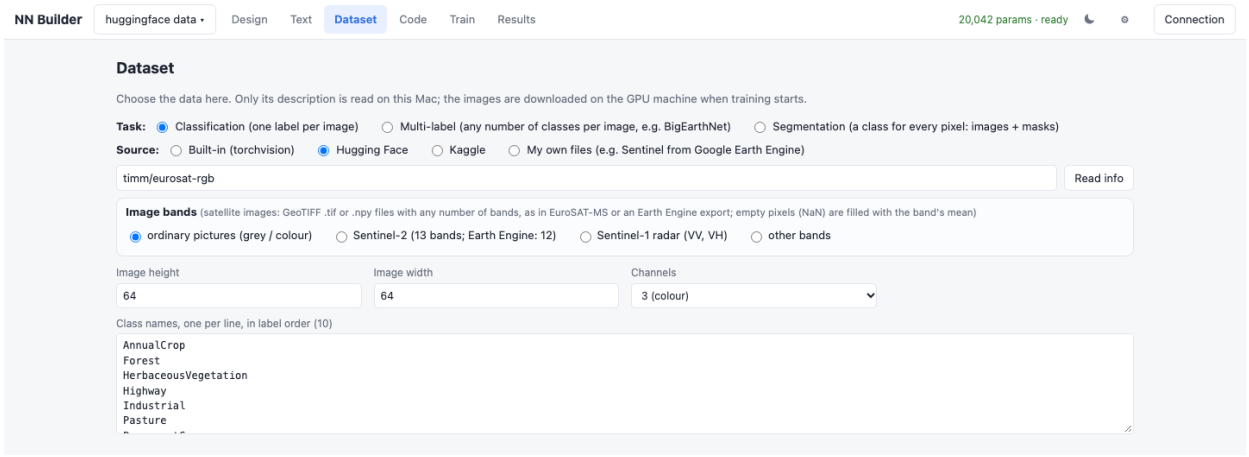


Figure 15: Dataset tab: a Hugging Face dataset (`timm/eurosat-rgb`, RGB satellite pictures in ten classes) before **Read info**.

1. **Source: Hugging Face**; type the id (`owner/name`) and press **Read info**.
2. Check the splits, the **Image column** and the **Label column** (its class names fill the list); **Show 8 example images** checks the choice.
3. Set the image size and channels the network should receive.

11.3 Pictures in class folders (Kaggle or your own files)

1. **Source: Kaggle** (type `owner/name`, **Read files**) or **My own files** (**Choose...**, **Read folder**).
2. **Training folder**: the folder whose subfolders are the classes (found by itself when there is one); optionally a **Test folder** with the same subfolders.
3. **Image bands**: **ordinary pictures**. Set the size and channels (1 grey, 3 colour); the class names are the folder names.

11.4 Segmentation: images and masks

1. **Task: Segmentation**. Hugging Face: choose the **Mask column**. Kaggle or own files: **Images folder** and **Masks folder** (a mask may add `_mask`; PNG, GeoTIFF or `.npy`).

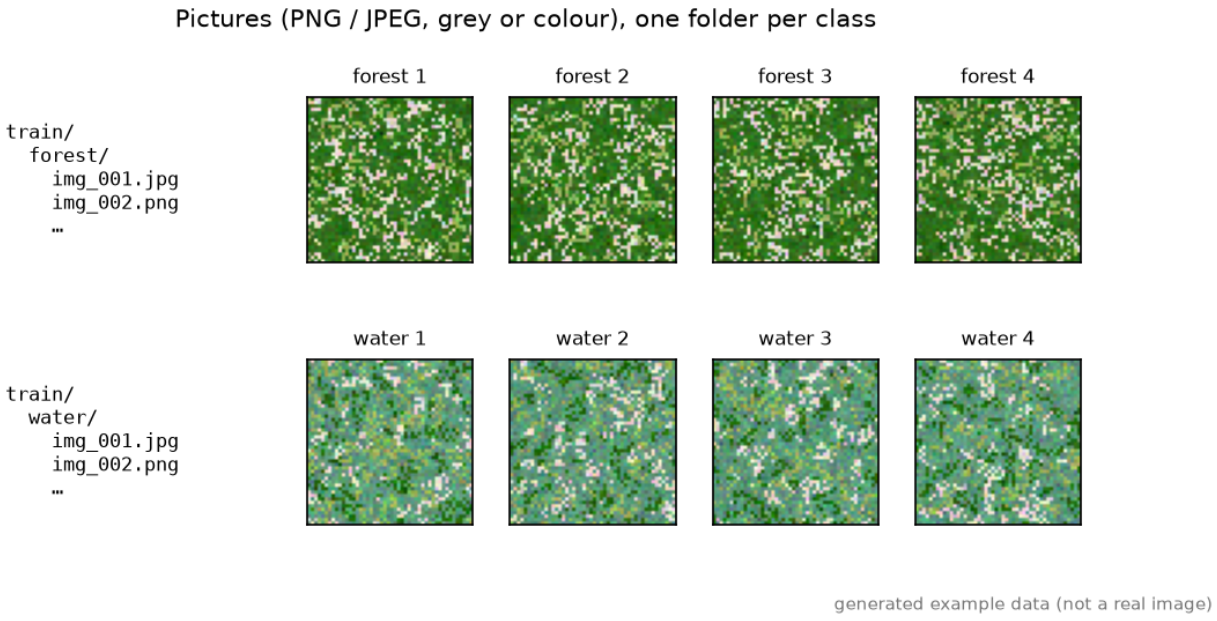


Figure 16: Pictures sorted into one folder per class (generated examples).

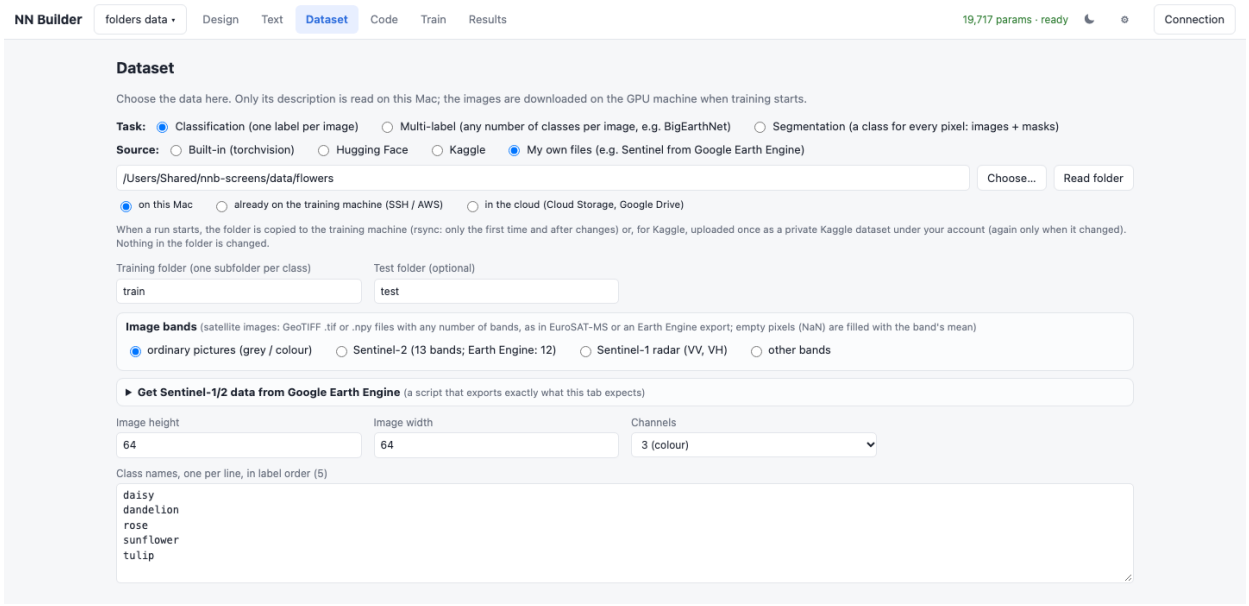


Figure 17: Dataset tab: your own folder with `train/` and `test/`, each with one subfolder per class.

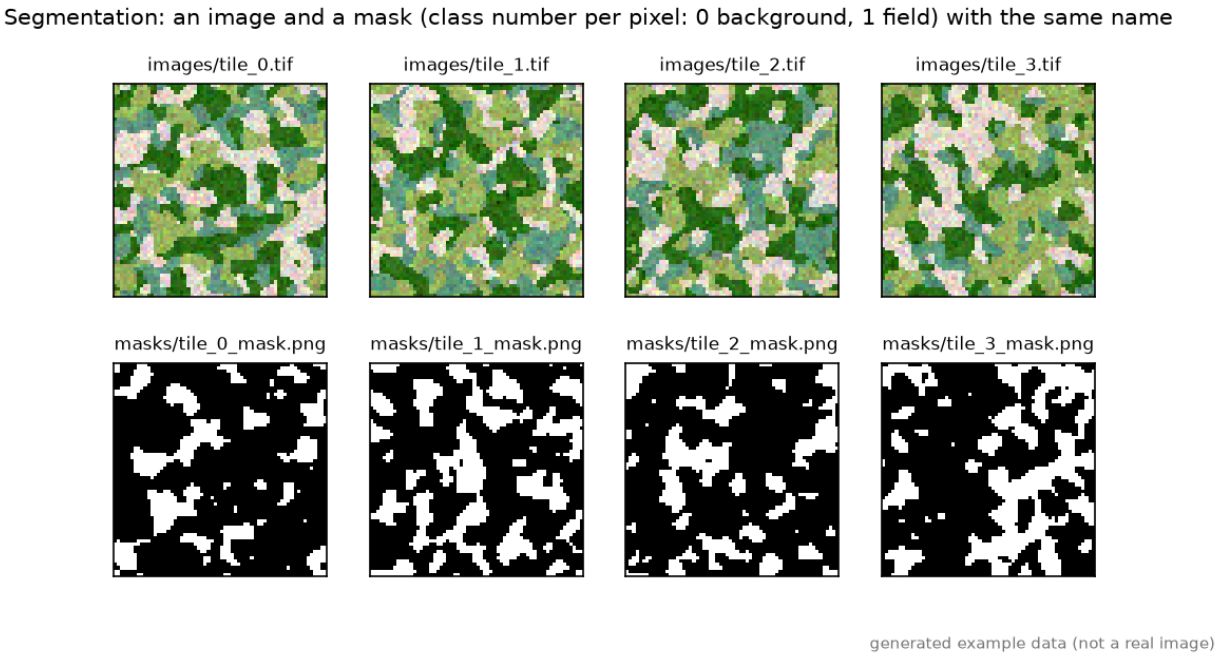


Figure 18: Images and masks with the same names; a mask holds the class number of every pixel (generated examples).

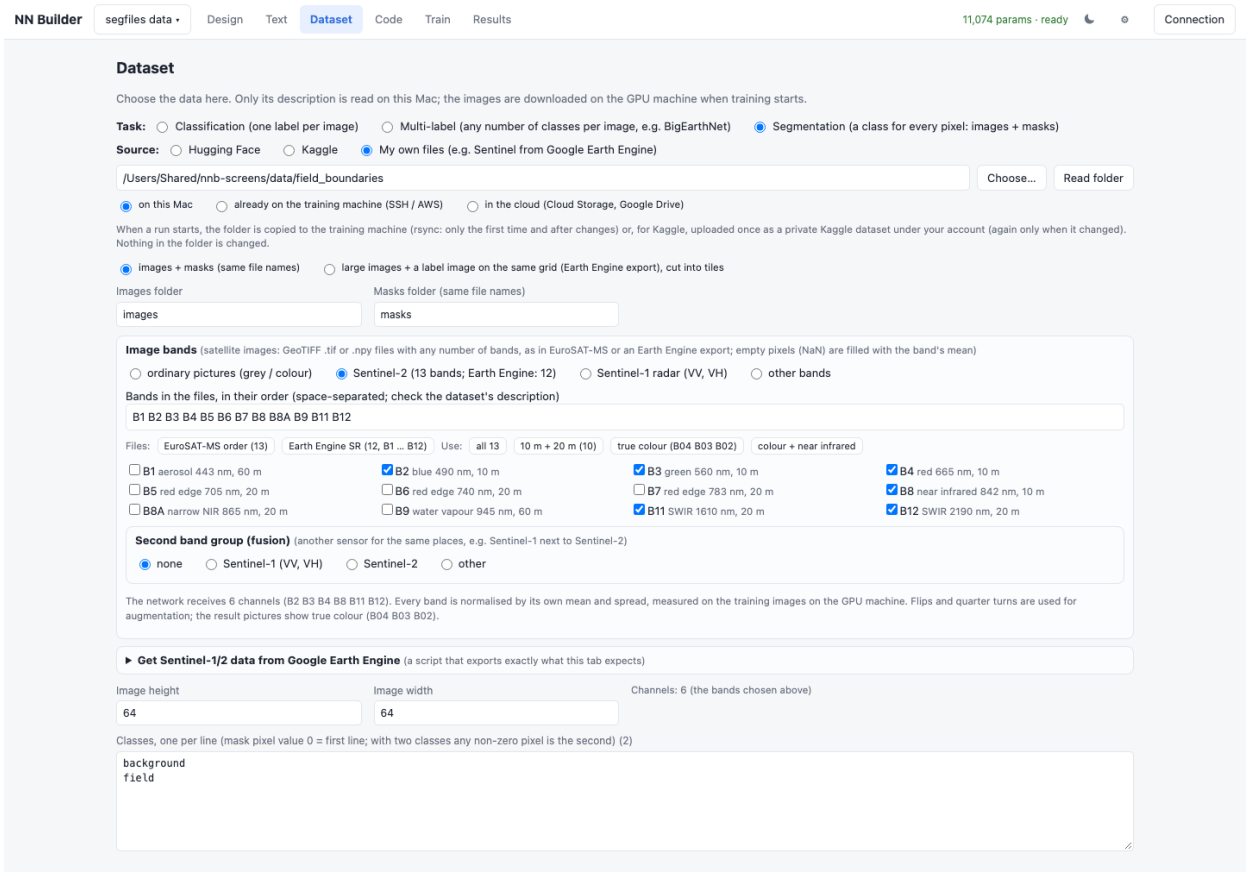


Figure 19: Dataset tab: multispectral images and masks in your own folder.

2. Pictures: **ordinary pictures**; satellite images: choose the bands as below.
3. The class list: mask value 0 is the first line, 1 the second, ...; other values (no data, 255) are not scored.
4. The design must end in **Segmentation output**, at the input's size (e.g. a U-Net).

11.5 Satellite images: Sentinel-2, Sentinel-1, any bands

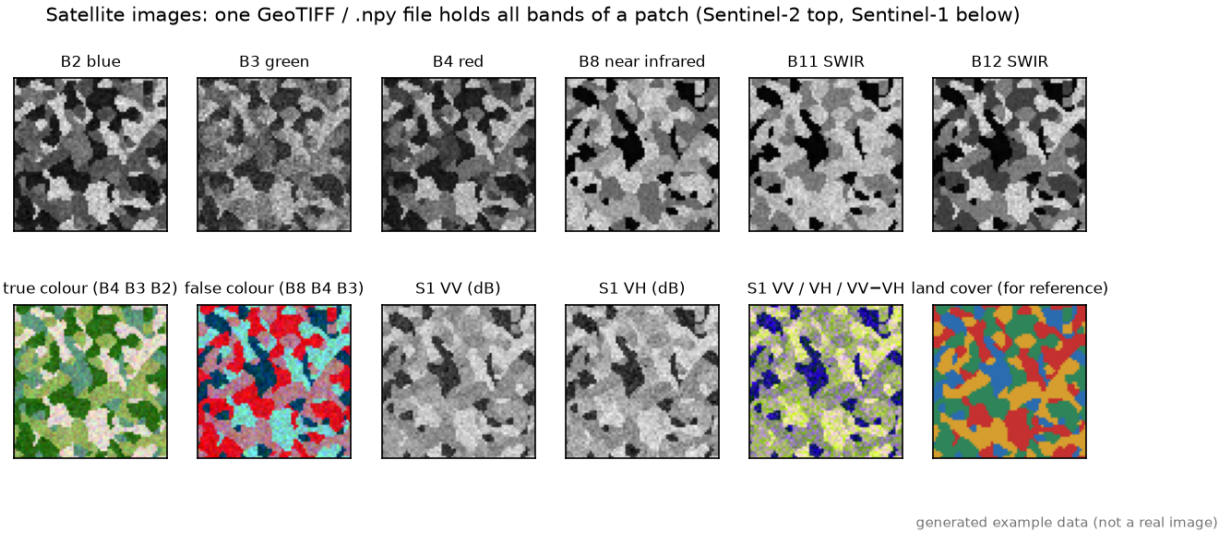


Figure 20: One patch, band by band: six Sentinel-2 bands, the true- and false-colour composites, and Sentinel-1 VV and VH in dB (generated example).

1. Kaggle or **My own files**; the files are GeoTIFF (`.tif`) or `.npy`, one per patch, in class folders (or images + masks, or a CSV for multi-label).
2. **Read folder** reads the band names from a GeoTIFF of the training folder when Earth Engine or GDAL wrote them; otherwise choose **Sentinel-2**, **Sentinel-1** or **other bands** and set **Bands in the files** (**EuroSAT-MS order** or **Earth Engine SR** fill it for Sentinel-2).
3. Tick the bands the network should see, or a preset (10 m + 20 m, true colour, colour + near infrared). Sentinel-1: tick **convert to decibels** only for linear values.
4. Set the image size; the channels are the ticked bands.

Details: Sections 10.6 and 10.7.

11.6 A large exported scene with a label image (Google Earth Engine)

1. **Task: Segmentation**, **Source: My own files**, the layout **large images + a label image**.
2. **Image file or folder** and **Label file or folder** (the same file names when the export is split into several files).
3. The bands, and the image size = the tile size (e.g. 128 × 128). The class list: label value 0 first.
4. No data yet? Open **Get Sentinel-1/2 data from Google Earth Engine**, set the dates, copy the script into the Earth Engine Code Editor, draw the region, run both exports, download the two Drive folders into one folder, and press **Set this tab to match the export**.

11.7 Several classes per image (multi-label)

1. **Task: Multi-label**. Own files or Kaggle: the **Labels file (CSV)** and, if the images are not next to it, the **Images folder**; **Read labels** fills the class list. Hugging Face: a label column

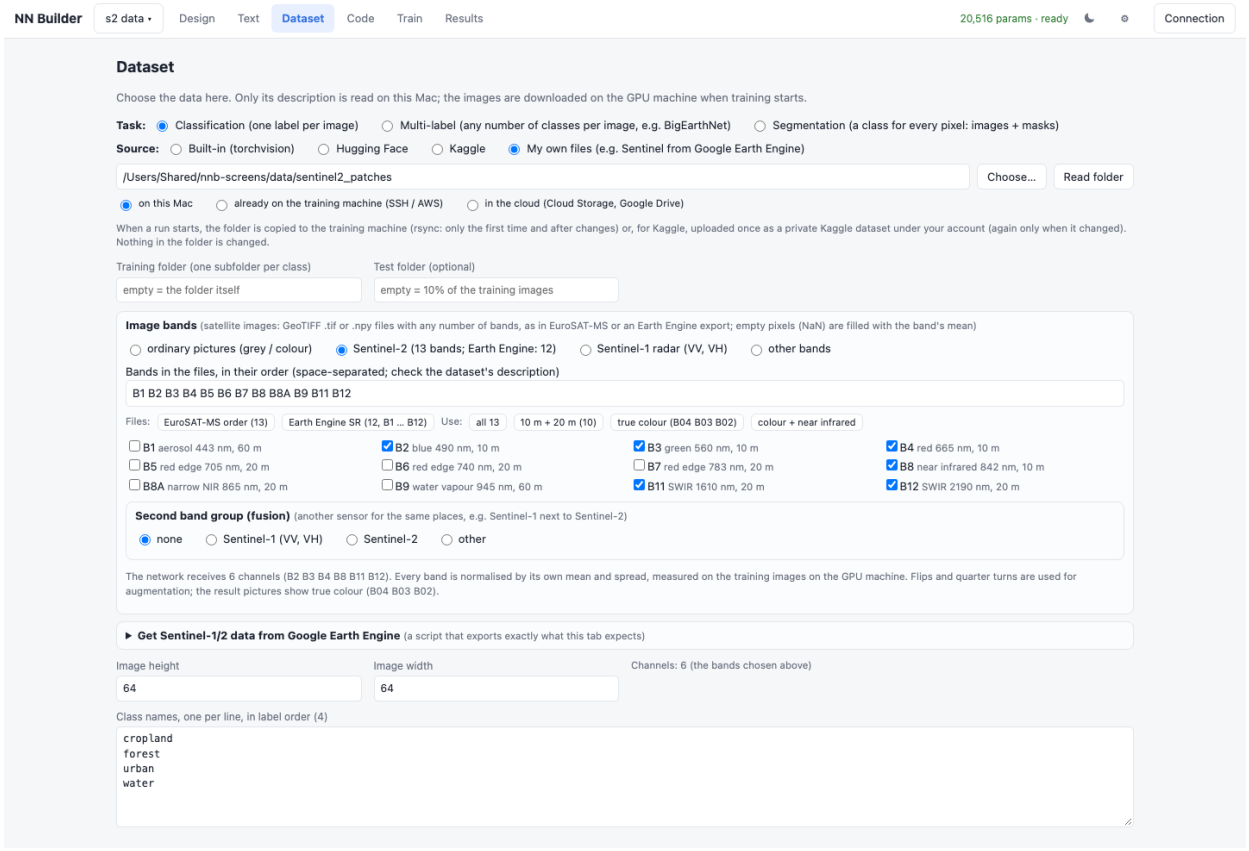


Figure 21: Dataset tab: Sentinel-2 patches exported from Earth Engine (band order B1 … B12), six bands chosen.

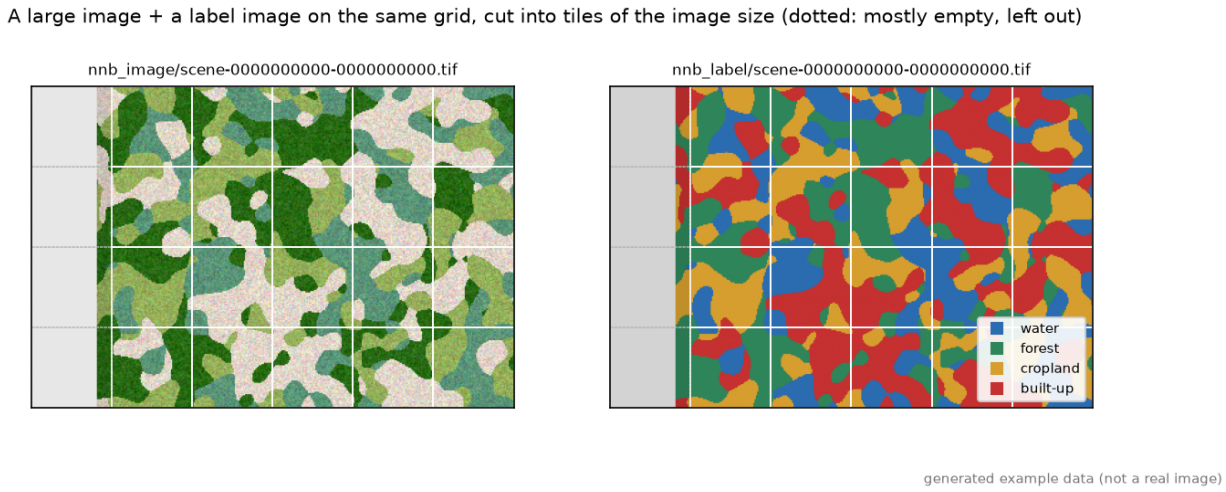


Figure 22: An export and its label image on the same grid, cut into tiles; mostly empty tiles (dotted) are left out (generated example).

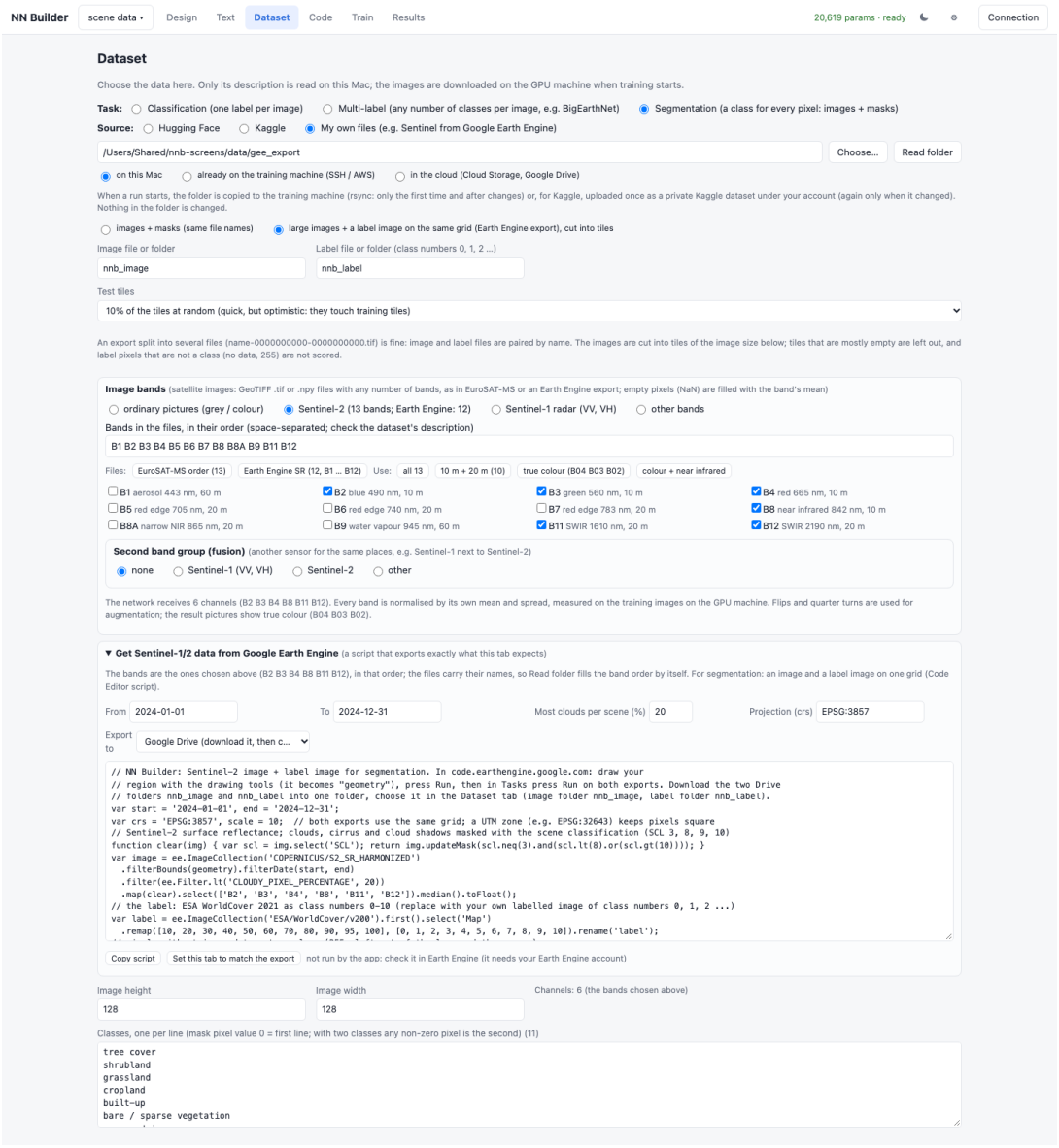


Figure 23: Dataset tab: a scene layout with the Earth Engine export script opened (it exports exactly these bands, and the WorldCover map as labels).

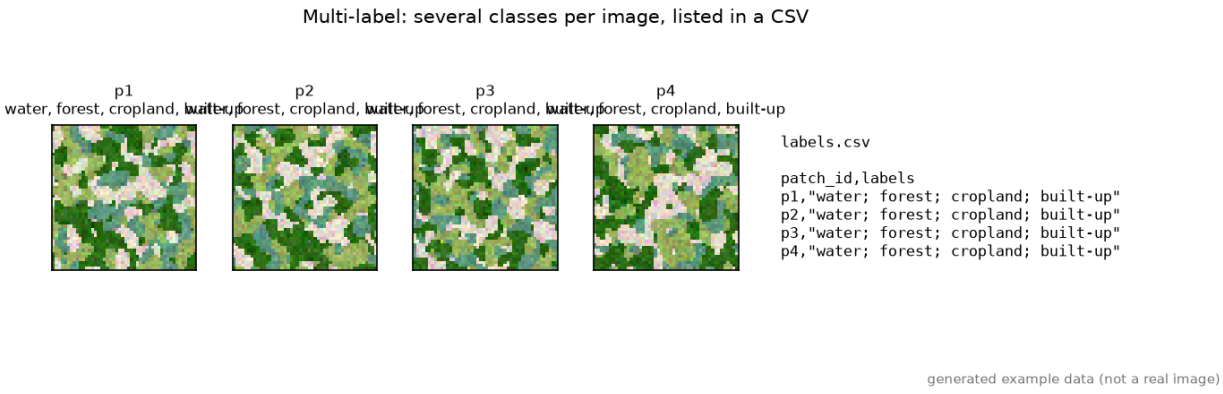


Figure 24: Patches with several classes each, and the CSV that lists them (generated examples).

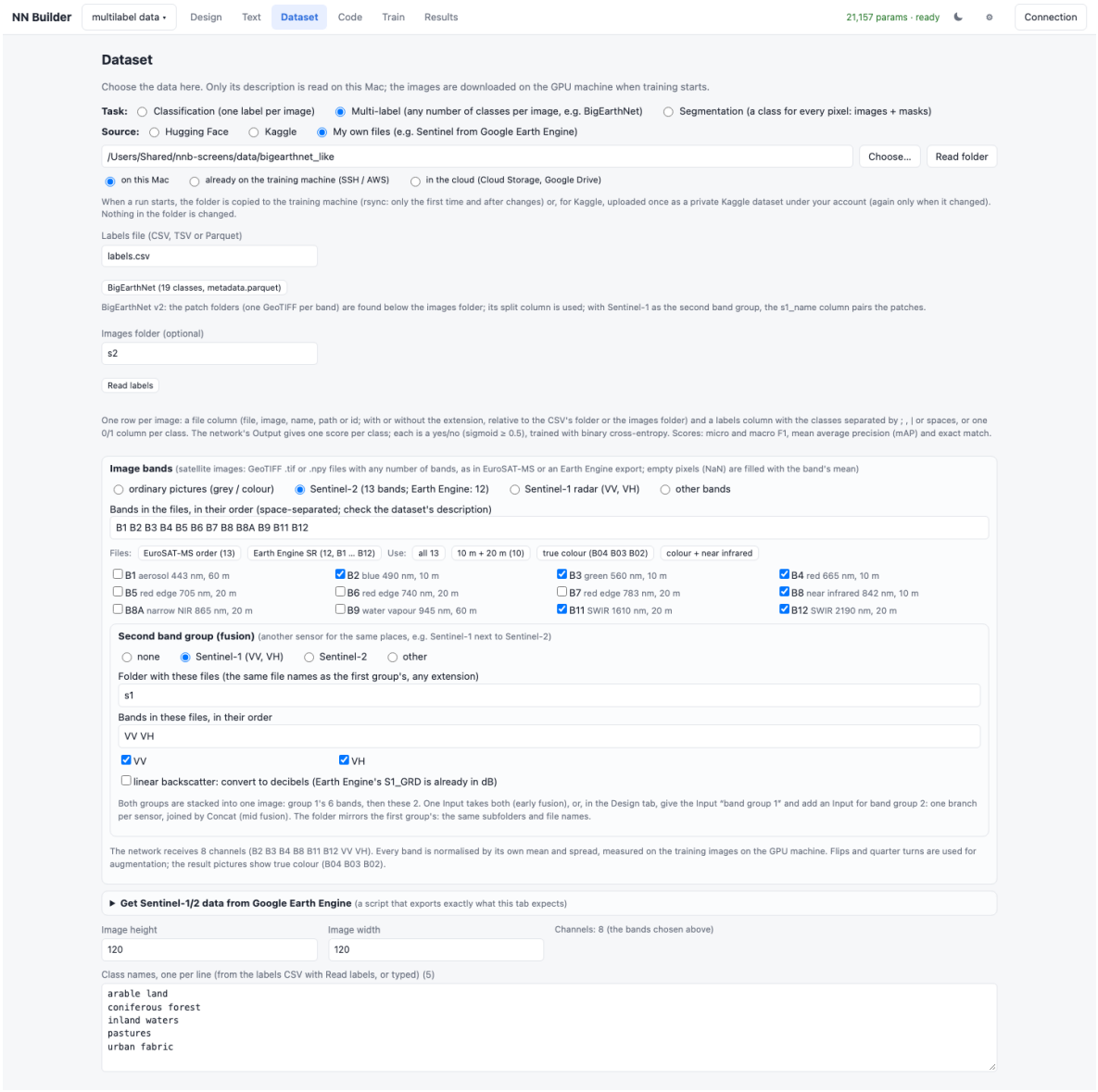


Figure 25: Dataset tab: a BigEarthNet-like folder: Sentinel-2 patches in `s2/`, Sentinel-1 in `s1/`, labels in `labels.csv`.

that is a list of classes.
2. Pictures or bands as above; the Output block gives one score per class.
3. The results are F1, mAP and exact match (Section 10.8).

11.8 Two sensors: Sentinel-2 and Sentinel-1 together (fusion)

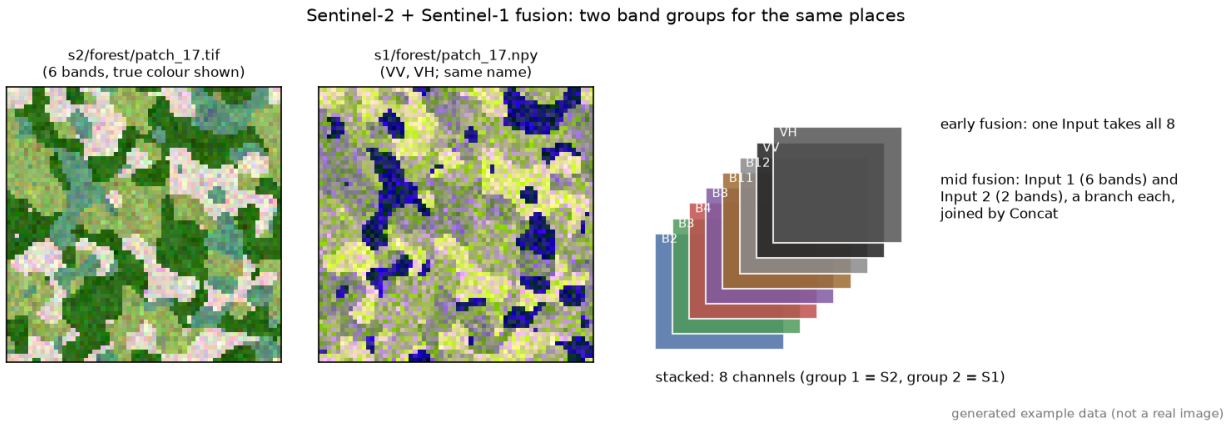


Figure 26: The same place from both sensors, in two folders with the same file names, stacked into one input of 8 channels (generated example).

1. Set up the first group (e.g. Sentinel-2) as above, with the **Training folder** pointing at its files (`s2`).
2. Under **Second band group (fusion)**: the sensor, the **Folder with these files** (`s1`, mirroring `s2`: the same subfolders and names) and its bands. It is a folder *inside* the dataset folder, never a full path: only the dataset folder is copied or uploaded to the training machine.
3. Design tab: keep one Input (early fusion), or select it, set **Bands** to group 1 and press **Add an Input for band group 2** (mid fusion): give each Input its own layers and join them with Concat. In text: `input 1` and `input 2`.

Section 10.14 has the details.

12 The Code tab: what the generated script does

The **Code** tab shows the complete `train.py`; **Copy** and **Export train.py** save it. It needs only PyTorch and torchvision (plus `datasets` or `kaggle` for those sources) and runs anywhere with `python train.py`. In order, it:

1. defines the network as an `nn.Module` whose `forward` follows the graph, and lists every layer's expected shape;
2. loads the data with the chosen transforms and augmentation (training images only), normalised with the dataset's statistics;
3. runs a **one-batch check** on the GPU machine: one batch forward and backward, comparing every layer's real shape with the application's, and stopping at once with a clear message on a difference;
4. trains with Adam [19], AdamW or SGD with momentum 0.9 and the chosen recipe (learning-rate schedule with warm-up, weight decay, label smoothing, gradient clipping), with automatic mixed precision on CUDA [29];

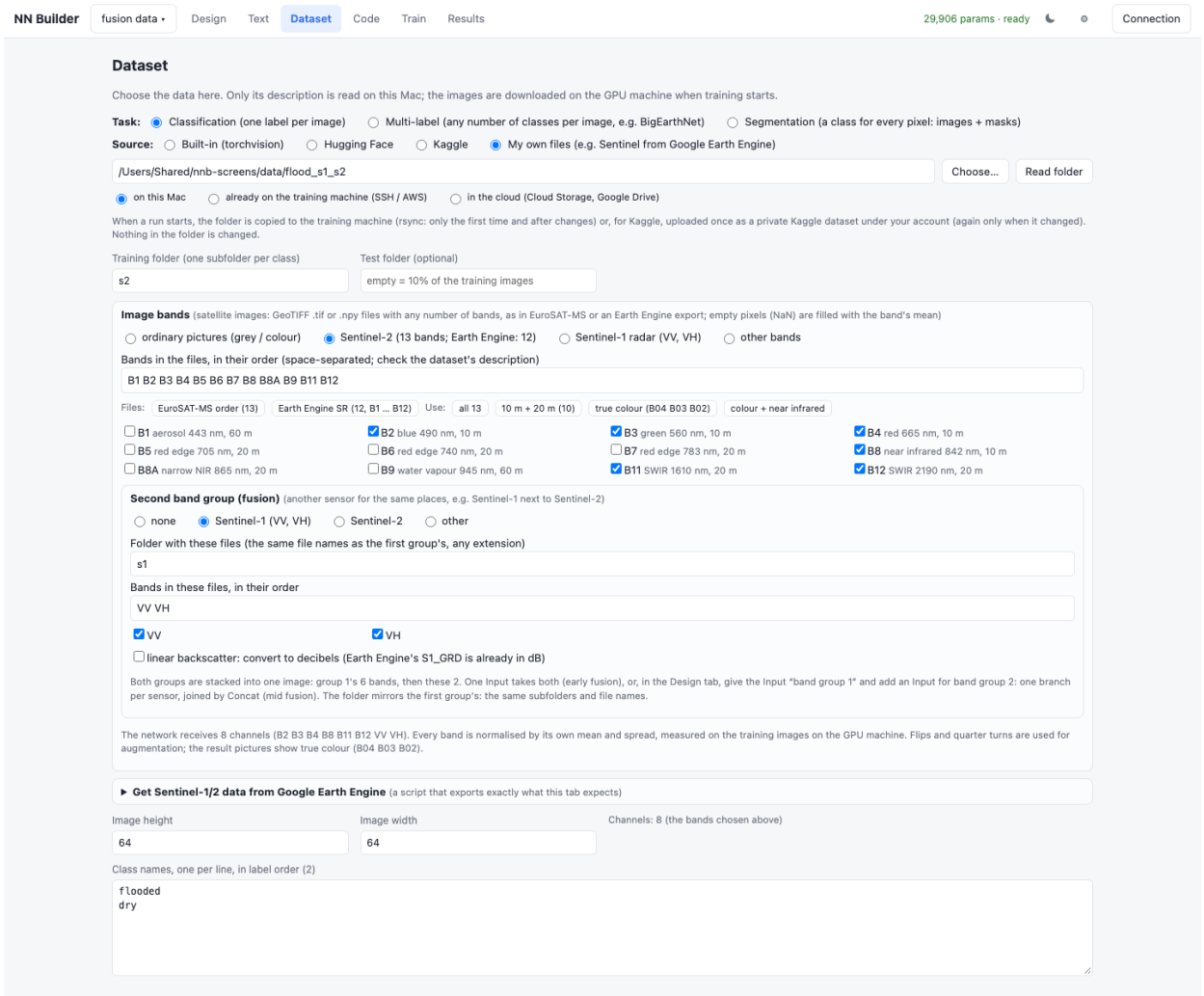


Figure 27: Dataset tab: Sentinel-2 (six bands, `s2/`) and, as the second band group, Sentinel-1 (VV, VH, `s1/`).

5. after each epoch evaluates on the test data and writes a **checkpoint** (`checkpoint.pt`; optionally also every few minutes inside an epoch); a restarted run **resumes** from it, in the middle of an epoch with the same batch order;

6. writes one JSON line per event to `metrics.jsonl` (start, batch, epoch, checkpoint, resume, interrupted, done, stopped, error) and finally `model.pt` (the weights) and `results.json` (test accuracy, confusion matrix, sample predictions, feature maps, first-layer filters, and per-class IoU for segmentation).

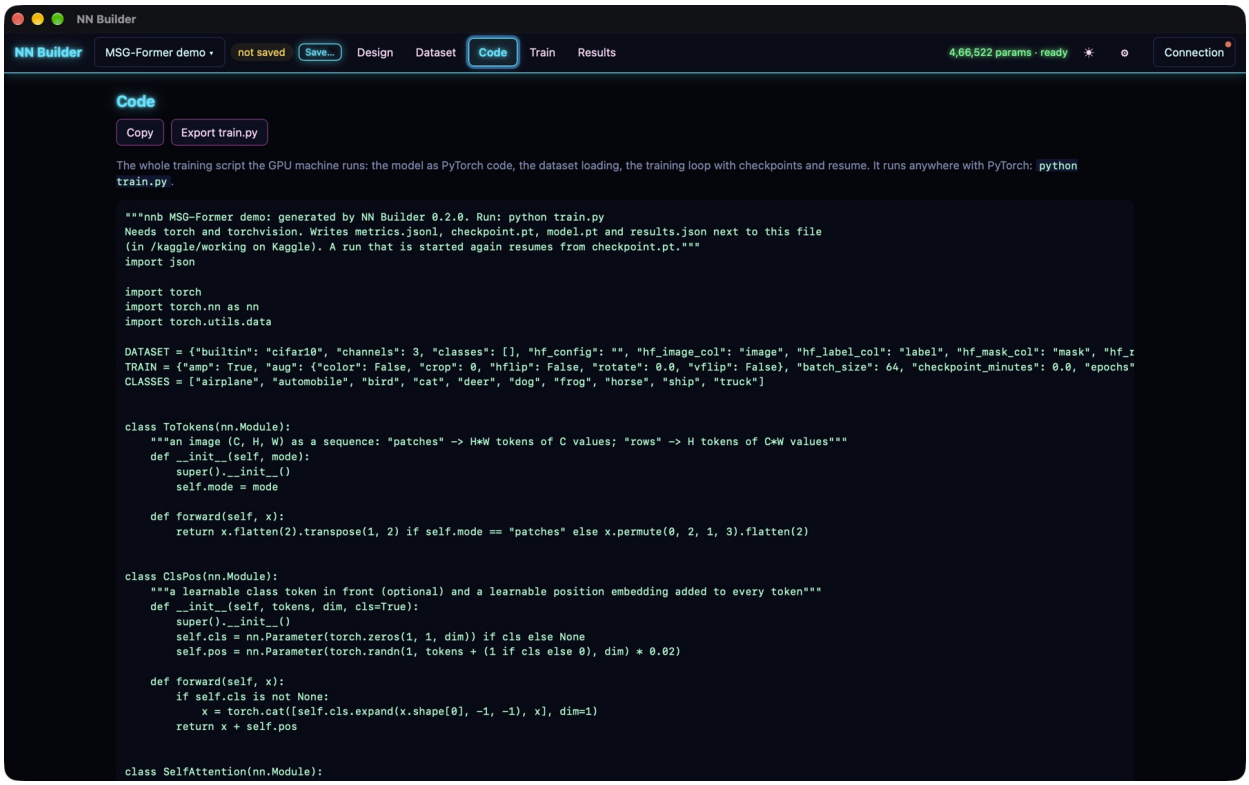


Figure 28: The Code tab: the generated `train.py` (here the start of MSG-Former's: the settings as Python literals and the transformer modules).

Signals: `TERM`/`INT` end the run cleanly (the model and results are still saved); on EC2, `USR1` means "AWS is taking the instance back" (checkpoint and exit to be resumed) and `USR2` means "checkpoint now" (Section 14.8).

13 The Connection tab

Tokens and keys are kept in the system keychain (macOS Keychain, Windows Credential Manager, Linux Secret Service) and handed to one call at a time; they are never written into projects, code or logs. When a run needs one on a remote machine, it is written there as a file readable only by its owner, through the SSH connection's input, never on a command line.

Kaggle. kaggle.com → profile → Settings → API → Create New Token. Paste the username and key (or the whole `kaggle.json`). GPU notebooks need a phone-verified account.

Hugging Face. Only for gated or private datasets: a "Read" token from huggingface.co → Settings → Access Tokens.

AWS. An IAM user's access key that may only start, stop and describe your instance (the page shows the policy to paste: `ec2:StartInstances` and `ec2:StopInstances` for

that instance, `ec2:DescribeInstances` and `ec2:DescribeInstanceAttribute`, plus `ec2:DescribeSpotInstanceRequests` for spot details). Before each start the app reads the instance's shutdown behaviour and refuses one set to Terminate (a run's end would delete it and its disk). When anything fails after the app started the instance, before training began, the app stops it again; an instance that was already running is left running.

Tools. Install tools makes a small private Python environment with the `kaggle` and `boto3` packages (about 60 MB) used to talk to Kaggle and AWS. No deep-learning library is installed.

13.1 Adding a machine, step by step

The form **Add a machine** is at the bottom of the Connection tab (Figure 29). The grey text in each field is only a hint (an example, or the default used when the field stays empty). A machine needs only a type, a name and a host (usually with a user); everything else may stay as it is.



Figure 29: The form *Add a machine* for a college or other SSH server. Grey text is only a hint; usually only Name, Host and User need to be filled in.

1. **Type. College / any SSH server** for a machine you reach with `ssh` (a college GPU server, a lab computer, a rented server); **AWS EC2 instance** for an instance the application should start and stop (the fields change to Region, Instance id, User and Key file, the AWS key above must be saved first).
2. **Name.** Your own short name, shown in **Train on** (for example `college-gpu`): letters, digits, spaces, `-`, `_` and `.`. Saving under a name already in the list replaces that machine.
3. **Host and User.** The server's address (`gpu01.college.edu` or an IP) and your login name on it, exactly as you would type `ssh user@host`. If you already have an entry for it in `~/.ssh/config`, put that alias in Host and leave User, Port, Jump host and Key file empty: `ssh` takes them from the file.
4. **Port.** 22 unless the server's administrator gave you another one.
5. **Login. Key / ssh config** when `ssh` logs you in without asking for a password (a key in `~/.ssh`); **Password (log in once in Terminal)** when it asks for one. The application never sees the password: you type it in a Terminal window and it reuses that connection.
6. **Jump host** (optional). When the server is reachable only through a gateway (`ssh -J user@gateway ...`), the gateway as `user@gateway.college.edu`. Not needed when this Mac is on the college network or VPN.
7. **Key file** (optional). A key that is not one of `ssh`'s default ones, e.g. `~/.ssh/id_ed25519`.

8. **Keep runs going with.** Leave it on **automatic**; it picks SLURM, `systemd-run`, `tmux` or `nohup` as the machine allows. Choose SLURM yourself on a cluster where jobs must go through the queue.
9. **Remote folder.** Where runs, data and the Python environment go, relative to your home folder on the machine (default `.nn_builder`). Change it when your home folder is small and the server has a larger scratch disk.
10. **Keep a copy of checkpoints on this Mac.** Off for a college server; turn it on for EC2 spot instances.
11. **Group** (optional). The same word (e.g. `lab`) on several machines makes them a pool: **Train on** then offers **Any free machine of lab**.
12. **Resume runs after a reboot.** Leave it on (the default for new machines) so a run cut off by a restart or power cut continues from its last checkpoint.
13. **Python with PyTorch on the machine** (optional). Leave it empty and the application makes its own environment with PyTorch in the remote folder, without administrator rights. Fill it in only to use a Python that already has PyTorch (for example `/opt/conda/bin/python`).
14. **Save**. The machine appears in the list above the form, and a message says what to do next.

After saving, in the machine's row:

1. With a password login: **Connect (log in once)**, then type the password in the Terminal window that opens.
2. **Test** checks that the application can log in. For an AWS machine, **Check instance** reports its state and shutdown behaviour instead.
3. **Prepare** looks at Python, the GPUs (`nvidia-smi`) and the disk, and installs PyTorch if needed (a few minutes the first time). The row then shows the PyTorch version, the GPUs and how runs are kept going. The first run prepares the machine by itself if you skip this.
4. The machine is now offered in the Training tab under **Train on**.

Edit changes a machine with the same form; **Delete** removes it from the list only (nothing on the machine is touched).

13.2 Machines

College / any SSH server. Host (or an alias from `~/.ssh/config`), user, port, login by key or *password*, optional jump host, remote folder (default `.nn_builder`), optional existing Python with PyTorch. With a password, **Connect (log in once)** opens a Terminal window where you type it; the application then reuses that connection and never sees the password. **Test** checks the login; **Prepare** checks Python, the GPUs (`nvidia-smi`), the disk, and installs PyTorch in the remote folder (no administrator rights needed). On an EC2 instance started from the AWS *Deep Learning AMI with PyTorch*, leave the Python field empty: **Prepare** finds the AMI's PyTorch (`/opt/pytorch`), uses it with its environment activated (the AMI keeps CUDA there) and adds only the few packages it lacks, so it takes about a minute instead of installing PyTorch. **Keep runs going with** chooses how a run survives logging out: SLURM (`sbatch`, the right way on a shared cluster), `systemd-run` (needs "lingering"), `tmux`, or `nohup`+`setsid`; *automatic* picks the best the machine allows.

AWS EC2 instance. Region, instance id, user (`ubuntu`) and the instance's key file (`.pem`). **Check instance** reports its state, type and shutdown behaviour (it must be *Stop*, not *Terminate*, or a

finished run would delete the instance), and for a spot instance its request type and interruption behaviour. The application starts the instance for a run and the run stops it at the end.

Keep a copy of checkpoints on this Mac. For either kind: every new checkpoint is copied back while the application is open, so a run can continue on another machine if its own machine is lost (essential for spot instances that may be terminated).

Group. Machines with the same group (for example `lab` for five lab computers with one GPU each) form a pool. **Train on** then offers **Any free machine of lab**: when the run starts, the application asks every machine of the group at once and takes the first that answers, has an idle GPU (under 1 GB in use and 10 % busy) and no NN Builder run going; it lists what it found for each (free, in use by someone, busy with a run, does not answer). With several seeds, each seed goes to another free machine. This Mac must be on the machines' network (the lab Wi-Fi or the VPN) to start the run; afterwards the run goes on without it.

Resume runs after a reboot (SSH machines, on for new ones). A lab computer may be restarted overnight, for updates or by a power cut. With this setting a run keeps a mark while it is unfinished, `train.py` saves a checkpoint every few minutes, and an `@reboot` line in your crontab on the machine starts the run again from its last checkpoint when the machine is back, also while this Mac is off. A clean restart (the system shutting down) is treated as an interruption, not as **Stop**. Under SLURM the scheduler requeues jobs itself, so this does not apply there. Without crontab on the machine, the run is started again only while the application is open and can reach the machine.

14 Training

14.1 Where to train

	Kaggle	**College GPU (SSH)**	**AWS EC2 g4dn.xlarge**
GPU	T4 or P100 (Kaggle's choice)	whatever the server has (e.g. RTX A6000 48 GB)	NVIDIA T4 16 GB, 4 vCPU
Cost	free (about 30 GPU hours a week)	free	per hour; spot is cheaper but may be interrupted
Limits	about 12 h per run; runs may queue	the server's rules (SLURM time limits)	your quota for G instances
Progress	from the notebook's log, every 30 s	live (every few seconds)	live
Needs	Kaggle token	SSH access	AWS key, key file, quota

14.2 Training settings

Setting	**Meaning and advice**
Train on	Kaggle, or a machine from the Connection tab
GPU on the machine	which GPU(s) to use on a shared server: empty = all of them, `0` or `0,1` = those, `free` = wait until enough GPUs are idle and take them (below)
Epochs	passes over the training data
Batch size	images per step; larger is faster but needs more memory (halve it on "out of memory")
Learning rate	0.001 suits Adam on small networks; very large networks (ViT-Base and bigger) need 0.0001 or less [39]

Setting	Meaning and advice
Optimizer	Adam [19], AdamW (weight decay applied separately [26]) or SGD with momentum 0.9 (often better for ResNet-style networks with a good schedule [9])
Training recipe	learning-rate schedule (constant; cosine decay to 0; or halve when the validation loss stalls for 3 epochs), linear warm-up over the first epochs, weight decay, label smoothing, gradient clipping. **Use recommended** sets AdamW, weight decay 0.05, 5 warm-up epochs then cosine, label smoothing 0.1 and clipping at 1.0, the usual recipe for transformers and a good one for CNNs [39, 9]. The defaults are plain constant-rate training, as in earlier versions
Training images (0 = all)	a quick trial on a subset
End the run after (hours of running)	a time limit, summed over resumes (protects the budget)
Checkpoint every (minutes)	also checkpoint inside an epoch; 0 = only at the end of each epoch (spot instances then use 10 minutes)
Automatic resumes	spot instances: how many interruptions are resumed automatically (default 20)
Validation part of the training data	held out from the training images (default 0%: the curves' "validation" is then the test data). Use 10–20% whenever results are reported: the test data is then touched only once, at the end
Early stopping	stop when the validation loss (or accuracy, or mIoU) has not improved for *patience* epochs; the best epoch's weights become `model.pt` and the test accuracy is measured on them. Switching it on sets a 10% validation part if there is none. Epochs is then the maximum
Random seed	fixes the initial weights, the derived splits and the batch order; run several seeds to report a mean and spread
Mixed precision	faster on T4/A100-class GPUs, half the activation memory [29]. GPUs with bfloat16 (A100, A6000, RTX 30xx and newer) use it by themselves (no loss scaling needed); networks with convolutions use the faster channels-last memory layout on any NVIDIA GPU
Split a batch that does not fit	on by default: when the GPU runs out of memory, the batch is processed in 2, 4, 8 ... parts whose gradients add up to the whole batch's, and the run continues instead of failing. Same batch, schedule and results (BatchNorm normalises over the smaller parts, the only difference); slower. The run card says when it happened
Save memory	activation checkpointing [3]: `train.py` cuts the network into about $\sqrt{n}$ parts at places where only one tensor passes, keeps only the parts' outputs and recomputes the inside of each part in the backward pass. The layer outputs then need roughly $2/\sqrt{n}$ of the memory, for about 30% more time; the results are the same
Stop the EC2 instance	stop the instance when the run ends (saves money)
Data augmentation	horizontal / vertical flip, random crop after padding, rotation, brightness/contrast; training images only [4]. Flips change the meaning of digits; vertical flips suit aerial images
Copy my token to the machine	needed for Kaggle data on a server/EC2 and for gated Hugging Face data

Setting	Meaning and advice
Start from ImageNet weights	transfer learning: torchvision's pretrained ResNet-18, VGG-16 or ViT-B/16 loaded into the matching demo, tensor by tensor (a new output head for your classes starts at random); optionally train only the new head. Images are then normalised with ImageNet's statistics. Usually far better than training from scratch on small datasets; use a smaller learning rate (e.g. 0.0001) when fine-tuning everything

A shared GPU server: wait for free GPUs. On a server that other people use, some GPUs may be full (memory almost used up, 90–100 % busy). Type `free` as the GPU on the machine. The run then waits on the server, looking every minute, until enough GPUs are idle: under 1 GB of memory in use and under 10 % busy, in two looks 15 s apart, so a GPU between someone's jobs is not taken. With **GPUs** = 2 it waits for two; with **GPUs** = all it starts as soon as one is idle and takes every idle one. The run card shows "waiting for 2 free GPU(s)" and then "GPU(s) 1,3 are free: training on them"; the log adds a line every ten minutes while it waits. Waiting does not count against the time limit, and **Stop** ends it. A run that has started keeps its GPUs: it does not add GPUs that become free later. Under SLURM there is no waiting here: the scheduler queues the job and hands out the GPUs itself.

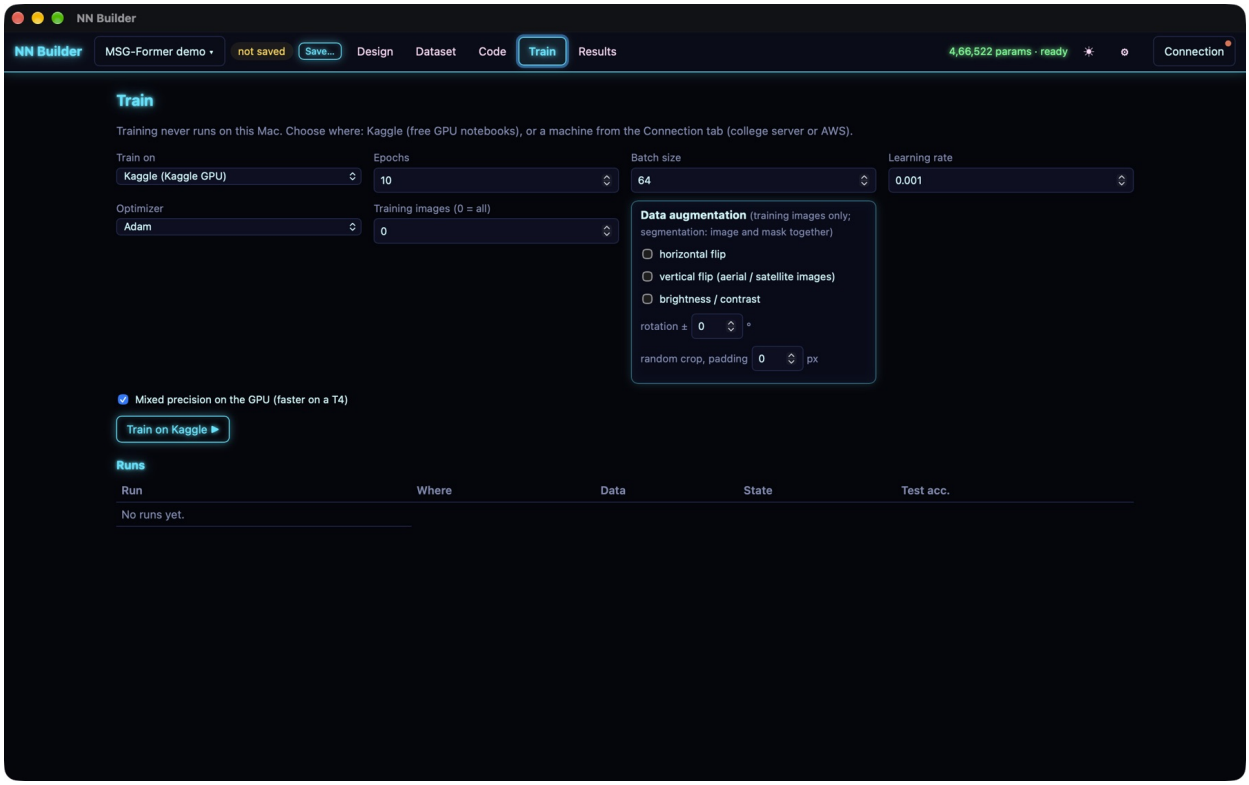


Figure 30: The Train tab: where to train, the main settings, augmentation, and the list of runs. Settings for SSH and EC2 machines (time limit, checkpoint interval, automatic resumes, stopping the instance) appear when such a machine is chosen.

Above the Train button the tab says what is still missing for the chosen target (tools, token, key) and gives a rough estimate of the GPU memory the training needs (16 bytes per parameter plus the layer outputs at the chosen batch size), compared with the target's GPU (Kaggle and EC2 g4dn: 16 GB; an SSH machine: as **Prepare** found it). It warns when the design is likely not to fit.

14.3 Unbalanced classes

Real land-cover data is rarely balanced (BigEarthNet's rarest labels are on under 1 % of the patches), and a network can then score well while ignoring the rare classes. The Dataset tab shows the images per class next to the class folders (Read folder / Read files), with a warning above a 3:1 ratio; **Check data** counts them on the machine for every source. Three settings in the Train tab's recipe help:

- **Unbalanced classes**: *weight by inverse frequency* gives class c the weight $N/(K\,n_c)$ in the loss (N training images, K classes, n_c of class c), so every class counts as much in total; *by its square root* is gentler. For multi-label, each class's positive examples weigh $(N - n_c)/n_c$ (its square root) against its negatives. Segmentation counts pixels. The weights are computed on the training machine and listed in the log and in Results.
- **multi-label: a threshold per class**: after training, each class gets the threshold (5 %–95 %) with the highest F1 on the validation data instead of 50 %; the test scores use them, Results lists them, and `thresholds.json` is saved next to the model. It needs validation data, so it sets the validation share to 10 % if it was 0.
- **Loss**: *cross-entropy* (binary cross-entropy for multi-label) is the default. *Focal* multiplies each image's (pixel's) term by $(1 - p)^\gamma$, where p is the probability given to the right answer, so the images the network already gets right count little and the rare, hard ones count more; $\gamma = 0$ is plain cross-entropy, 2 is usual. For segmentation and multi-label there is also *Dice*: $1 - \frac{1}{K}\sum_c \frac{2|P_c \cap T_c|}{|P_c| + |T_c|}$ on the batch's predicted probabilities P_c and true pixels (images) T_c of class c, so a small class counts as much as a large one; *Dice + cross-entropy* adds the two and learns more steadily at the start. Unlabelled pixels are left out of every loss, and class weights apply to all of them (Dice: a weighted mean over the classes). Dice for one class per image is refused before the run starts. Results names the loss used.

Look at the per-class scores (precision, recall, F1) rather than the accuracy alone.

14.4 Check data before training

Check data on ..., next to the Train button, starts a short run on the same target that loads the data exactly as training would, reports it, and sends one batch through the network, but does not train: minutes instead of a GPU hour lost to a wrong band order or a bad labels file. The run is listed as a *data check* ("looks right", or how many things to look at), and the Results tab shows:

- the number of training, validation and test images, and the training images per class (pixels per class for segmentation, counted on a sample), with a bar per class;
- for satellite images, every band's raw values in the files (minimum, maximum, mean, spread, empty and zero pixels) from a sample of training images;
- twelve test images as the network receives them (composites for satellite images) with their labels;
- warnings: a band that is empty or constant (often a wrong band order), many empty pixels, Sentinel-1 in linear units without **convert to decibels**, Sentinel-2 values far above 10 000, a class without any images, strongly unbalanced classes, very few images, rows of a labels CSV without a file, and anything the data loading itself reported.

Nothing is saved in the project's settings: the next **Train on ...** trains as usual.

14.5 Following a run

Train on ... starts a background task whose log appears at the bottom right. The run card then shows the state (starting, running with the epoch, interrupted, done, stopped, crashed, failed), the GPU, the training set size, the loss curves (per batch, training and validation per epoch) and the accuracy curves (and mIoU for segmentation), the last checkpoint, and the log. **Stop** ends the run

cleanly (the current epoch ends early, the model and results are saved; the data-loading helpers ignore the stop signal and leave it to the main process, so they cannot end the run with an error first). **Fetch results** copies the outputs back (for a stopped EC2 instance it starts the instance, copies, and stops it again). The **Runs** table lists every run of the project with its seed and test accuracy.

14.6 Several seeds, comparing runs, exporting results

A single training run is one sample: the same design and settings with another random seed give a slightly different accuracy. Next to the Train button, **seeds** starts 3 or 5 runs at once with seeds $s, s{+}1, \ldots$ (one group; on Kaggle one notebook per seed, which Kaggle may queue; on a server they share its GPU). Tick runs in the **Runs** table (or select a whole group of seeds) and press **Compare selected**: their validation curves are overlaid, and a table gives each run's recipe, epochs, best epoch, test accuracy (and top-5), training time, and for each group the mean $\pm$ sample standard deviation. **Export CSV + LaTeX** writes the table to the project's `exports` folder as `runs-<time>.csv` and `runs-<time>.tex` (a `booktabs` tabular with a "Mean $\pm$ s.d." row, ready to `\input` into a paper).

14.7 Independence from the connection

On a server or EC2, the run folder is copied with `rsync` and started detached (SLURM, systemd, tmux or nohup). The application only looks at it: every few seconds while the run is open it copies the new metric lines and the end of the log. Closing the application, sleeping the laptop or losing the network does not affect the run; the application catches up when it can reach the machine again.

14.8 EC2 spot instances

Spot instances use spare EC2 capacity at a lower price, but AWS may take one back with a two-minute notice. NN Builder recognises a spot instance by itself. The recommended setup is a *persistent* spot request with interruption behaviour *stop* (AWS restarts the instance when capacity returns, with its disk), plus **Keep a copy of checkpoints on this Mac**.

Situation	What NN Builder does
Two-minute interruption notice	a watcher on the instance sees it in the instance metadata; the script saves a mid-epoch checkpoint at once and exits to be resumed
Rebalance recommendation (earlier warning)	a checkpoint is saved now; training continues
Interruption without notice	at most the checkpoint interval of training is lost (10 minutes by default on spot)
Interrupted while writing a checkpoint, or disk full	checkpoints are written to a temporary file and swapped in: the previous one is never damaged
Instance restarted while the laptop is off	a boot-time job on the instance resumes the run: no laptop needed
Instance back while the application is open	the application follows the instance's new address and restarts the run if needed
Two starters at once	a lock lets only one copy of a run train
Resumed in the middle of an epoch	the remaining batches of that epoch, in the same order; optimiser, mixed-precision scaler and random state restored

Situation	What NN Builder does
Interrupted again and again	at most the set number of automatic resumes; the time limit counts running time over all resumes
Instance terminated	shown as "terminated"; with the copy on the Mac, **Continue on another machine** starts a new run from it (for example on a college GPU)
Old instance returns after moving	when that run is opened and the instance answers, the run there is stopped (cancel its spot request in the AWS console to be sure)
Stop while the instance is down	remembered: the run is stopped instead of resumed when the instance answers

14.9 Can my GPU train this network?

Use the 16-bytes-per-parameter rule of Section 7 plus room for activations. A 48 GB RTX A6000 can train ViT-g/14 (about 18 GB of fixed state) with batches of roughly 16 to 32 images at 224×224; a 16 GB T4 cannot. Large networks trained from scratch on small datasets overfit: ViT-g was trained on billions of images [47], and even ViT-Base needs strong augmentation and regularisation, or pre-training, to do well on small data [39, 41].

With **Save memory** the layer outputs shrink to a small fraction (the estimate on the Train tab takes it into account), so a network whose weights and optimizer state fit can usually train with a reasonable batch; the weights and optimizer state themselves (16 bytes per parameter) do not shrink. After the first epoch the run card shows the memory really used (the peak of the GPU allocator), the precision, and whether batches were split; it also warns when the GPU waited for data more than 30% of an epoch (loading and augmenting images is then the bottleneck: the data workers already use one CPU core each, up to 8). Halfway through every epoch the log (`log.txt`, or the notebook's log on Kaggle) also has a plain line such as `GPU use (epoch 25, step 781/1563): training on 2 of 2 GPU(s) · GPU 0 Tesla T4: 3.2 / 15.0 GB, 87 % busy · GPU 1 ...`: how many GPUs train, and for each the memory in use (all processes on it) and how busy it is, read with `nvidia-smi`.

14.10 How large a design the application handles

The application never holds weights, so the number of parameters does not matter to it; the number of *blocks* does. Measured on a Mac (designs written as text: ViT-style encoders, 224×224 input):

Design	parameters	blocks	window memory	analysis per edit
none open	–	–	56 MB	–
ViT-g/14 (40 layers, width 1 408)	1.0 B	287	164 MB	1–2 ms
width 8 192, 32 layers	25.8 B	231	151 MB	1 ms
width 12 288, 55 layers	99.7 B	392	192 MB	1 ms
width 4 608, 100 layers	25.5 B	707	287 MB	3 ms
width 2 048, 500 layers	25.2 B	3 507	965 MB	21 ms

The canvas needs about 0.3 MB per block drawn. **Settings** → **Design size** sets the most blocks a design may have: recommended from the computer's memory (a quarter of it at 0.3 MB per block, e.g. 13 000 blocks with 16 GB, 105 000 with 124 GB) and editable; it concerns the computer running NN Builder, not the training machine. Above 500 blocks the canvas says so and offers **Collapse repeated blocks (×N)**, which draws each repeated group once; the Text tab edits a

design of any size without drawing it. Conv2D filters and attention heads go up to 65 536 and 512; a setting outside its range is an error on its block (it is never silently changed).

15 The Results tab

What it shows depends on the task: *multi-label*: micro / macro F1, mAP, exact match and every class's precision, recall, F1 and AP; *regression*: R^2, MAE and RMSE per target and a predicted-against-true scatter plot; *detection*: mAP@0.5, mAP@0.5:0.95, AP per class and test images with the boxes found and the true ones; *extra outputs*: a table of every output's score; *early exits*: each exit's accuracy and what stopping confident images early would give; a *data check*: the data report. For classification and segmentation, as follows. For a finished run (open it in the Train tab): the test accuracy (and top-5 accuracy with more than five classes, and mIoU), precision, recall and F1 of every class with their macro averages and the hardest classes, the confusion matrix (rows: true class, columns: predicted; the diagonal is correct), sample predictions with their confidence (wrong ones marked), the feature maps of the first test image (first eight channels of early layers), the first layer's filters as small pictures, and for segmentation the true and predicted masks and the IoU of every class; **Grad-CAM** heat maps on the sample images (where the network looked: the last image-shaped layer's feature maps weighted by how strongly the predicted class depends on them [36]); and a gallery of up to 16 **misclassified** test images with their true and predicted class. **Show files** reveals `model.pt`, `train.py` and `results.json`, and with **Also save the trained model as TorchScript and ONNX** also `model.ts` (load with `torch.jit.load`, no NN Builder code needed) and `model.onnx` (for ONNX Runtime and other frameworks). The weights load in PyTorch with `Net().load_state_dict(torch.load("model.pt"))` using the `Net` class from the same `train.py`.

16 The demo architectures

Each demo opens unsaved on the canvas (Projects page, **File** → **New from Demo**, or **Famous architectures** in the Design tab) at the input size it was made for. Parameter counts below are the application's, with a 10-class output (2 classes for the segmentation networks); they change with the number of classes. Where a demo differs from the paper, the difference is stated.

Demo	Year	Task	Input	Main idea	Parameters
LeNet-5	1998	classify	1×28×28	convolution + subsampling	61 706
Small CNN (CIFAR-10)	2015	classify	3×32×32	VGG-style blocks + BatchNorm	551 018
AlexNet	2012	classify	3×224×224	deep CNN on GPUs, ReLU, dropout	58 322 314
VGG-16	2014	classify	3×224×224	only 3×3 convolutions	134 301 514
GoogLeNet	2014	classify	3×224×224	Inception modules	5 983 802
ResNet-18	2015	classify	3×224×224	residual connections	11 181 642
MobileNet v1	2017	classify	3×224×224	depthwise separable convolutions	3 217 226
DenseNet (small)	2017	classify	3×32×32	dense connectivity	76 810
Transformer encoder	2017	classify	1×28×28	self-attention on image rows	71 498
ViT-Base/16	2020	classify	3×224×224	image patches as tokens	85 806 346
ViT-Tiny (CIFAR-10)	2020	classify	3×32×32	a small ViT	1 806 538
ViT-g/14	2022	classify	3×224×224	scaling a ViT to 1 B parameters	1 011 216 522
U-Net	2015	segment	3×128×128	encoder–decoder with skips	7 760 130
Attention U-Net	2018	segment	3×128×128	attention gates on the skips	7 847 894

Demo	Year	Task	Input	Main idea	Parameters
MoE Transformer (GShard)	2020	classify	3×32×32	mixture of experts in every other layer	2 198 306
MSG-Former	2026	classify	3×32×32	NN Builder original (hybrid)	466 522

16.1 LeNet-5 (1998)

Background. LeCun et al. trained LeNet-5 to read handwritten digits on bank cheques [23]; it established the pattern every CNN since has followed: convolutions that detect local features, subsampling that makes them tolerant to small shifts, and fully connected layers that classify.

As built in NN Builder.

Block	Settings	Output	Parameters
Conv2D C1	6 filters, 5×5, padding 2, tanh	6×28×28	156
Pool S2	average 2×2	6×14×14	0
Conv2D C3	16 filters, 5×5, valid, tanh	16×10×10	2 416
Pool S4	average 2×2	16×5×5	0
Flatten		400	0
Dense C5, F6	120 then 84 units, tanh	84	48 120 + 10 164
Output	10 classes	10	850

Differences from the paper. The paper used 32×32 inputs with no padding (padding 2 on 28×28 MNIST gives the same 28×28 maps), trainable subsampling coefficients, a hand-designed partial connection table in C3, and Gaussian output units; the demo uses plain average pooling, full connections and a softmax output.

Training it. Dataset: built-in MNIST (no resizing needed). Adam, learning rate 0.001, batch 64, 5–10 epochs; it reaches about 99% test accuracy on any GPU in a few minutes. Look at the first-layer filters in Results: they become stroke and edge detectors.

16.2 Small CNN for CIFAR-10

Background. Not one paper but the standard "VGG-style" baseline for 32×32 colour images: stacks of 3×3 convolutions [37] with BatchNorm [18] and dropout [38].

As built. Three blocks with 32, 64 and 128 filters; each block is two times (Conv2D 3×3 without bias → BatchNorm → ReLU), then max pooling 2×2 and dropout (0.2, 0.3, 0.4). The image goes 32 → 16 → 8 → 4; Flatten gives 2 048 values, then Dense 128 (ReLU), Dropout 0.5 and the output. The classifier holds 263 562 of the 551 018 parameters.

Training it. Built-in CIFAR-10; Adam 0.001, batch 128, 30–50 epochs, with horizontal flips and a random crop with 4 pixels of padding: such networks commonly reach the mid to high 80s in test accuracy. Without augmentation the training accuracy runs far ahead of the validation accuracy (overfitting): compare the two curves in the Train tab.

16.3 AlexNet (2012)

Background. Krizhevsky, Sutskever and Hinton won the ImageNet challenge [35] of 2012 by a wide margin with a deep CNN trained on two GPUs [21]; its ingredients (ReLU instead of tanh, dropout, data augmentation, GPU training) started the deep-learning era in computer vision.

As built. Conv2D 96 filters 11×11 stride 4 padding 2 (224 → 55) → max pool 3×3 stride 2 (overlapping, 55 → 27) → Conv2D 256 5×5 → pool (13) → Conv2D 384, 384, 256 (3×3) → pool

(6) → Flatten (9 216) → Dropout 0.5 → Dense 4 096 → Dropout 0.5 → Dense 4 096 → Output. Of the 58.3 million parameters, 37.8 million sit in the first Dense layer: the layer table's Classifier group shows this at a glance.

Differences from the paper. No local response normalisation, and no split of the layers into two groups (one per GPU); with 1 000 classes the demo has about 62 million parameters (the paper reports 60 million). A later note by Krizhevsky describes the single-GPU variant [22].

Training it. AlexNet expects 224×224 images: use CIFAR-10 with **Resize** (slow and not ideal: tiny photos enlarged) or a Hugging Face dataset of larger photos. Use a smaller learning rate (Adam 0.0001) and flips. It fits a T4 at batch 64–128.

16.4 VGG-16 (2014)

Background. Simonyan and Zisserman showed that depth matters and that small filters suffice: two 3×3 convolutions see as far as one 5×5, three as far as one 7×7, with fewer parameters and more non-linearities [37].

As built. Five blocks of 3×3 convolutions (ReLU, padding same) with 64, 64 | 128, 128 | 256 ×3 | 512 ×3 | 512 ×3 filters, each block followed by 2×2 max pooling (224 → 7); Flatten (25 088), two Dense 4 096 with dropout 0.5, Output. 134 301 514 parameters, exactly torchvision's `vgg16` with a 10-class head; 102.8 million of them are in the first Dense layer.

Training it. Like AlexNet: 224×224 images, Adam 0.0001 or SGD 0.01 with momentum, batch 32–64 on a T4. Watch the receptive field grow in the inspector: 3×3 convolutions and poolings reach 212×212 pixels at the last convolution.

16.5 GoogLeNet / Inception v1 (2014)

Background. Szegedy et al. asked which filter size to use, and answered "all of them, side by side": the Inception module runs 1×1, 3×3 and 5×5 convolutions and a pooling branch in parallel and concatenates them [40]. 1×1 convolutions first reduce the channels (the idea comes from Network in Network [25]), keeping the cost low: GoogLeNet has twelve times fewer parameters than AlexNet and was more accurate.

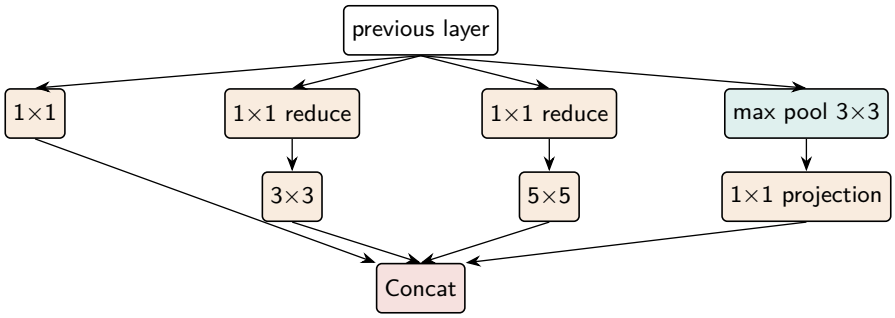


Figure 31: An Inception module: four branches side by side (all keep the height and width); Concat stacks their channels.

As built. Stem: Conv2D 64 7×7 stride 2, max pool, Conv2D 1×1 64, Conv2D 3×3 192, max pool (224 → 28). Then nine Inception modules with the paper's channel numbers: 3a (output 256 channels), 3b (480), max pool, 4a–4e (512, 512, 512, 528, 832), max pool, 5a (832), 5b (1 024); global average pooling, dropout 0.4, output. 5 983 802 parameters.

Differences from the paper. The two auxiliary classifiers used during training are left out (later work found them of little use), and there is no local response normalisation.

Training it. 224×224 inputs; without BatchNorm, deep plain networks like this train slowly: use

Adam 0.0001–0.0003 and patience, or add BatchNorm after the convolutions yourself.

16.6 ResNet-18 (2015)

Background. Deeper plain networks became *harder to train*, not just to generalise: adding layers raised the training error. He et al. let every block learn a correction to its input, $y = x + F(x)$, through a shortcut [10]; gradients then flow through the shortcuts, and networks of 50, 101, even 1 000 layers train well [11].

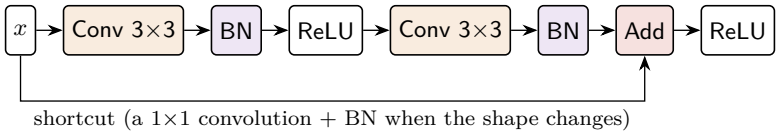


Figure 32: A basic residual block as built on the canvas: $y = \mathrm{ReLU}(x + F(x))$.

As built. Stem: Conv2D 64 7×7 stride 2 + BatchNorm + ReLU, max pool 3×3 stride 2 (224 → 56). Four stages of two basic blocks with 64, 128, 256 and 512 filters; the first block of stages 2–4 has stride 2 and a 1×1 stride-2 convolution with BatchNorm on the shortcut (56 → 28 → 14 → 7). Global average pooling and the output. 11 181 642 parameters, exactly torchvision's `resnet18` with a 10-class head.

Training it. The most forgiving of the large demos thanks to BatchNorm and the shortcuts. At 224×224 it fits a T4 at batch 64–128. For CIFAR-10 at 32×32 the stem shrinks the image too early (32 → 8 before the first block); the usual CIFAR variant replaces the stem with a 3×3 stride-1 convolution and drops the max pooling, an easy edit on the canvas.

16.7 MobileNet v1 (2017)

Background. For phones, Howard et al. factorised a convolution into a *depthwise* 3×3 convolution (one filter per channel, spatial only) and a *pointwise* 1×1 convolution (mixing channels) [15]. The cost drops by a factor of about $1/N + 1/k^2$ (N output channels, $k = 3$): roughly eight to nine times fewer computations.

As built. Conv2D 32 3×3 stride 2 + BN + ReLU, then 13 depthwise separable blocks (Depthwise Conv2D 3×3 + BN + ReLU, Conv2D 1×1 + BN + ReLU) with 64, 128, 128, 256, 256, 512 (×6), 1 024, 1 024 channels; five of them have stride 2 (224 → 7); global average pooling and the output. 3 217 226 parameters (4.2 million with 1 000 classes, as in the paper). Width and resolution multipliers of the paper can be imitated by changing the filter counts and the input size.

Training it. Light and fast; SGD or Adam 0.001 with BatchNorm works well. Depthwise convolutions are memory-bound on GPUs, so it is not proportionally faster to *train*; its advantage is at inference on small devices.

16.8 DenseNet, small CIFAR version (2017)

Background. Huang et al. connected each layer to *all* earlier layers of its block by concatenation [17]: features are reused instead of re-learnt, so each layer can be narrow (the "growth rate" k, here 12 channels).

As built. Conv2D 24 3×3; three dense blocks of four layers (BatchNorm → ReLU → Conv2D 12 3×3, concatenated to everything before); between blocks a transition (BatchNorm → 1×1 Conv2D halving the channels → average pool 2×2). The channels go 24 → 72 → 36 → 84 → 42 → 90 while the image goes 32 → 16 → 8; BatchNorm, ReLU, global pool and output. Only 76 810 parameters.

Differences from the paper. The paper's CIFAR models are much deeper (for example 100 layers with bottleneck 1×1 convolutions inside each layer, about 0.8 million parameters); the demo is a small teaching version. The paper's transitions have a ReLU after their BatchNorm; the demo's do not.

Training it. CIFAR-10 built in, Adam 0.001, flips and crops, 50+ epochs; small but deep, so batch 64–128 fits easily.

16.9 Transformer encoder on image rows (2017)

Background. The Transformer of Vaswani et al. replaced recurrence with self-attention: every token computes a weighted mix of all tokens, the weights coming from query–key similarities [42]. Its encoder is a stack of attention and two-layer MLP blocks with residual connections and layer normalisation [1].

As built. A teaching version on MNIST: To tokens (*rows*) makes each of the 28 rows a token of 28 values; Dense 64 embeds them; Class token + position gives 29 tokens of 64; two pre-norm encoder blocks with 4 heads, an MLP of 128 (GELU [13]) and dropout 0.1; LayerNorm, the class token, output. 71 498 parameters.

Training it. MNIST, Adam 0.001, 10–20 epochs: it reaches high-90s accuracy and shows that attention can learn image structure from a sequence of rows. Change the number of heads (64 must stay divisible by it) or blocks and compare.

16.10 ViT-Base/16 (2020)

Background. Dosovitskiy et al. showed that a plain Transformer encoder on image patches, "an image is worth 16×16 words", matches the best CNNs when pre-trained on enough data [7].

As built. Conv2D 768 filters 16×16 stride 16 (the patch embedding: 224 → 14×14), To tokens (196 tokens of 768), Class token + position (197 tokens), 12 encoder blocks (12 heads, MLP 3 072), LayerNorm, class token, output. 85 806 346 parameters, exactly torchvision's `vit_b_16` with a 10-class head (86.6 million with 1 000 classes).

Training it. ViTs lack the built-in assumptions of convolutions (locality, translation equivariance) and need much data or strong augmentation and regularisation [39, 41]. From scratch on a small dataset it does worse than ResNet-18. Use a low learning rate (0.0001 or less; AdamW with warm-up is the usual recipe [26, 9], not built in), batch 32–64 on a T4 with mixed precision.

16.11 ViT-Tiny for CIFAR-10

Background. Small ViTs in the spirit of DeiT-Ti [41] (embedding 192, 3 heads) make transformers trainable on small images in minutes.

As built. Conv2D 192 4×4 stride 4 (32 → 8×8 = 64 patches), To tokens, Class token + position (65 tokens), 6 encoder blocks with 3 heads, MLP 384 and dropout 0.1, LayerNorm, class token, output. 1 806 538 parameters. (DeiT-Ti itself has 12 blocks and an MLP of 768.)

Training it. CIFAR-10 with flips and crops (augmentation matters a lot for ViTs), Adam 0.001 or a little lower, 50+ epochs. It usually stays below the Small CNN on CIFAR-10 when trained from scratch for a short time: a useful experiment on the data hunger of transformers.

16.12 MoE Transformer encoder (GShard, 2020)

Background. Lepikhin et al. scaled a translation Transformer to 600 billion parameters by replacing the feed-forward network of every other layer with a mixture of experts [6]: a gating

router sends each token to its top 2 of E experts, so the weights grow with E while the work per token stays that of about two feed-forward networks.

As built. Conv2D 128 4×4 stride 4 (64 patches), To tokens, Class token + position (65 tokens), then 6 layers in the paper's post-norm order: Self-attention (4 heads, dropout 0.1) → Add → LayerNorm, then the feed-forward part → Dropout → Add → LayerNorm. Layers 1, 3 and 5 use a **Mixture of experts** (8 experts, top 2, 256 hidden, capacity 1.25, random second expert), layers 2, 4 and 6 a dense feed-forward network (256 GELU, 128). The Text tab shows it as `repeat 3` of a pair of layers. 2 198 306 parameters, 72 % of them in the experts.

What differs from the paper. The paper's third panel places each expert on its own device and exchanges tokens all-to-all; here every GPU holds all experts (see the Mixture of experts block above). Images instead of text, and 6 layers of width 128 instead of 36 of 1024.

16.13 ViT-g/14, one billion parameters (2022)

Background. Zhai et al. studied how ViTs scale with model size, data and compute, training up to two billion parameters on billions of images [47]. ViT-g/14 ("giant") is their one-billion-parameter model.

As built. Conv2D 1 408 filters 14×14 stride 14 (224 → 16×16 = 256 patches), Class token + position (257 tokens), 40 encoder blocks with 16 heads and MLP 6 144, LayerNorm, class token, output: 1 011 216 522 parameters. Each encoder block alone has 25 250 176 parameters (attention $4d^2 + 4d = 7\,935\,488$; MLP $2 \cdot 1408 \cdot 6144 + 6144 + 1408 = 17\,309\,056$; two LayerNorms 5 632). The canvas holds about 290 blocks; the layer table groups them into Embedding, 40 encoder blocks, Final norm and Classifier.

Training it. About 16 GB of fixed state for Adam (Section 7): it does *not* fit a 16 GB T4 (Kaggle, g4dn.xlarge). A 48 GB RTX A6000 fits it with batches of roughly 16–32 at 224×224 (estimate), at around 1.5 TFLOP of compute per training image; a checkpoint with the optimiser state is about 12 GB and the final `model.pt` about 4 GB, both copied over the network. From scratch on a small dataset it overfits badly and scores below ResNet-18; its value here is to see how a transformer scales (compare it block by block with ViT-Base) and to measure memory and speed.

16.14 U-Net (2015)

Background. Ronneberger et al. designed U-Net for biomedical image segmentation with few training images [34]: a contracting path captures context, an expanding path recovers resolution, and skip connections hand the fine detail of each encoder level to the decoder level of the same size.

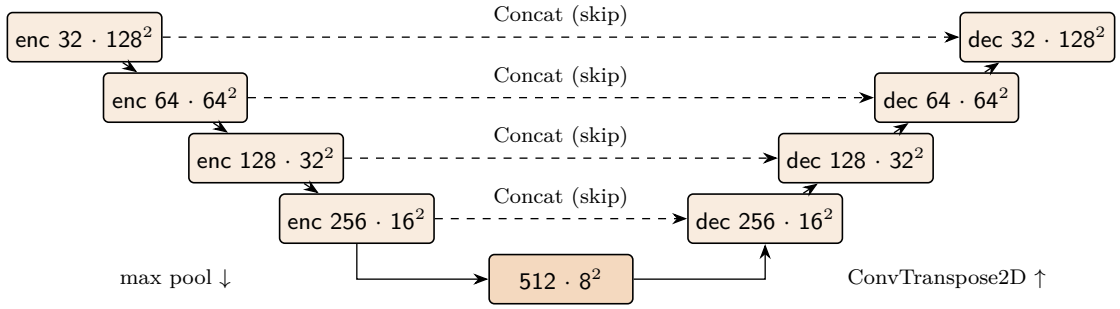


Figure 33: The U shape as built: channels · image size per level, 128 → 8 pixels and back, with a skip at every level.

As built. Input 3×128×128. Encoder: four levels of two 3×3 convolutions (ReLU) with 32, 64, 128, 256 filters, each followed by 2×2 max pooling; bottleneck: two convolutions with 512 filters at

8×8. Decoder: at each level a 2×2 stride-2 ConvTranspose2D halving the channels, Concat with the encoder output of the same size, two 3×3 convolutions; Segmentation output (1×1) with two classes. 7 760 130 parameters.

Differences from the paper. The paper starts at 64 filters (1 024 at the bottom, about 31 million parameters) and uses unpadded convolutions, cropping the encoder maps before concatenating; the demo uses padding "same", so sizes match without cropping.

Training it. Dataset → Segmentation, with an image column and a mask column (Hugging Face) or images and masks folders (Kaggle); the class list maps mask values to classes. Flips and rotations are applied to the image and its mask together. Watch the mIoU curve and the predicted masks in Results.

16.15 Attention U-Net (2018)

Background. Oktay et al. added *attention gates* to U-Net's skip connections: the decoder's coarser signal decides which regions of the encoder features matter, suppressing irrelevant background [30].

As built. U-Net as above, but before each Concat the skip features x pass a gate: g (the up-sampled decoder signal) and x are each projected by a 1×1 convolution to half the channels, added, passed through ReLU and a 1×1 convolution to one channel with a sigmoid; the resulting map $\psi \in [0,1]^{H \times W}$ multiplies x (Multiply, one-channel map times image). 7 847 894 parameters.

Difference from the paper. The paper takes the gating signal from the coarser decoder level and resamples it; the demo uses the up-sampled decoder features, which already have the skip's size.

16.16 MSG-Former: an NN Builder original

MSG-Former (*Multi-Scale Gated transFormer*) is the one demo that does not come from a paper: it was designed for NN Builder as a small, fast network for 32×32 colour images that combines three published ideas in one design. It is an untested proposal, not a benchmark result: every number below (shapes, parameters, receptive fields, compute) is exact, but how well it classifies is for its first training runs to show.

16.16.1 The idea

Convolutional networks and transformers are good at different things. A convolution looks at a small neighbourhood and is very efficient at finding local texture and edges, but needs many layers before one output can relate two distant parts of the image. Self-attention relates every position to every other in one step, but on raw pixels it is expensive and, without the built-in assumptions of convolutions, needs much more data [7, 41]. MSG-Former therefore works in two phases:

1. **Local, multi-scale, gated** (32×32 and 16×16 pixels): convolution stages that look at three scales at once, weigh their channels by how useful they are, and keep a residual path;
2. **Global** (an 8×8 grid): the feature map becomes 64 tokens, and two Transformer encoder blocks let every region attend to every other before the class token makes the prediction.

The same convolution-first, attention-later order is used by hybrid models such as the ResNet–ViT hybrid of the ViT paper [7], CoAtNet [5] and MobileViT [28]; MSG-Former's own contribution is the form of its convolution stage.

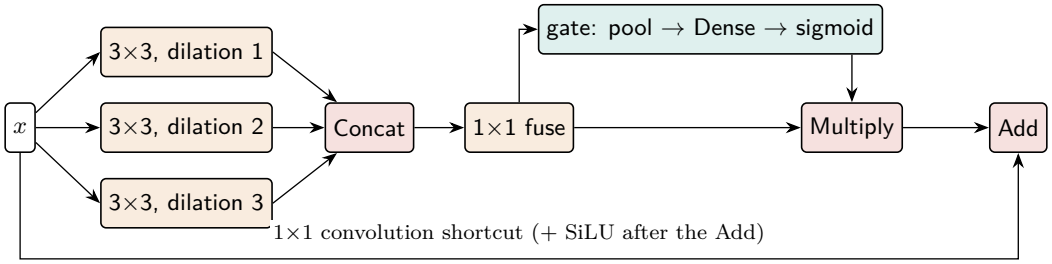


Figure 34: The multi-scale gated residual block. Each 3×3 branch is Conv2D (no bias) → BatchNorm → SiLU; the fuse is a 1×1 Conv2D → BatchNorm → ReLU; the gate is Global pool → Dense ($C/4$, ReLU) → Dense (C, sigmoid).

16.16.2 The multi-scale gated residual block

For an input x with C_{in} channels, a branch width b and an output width C, the block computes

$$u_r = \text{SiLU}(\text{BN}(W_r *_r x)), \qquad r \in \{1,2,3\} \quad (*_r :\ 3\times3 \text{ convolution with dilation } r), \tag{4}$$

$$v = \text{ReLU}(\text{BN}(W_f * [u_1; u_2; u_3])) \qquad (1\times1 \text{ fuse of the } 3b \text{ concatenated channels to } C), \tag{5}$$

$$g = \sigma(W_2\ \text{ReLU}(W_1\ \text{GAP}(v) + c_1) + c_2) \in (0,1)^C, \tag{6}$$

$$y = \text{SiLU}(g \odot v + W_s * x) \qquad (W_s :\ 1\times1 \text{ shortcut}, C_{\text{in}} \to C). \tag{7}$$

Each part has a reason:

Three dilations, same cost. A 3×3 kernel with dilation r covers $(2r+1)\times(2r+1)$ pixels with the same nine weights: 3×3, 5×5 and 7×7 here [46]. The block sees fine detail and wider context at the same resolution and the price of three ordinary 3×3 convolutions, like an Inception module [40] without the cost of large kernels, or the atrous spatial pyramid of DeepLab [2] inside a classifier. Mixing the rates 1, 2 and 3 (rather than only large ones) keeps every pixel covered and avoids the "gridding" gaps that stacked dilated convolutions can leave [43].

1×1 fuse. Concatenation gives $3b$ channels; the 1×1 convolution mixes the scales and sets the block's width.

Squeeze-and-excitation gate. Global average pooling summarises each channel over the whole image; two small Dense layers (reduction 4) turn that summary into one weight in $(0,1)$ per channel, and Multiply scales the channels with it, so the block can emphasise the scales and features that matter for this image [16]. The original SE networks use reduction 16; with 64 and 128 channels, 4 keeps the bottleneck from becoming too narrow (16 and 32 units).

Residual shortcut. As in ResNet [10], the block learns a correction to its input; because the channel count changes ($32 \to 64$, $64 \to 128$), the shortcut is a 1×1 projection.

SiLU. A smooth activation [33] in the branches and after the sum; the fuse keeps ReLU.

16.16.3 The whole network

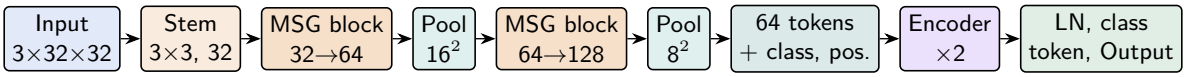


Figure 35: MSG-Former from left to right: local multi-scale stages at 32×32 and 16×16, then global attention on an 8×8 grid.

Table 15 lists every layer as the application computes it (with 10 classes). The receptive field (RF) is given in input pixels.

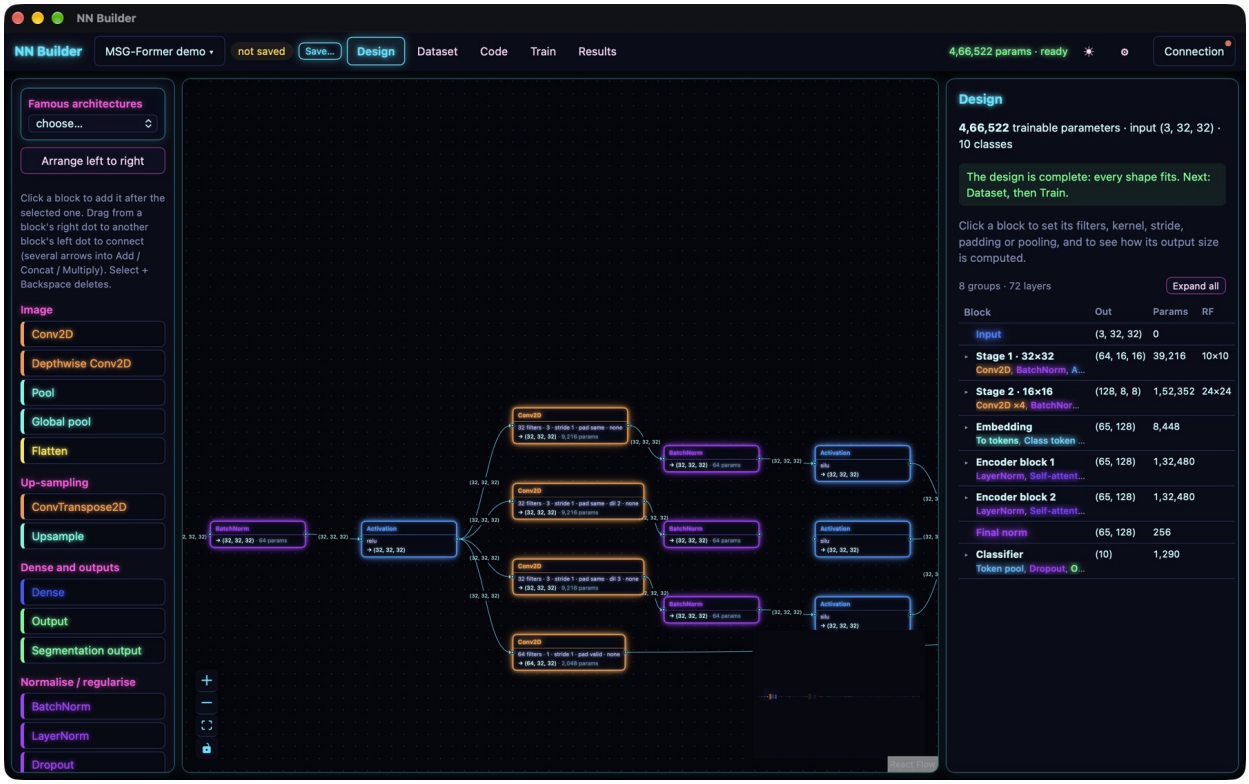


Figure 36: MSG-Former on the canvas, zoomed in on the first block: the stem's activation feeds the three dilated 3×3 convolutions (dilation 1, 2, 3, each with BatchNorm and SiLU) and, at the bottom, the 1×1 shortcut; the branches meet in Concat (96 channels). Every arrow shows its tensor shape.

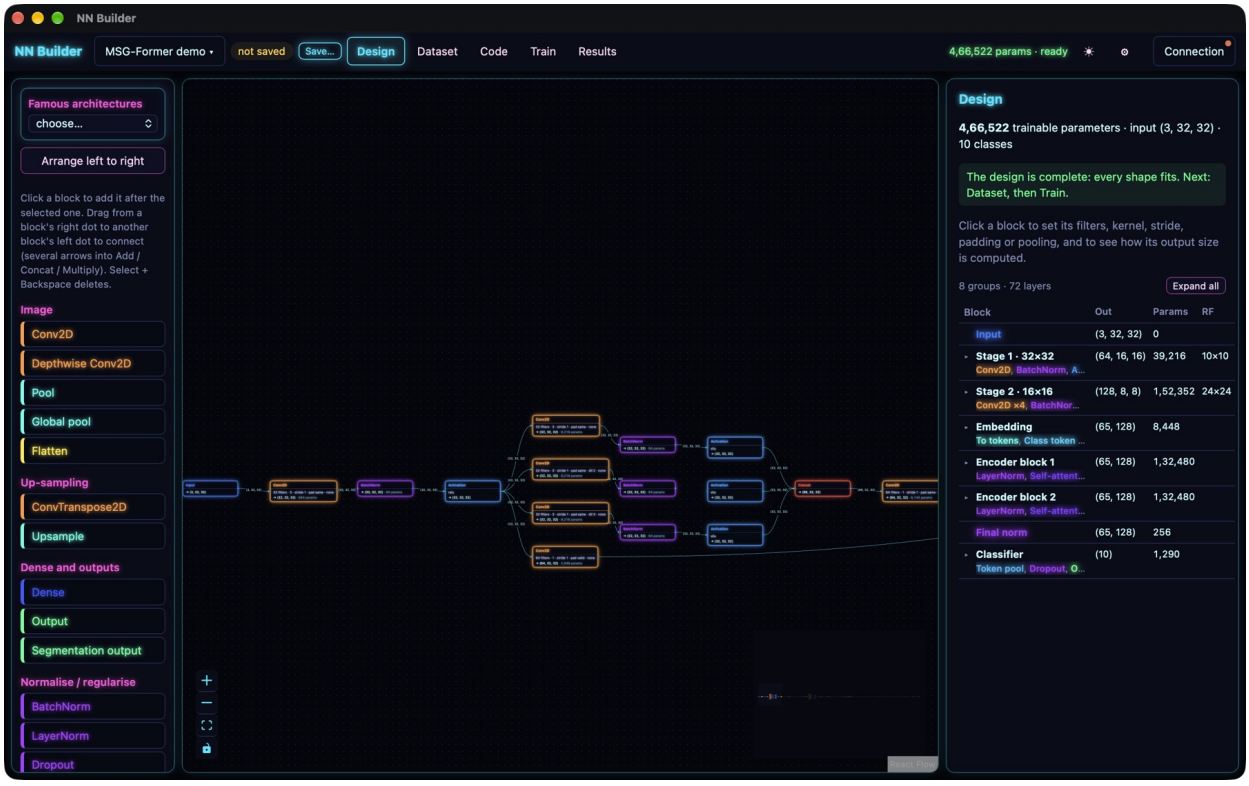


Figure 37: The whole design opened from the demo: 72 blocks, 466 522 parameters, "ready". The layer table groups them as Input, Stage 1 · 32×32, Stage 2 · 16×16, Embedding, Encoder block 1 and 2, Final norm and Classifier, each with its output, parameters and receptive field.

Table 15: MSG-Former layer by layer.

Stage	Layer	Output	Parameters	RF
Input	image	3×32×32	0	1
Stem	Conv2D 32, 3×3, no bias; BatchNorm; ReLU	32×32×32	864 + 64	3
Block 1	branch $r = 1$: Conv2D 32 3×3; BatchNorm; SiLU	32×32×32	9 216 + 64	5
	branch $r = 2$: Conv2D 32 3×3 dilation 2; BatchNorm; SiLU	32×32×32	9 216 + 64	7
	branch $r = 3$: Conv2D 32 3×3 dilation 3; BatchNorm; SiLU	32×32×32	9 216 + 64	9
	Concat	96×32×32	0	9
	fuse: Conv2D 64 1×1; BatchNorm; ReLU	64×32×32	6 144 + 128	9
	gate: Global pool; Dense 16 ReLU; Dense 64 sigmoid	64	1 040 + 1 088	
	Multiply (gate × fused map)	64×32×32	0	
	shortcut: Conv2D 64 1×1, no bias	64×32×32	2 048	
	Add; SiLU	64×32×32	0	
	Pool max 2×2	64×16×16	0	10
Block 2	branches $r = 1, 2, 3$: Conv2D 64 3×3; BatchNorm; SiLU	3 × 64×16×16	3 × (36 864 + 128)	14, 18, 22
	Concat	192×16×16	0	22
	fuse: Conv2D 128 1×1; BatchNorm; ReLU	128×16×16	24 576 + 256	22
	gate: Global pool; Dense 32 ReLU; Dense 128 sigmoid	128	4 128 + 4 224	
	Multiply; shortcut Conv2D 128 1×1; Add; SiLU	128×16×16	8 192	
	Pool max 2×2	128×8×8	0	24
Embedding	To tokens (patches): every cell of the 8×8 map	64×128	0	
	Class token + position embeddings	65×128	8 448	
Encoder 1	LayerNorm; Self-attention (4 heads, dropout 0.1); Dropout; Add	65×128	256 + 66 048	global
	LayerNorm; Dense 256 GELU; Dropout; Dense 128; Dropout; Add	65×128	256 + 33 024 + 32 896	
Encoder 2	the same	65×128	132 480	
Head	LayerNorm; Token pool (class token); Dropout 0.2; Output	10	256 + 1 290	
	Total		**466 522**	

Where the parameters are.

Part	Parameters	Share
Stem	928	0.2%
Block 1 (branches 27 840, fuse 6 272, gate 2 128, shortcut 2 048)	38 288	8.2%
Block 2 (branches 110 976, fuse 24 832, gate 8 352, shortcut 8 192)	152 352	32.7%
Class token and positions	8 448	1.8%
Two encoder blocks (attention 66 048, MLP 65 920, norms 512 each)	264 960	56.8%
Final LayerNorm and output	1 546	0.3%
Total	466 522	100%

These counts were derived by hand, agree with the application's engine, are pinned in its auto-

matic tests, and are confirmed against a real PyTorch build of the network on every change (the comparison on GitHub, Section 7).

Where the computation is. Counting multiply–accumulate operations (MACs) of the convolutions, the Dense layers and attention for one 32×32 image (BatchNorm, activations and pooling are negligible):

Part	MACs per image
Stem ($9 \cdot 3 \cdot 32 \cdot 32^2$)	0.88 M
Block 1: branches $3 \cdot (9 \cdot 32 \cdot 32 \cdot 32^2)$ 28.3 M, fuse 6.3 M, shortcut 2.1 M	36.7 M
Block 2: branches $3 \cdot (9 \cdot 64 \cdot 64 \cdot 16^2)$ 28.3 M, fuse 6.3 M, shortcut 2.1 M	36.7 M
Encoders: per block projections $4Nd^2$ 4.3 M, attention $2N^2d$ 1.1 M, MLP $2Nd \cdot 256$ 4.3 M ($N = 65$, $d = 128$)	19.2 M
Total	about 93.5 M

About 94 million multiply–accumulates (roughly 0.19 GFLOP) per image: the convolution stages do about 80% of the work with about 41% of the parameters, the encoders the reverse (21% of the work, 57% of the parameters). Because pooling halves the image before block 2 while the width doubles, both blocks cost almost exactly the same. Attention is cheap here because there are only 65 tokens: its N^2 term is 1.1 M MACs per block. Training costs about three times the forward pass, so one epoch of CIFAR-10 (50 000 images) is roughly 1.4×10^{13} MACs, a few minutes on a T4.

Receptive field. Before the transformer each token sees a 24×24 pixel region of the 32×32 image (the RF column of Table 15), and neighbouring tokens are 4 pixels apart, so their regions overlap heavily. The first encoder block then makes the receptive field global: every token, and the class token, can draw on every part of the image.

16.16.4 Training it

1. Open MSG-Former from the Projects page and save it (Cmd+S). It opens with built-in CIFAR-10 at its own size.
2. Train tab: Adam, learning rate 0.001 (0.0005 if the loss is unstable), batch 128, 30–60 epochs; augmentation: horizontal flip and random crop with 4 pixels of padding (transformer layers in particular profit from augmentation [39]).
3. Train on Kaggle or any GPU: by a rough estimate it needs about 1 GB of GPU memory at batch 128 (mostly the activations of the two convolution stages kept for the backward pass), so even small GPUs are enough.
4. Compare with the Small CNN (551 018 parameters, convolution only) and ViT-Tiny (1 806 538 parameters, attention only), trained with the same settings, in the Runs table.

16.16.5 Experiments to run (ablations)

Each is a small edit on the canvas, and together they show which part of the design earns its place:

	Change on the canvas	Parameters
A	Remove both gates (delete Global pool, the two Dense layers and Multiply; connect the fuse to Add)	456 042
B	All dilations 1 (three identical-scale branches)	466 522
C	No transformer: Global pool after the second pooling, then Output	192 858
D	A third encoder block	598 970
E	Token pool *mean* instead of the class token	466 522
F	Wider: branch filters 64 and 128	larger

(Counts for A, C and D follow from the tables above: A removes $2\,128 + 8\,352$; C removes the embedding, encoders, final norm and the 128-input head and adds a 128-input head; D adds one encoder block of 132 480. The application shows the exact count after each edit.) If B matches the full model, the multi-scale branches are not helping on this data; if C matches it, the attention is not; and so on.

16.16.6 Adapting it

- **Larger images** (for example 64×64 or 224×224): add a stride-2 stem (or a third block with pooling) so that the grid reaching the transformer stays around 8×8 to 14×14; otherwise the number of tokens, and the N^2 attention cost, grows with the image.
- **More classes:** only the output layer changes ($128 \cdot K + K$ parameters).
- **Segmentation:** the two convolution stages can serve as an encoder in a U-Net-style design, with ConvTranspose2D decoders and skips from the blocks.

16.16.7 Status

The design, its shapes and its parameter count are verified (by hand, by the application's engine and against PyTorch). Its accuracy is not: no training results exist yet. Treat it as a starting point for experiments, and record the results of the comparisons above.

17 The test architectures

On 6–7 November 2026 every feature of the application was tried by hand on real machines (the checklist is `TEST_APP.md` in the repository). For that, eighteen small projects were built, *TEST 1* to *TEST 19* (there is no TEST 18: that number was a project saved and deleted to test **Delete**). Each is a working example of one kind of network, and most were trained on a Kaggle GPU or an AWS EC2 g4dn.xlarge. This section shows each design as it looks on the Design screen, explains why it is built that way, and lists every block with its output shape and parameter count. The counts are the application's own; they are the same numbers PyTorch reports (checked for every block kind by the automatic tests).

Most of these networks are deliberately tiny: they test that a feature works end to end (data, shapes, training, results), not how good a network can be. Projects 2–10, 14 and 15 use *generated* data (patterns drawn by a script, so the right answer is known); 1, 11, 12, 13, 16, 17 and 19 use real data (EuroSAT, CIFAR-10, and Sentinel-1/2 exported from Google Earth Engine). TEST 24 (10 November) came later: one CIFAR-10 network that uses every block a classifier can use, each exactly once, built in the application window (Section 17.19).

How the parameters are counted. The counts follow from a few rules (Section 8 has every block):

- Conv2D: $k_h \cdot k_w \cdot C_{in} \cdot C_{out}$ weights $+ C_{out}$ biases. A 3×3 convolution from 16 to 32 channels: $3 \cdot 3 \cdot 16 \cdot 32 + 32 = 4\,640$. Conv3D adds the date kernel: $k_t \cdot k_h \cdot k_w \cdot C_{in} \cdot C_{out} + C_{out}$.
- Dense: $n_{in} \cdot n_{out} + n_{out}$. BatchNorm, LayerNorm: $2C$ (scale and shift); RMSNorm: C.
- Self-attention with dimension d: four $d \times d$ projections with biases, $4d^2 + 4d$.
- LSTM with h units on n inputs: $4(nh + h^2 + 2h)$ per direction; GRU: $3(nh + h^2 + 2h)$.
- LoRA of rank r on a $n_{in} \times n_{out}$ layer adds $r(n_{in} + n_{out})$ trainable weights; the original weights are *frozen* (kept, not trained). *Frozen* blocks keep their weights fixed too; they still count in the total.
- Pooling, activations, Add, Concat, Subtract, Flatten, Dropout and *Shared weights* have no parameters of their own (a shared block uses the weights of the block it copies, so they are counted once).

The screenshots were taken from copies of the projects in a neutral settings folder; the block ids in the tables (`t2`, `n10` ...) are the ids in the project's `model.json` and on the Text tab.

Test	What it tests	Input	Blocks	Parameters	Result
1	every new block kind in one network, real EuroSAT-MS	13 × 64 × 64	22	170 904	83.7 % (early exit 79.3 %), Kaggle T4
2	numbers per image (regression), DistributedDataParallel	2 × 16 × 16	5	5 010	R^2 0.998, Kaggle T4×2
3	extra outputs (multi-task): a class, two numbers, a second class	2 × 16 × 16	9	6 198	class 100 %, R^2 0.987, second class 97.5 %
4	several classes per image, S2 + S1 early fusion	5 × 24 × 24	6	10 819	micro-F1 1.0
5	mid fusion, Sentinel-2 at 32×32 + Sentinel-1 at 16×16	6 × 32 × 32	9	7 754	100 %
6	shared weights (twin branch), $\lvert a - b \rvert$, custom loss	4 × 24 × 24	9	2 658	100 %
7	dates as a sequence: Conv1D, GRU, LSTM	2 × 6 × 16 × 16	7	6 610	100 %
8	segmentation of a time series: Conv3D + temporal attention	2 × 6 × 16 × 16	6	10 196	95.0 % pixels, mIoU 0.869
9	boxes (CenterNet-style detection head)	3 × 64 × 64	9	26 102	mAP@0.5 0.951
10	200 : 20 images, class weights and focal loss	2 × 16 × 16	5	2 658	100 % (weights 0.55 / 5.5)
11	ImageNet weights + LoRA rank 4 on attention, real EuroSAT RGB	3 × 224 × 224	91	86 027 530	93.6 % training 228 874 weights
12	the same ViT, every weight trained	3 × 224 × 224	91	85 806 346	96.8 % training 85.8 M weights
13	`everything.nnb`: all block kinds; recipe, 3 seeds, export	3 × 32 × 32	97	217 396	50.95 ± 1.62 % (2 epochs, 3 seeds)
14	the Text tab: `timeseries.nnb` pasted and edited	4 × 6 × 32 × 32	18	27 684	built and checked (window test; not trained)
15	a scene + label image cut into tiles, strip test split	2 × 16 × 16	4	2 675	95.9 % pixels, mIoU 0.919
16	Small CNN (CIFAR-10) demo on an EC2 g4dn.xlarge	3 × 32 × 32	29	551 018	40.5 % (2 epochs), 79.5 % (10 epochs), Stop: 75.7 %
17	40 real S2 L1C + S1 exports, fused	15 × 64 × 64	7	23 562	data check passed (no training)

Test	What it tests	Input	Blocks	Parameters	Result
19	the app's own GEE script: S2 + WorldCover, from a Drive link	$4 \times 64 \times 64$	5	10 859	79.8 % pixels (built-up IoU 0.71, trees 0.67)
24	every classifier block once (36 kinds), set by hand, CIFAR-10	$3 \times 32 \times 32$	36	94 777	built and checked in the window (not trained)

17.1 TEST 1: a hybrid CNN + transformer on 13 Sentinel-2 bands

TEST 1 puts nearly every block added in the architecture steps into one network and trains it on real data: EuroSAT-MS (Sentinel-2, all 13 bands, 64×64 pixels, 10 land-cover classes), with focal loss and class weights.

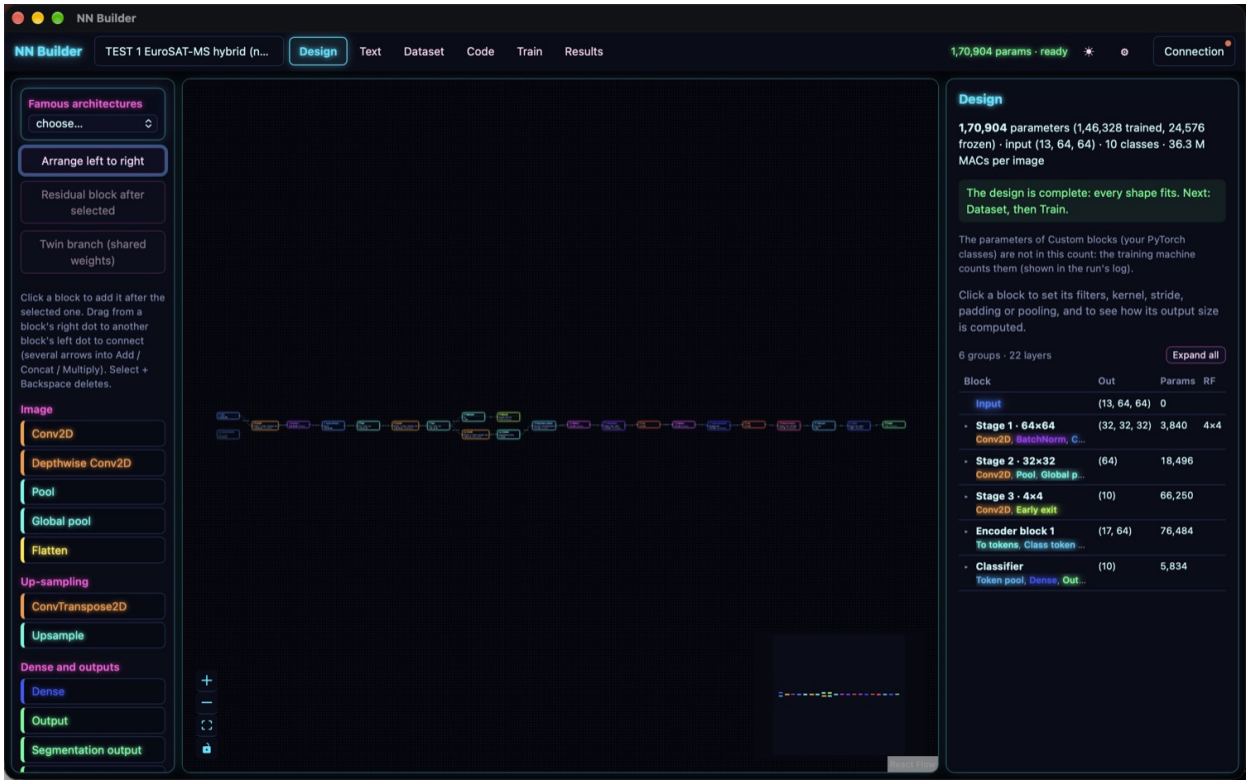


Figure 38: TEST 1 on the Design screen: a convolutional front end, an early exit, then a transformer encoder with LoRA, SwiGLU and a mixture of experts.

The first convolution (32 filters, 3×3 on 13 bands: $3 \cdot 3 \cdot 13 \cdot 32 + 32 = 3\,776$) is *frozen*, as if it came from an earlier model. A Custom block `SE` (squeeze-and-excitation, written in PyTorch on the Text tab) reweights the 32 channels. After two pooling steps the network branches: a global average pool feeds an *early exit* (a small classifier, $64 \cdot 10 + 10 = 650$, trained with loss weight 0.3) that can answer for easy images, while a 4×4 convolution with stride 4 ($4 \cdot 4 \cdot 64 \cdot 64 + 64 = 65\,600$, the largest block) turns the 16×16 map into 16 patches of 64 numbers. The transformer part adds a class token (sinusoidal positions have no weights, so only the token's 64 count), RMSNorm, self-attention with rotary positions and LoRA (base $4 \cdot 64^2 + 4 \cdot 64 = 16\,640$ frozen, LoRA rank 4 on q, k, v: $3 \cdot 4 \cdot (64 + 64) = 1\,536$), a SwiGLU feed-forward ($3 \cdot 64 \cdot 128 = 24\,576$, no biases) and a mixture of 4 experts (router $64 \cdot 4 + 4 = 260$, each expert two 64×64 layers, $4 \cdot 8\,320$, so 33 540; each token uses its best 2). The classifier is a LoRA Dense (64 units, base 4 160 frozen + rank 8: 1 024) and the Output. Of 170 904 parameters 24 576 are frozen, so 146 328 are trained; the SE block's 552 weights are counted on the training machine, not by the app.

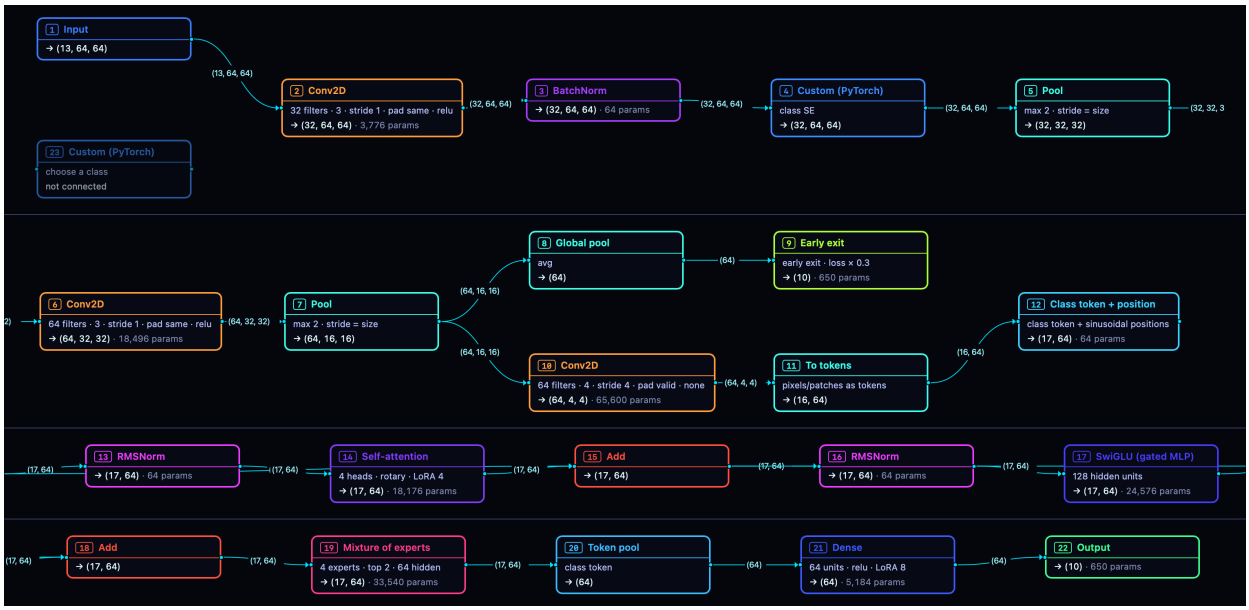


Figure 39: TEST 1 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

Id	Block	Settings	Output	Parameters
t1	Input	all bands	$13 \times 64 \times 64$	0
t2	Conv2D	32 filters, 3×3, same, relu, frozen	$32 \times 64 \times 64$	3 776 (3 776 frozen)
t3	BatchNorm		$32 \times 64 \times 64$	64
t4	Custom (PyTorch)	class SE	$32 \times 64 \times 64$	0
t5	Pool	max 2×2	$32 \times 32 \times 32$	0
t6	Conv2D	64 filters, 3×3, same, relu	$64 \times 32 \times 32$	18 496
t7	Pool	max 2×2	$64 \times 16 \times 16$	0
t8	Global pool	avg	64	0
t10	Conv2D	64 filters, 4×4, stride 4, pad valid	$64 \times 4 \times 4$	65 600
t9	Early exit	loss weight 0.3	10	650
t11	To tokens	patches	16×64	0
t12	Class token + positions	positions sinusoidal	17×64	64
t13	RMSNorm		17×64	64
t14	Self-attention	4 heads, rotary, LoRA rank 4	17×64	18 176 (16 640 frozen)
t15	Add		17×64	0
t16	RMSNorm		17×64	64
t17	SwiGLU	128 hidden units	17×64	24 576
t18	Add		17×64	0
t19	Mixture of experts	4 experts, top 2, 64 units	17×64	33 540
t20	Token pool	cls	64	0
t21	Dense	64 units, relu, LoRA rank 8	64	5 184 (4 160 frozen)
t22	Output	main output	10	650
	Total			**170 904**

The design also holds one Custom block that is not connected (left from a test of the code editor); it has no shape and no parameters and is left out of the table.

17.2 TEST 2: regression on two GPUs

TEST 2 predicts two numbers per image (regression, Huber loss) instead of a class, and was trained on Kaggle's two T4 GPUs at once (the run split each batch between them).

The design is the smallest useful image regressor: a 3×3 convolution from 2 bands to 16 channels ($3 \cdot 3 \cdot 2 \cdot 16 + 16 = 304$), one from 16 to 32 (4 640), a global average pool that turns each 16×16 map into one number, and an Output with one unit per target ($32 \cdot 2 + 2 = 66$). The Output of a regression network has no softmax: its two numbers are the predictions (the app standardises the targets for training and converts back). This run also found a real bug: on two GPUs the training

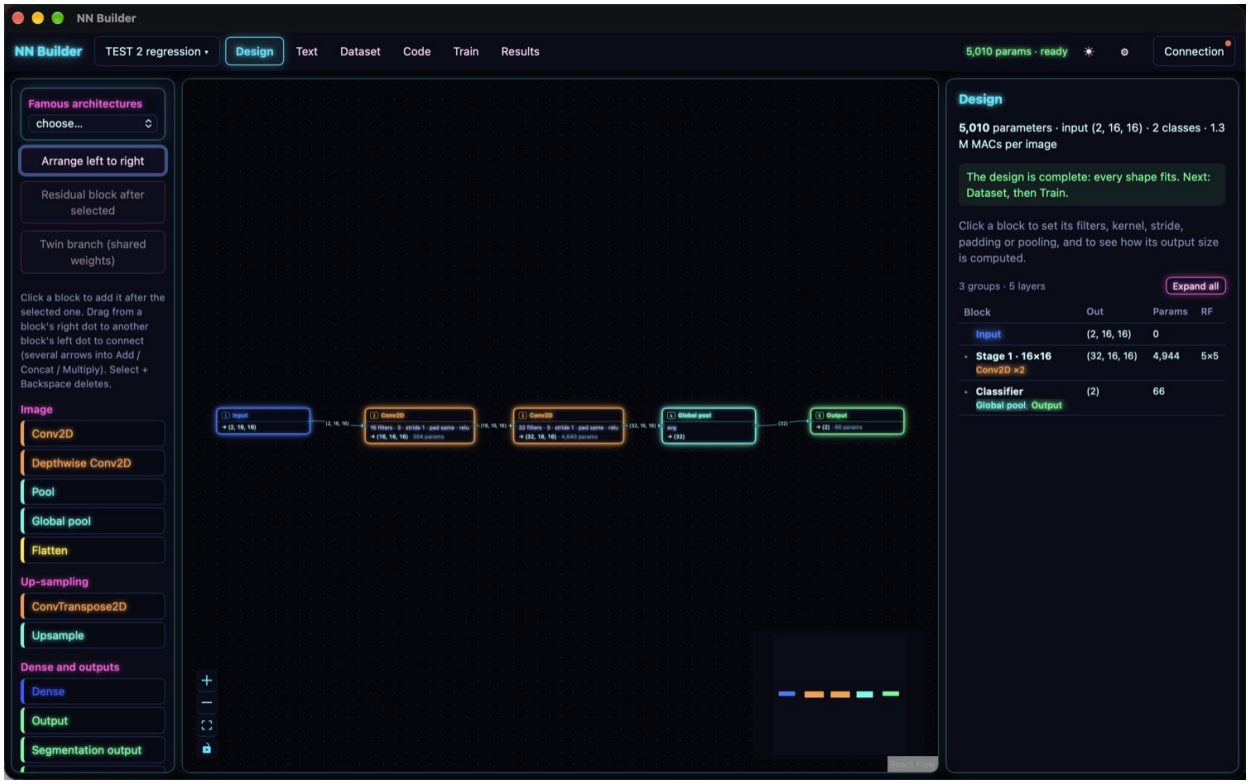


Figure 40: TEST 2 on the Design screen: two convolutions, a global pool and an Output with one number per target.



Figure 41: TEST 2 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

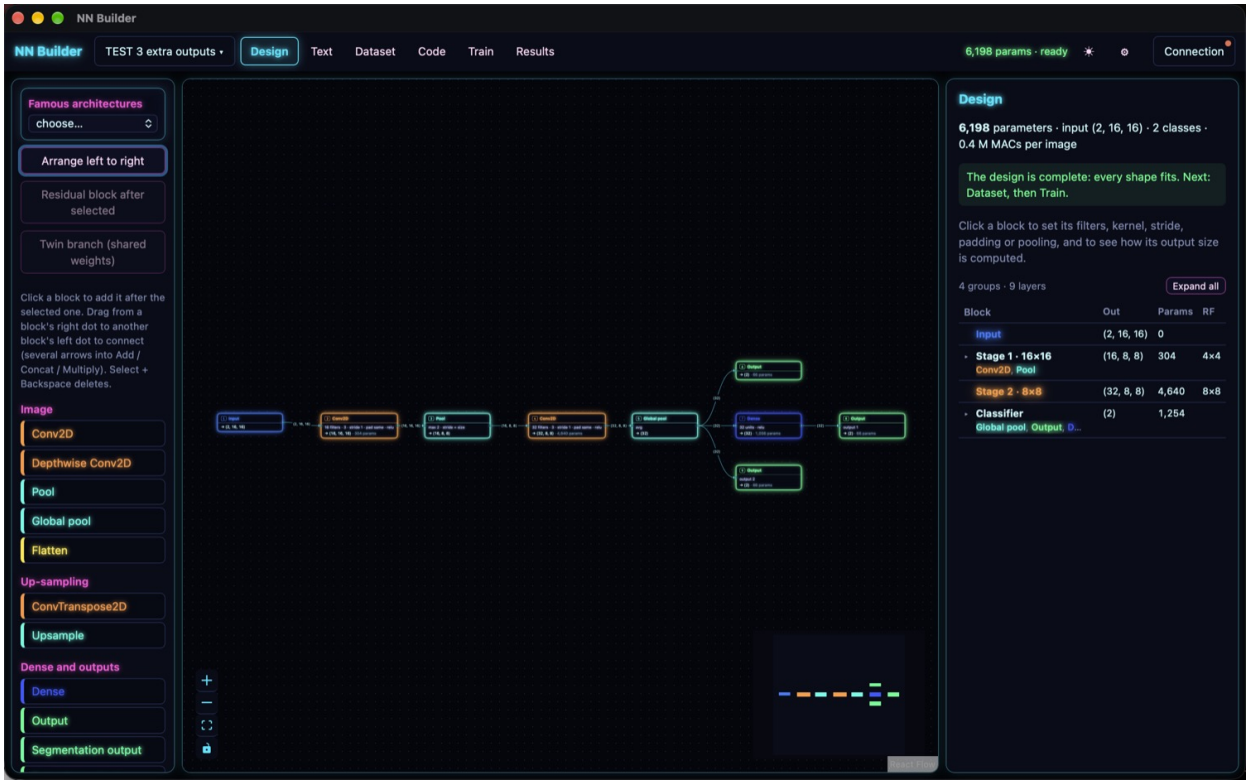


Figure 42: TEST 3 on the Design screen: one trunk, three Output blocks: the main class, an extra output with two numbers and an extra output with a second class.



Figure 43: TEST 3 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

The shared trunk (304 + 4 640 parameters) learns features useful for all three tasks. The main Output and extra output 1 sit directly on the pooled 32 numbers ($32 \cdot 2 + 2 = 66$ each); extra output 2 has its own Dense layer of 32 units first ($32 \cdot 32 + 32 = 1\,056$), the usual way to give one task a little more capacity. In the Output inspector, **Output for** chooses which task an Output serves.

Id	Block	Settings	Output	Parameters
t1	Input	all bands	$2 \times 16 \times 16$	0
t2	Conv2D	16 filters, 3×3, same, relu	$16 \times 16 \times 16$	304
t3	Pool	max 2×2	$16 \times 8 \times 8$	0
t4	Conv2D	32 filters, 3×3, same, relu	$32 \times 8 \times 8$	4 640
t5	Global pool	avg	32	0
t6	Output	main output	2	66
t7	Dense	32 units, relu	32	1 056
t9	Output	extra output 2	2	66
t8	Output	extra output 1	2	66
	Total			**6 198**

17.4 TEST 4: multi-label with Sentinel-1 + Sentinel-2

TEST 4 gives each image any number of classes at once (multi-label, a labels table) and stacks two sensors in one Input: three Sentinel-2 bands and the two Sentinel-1 radar bands (early fusion).

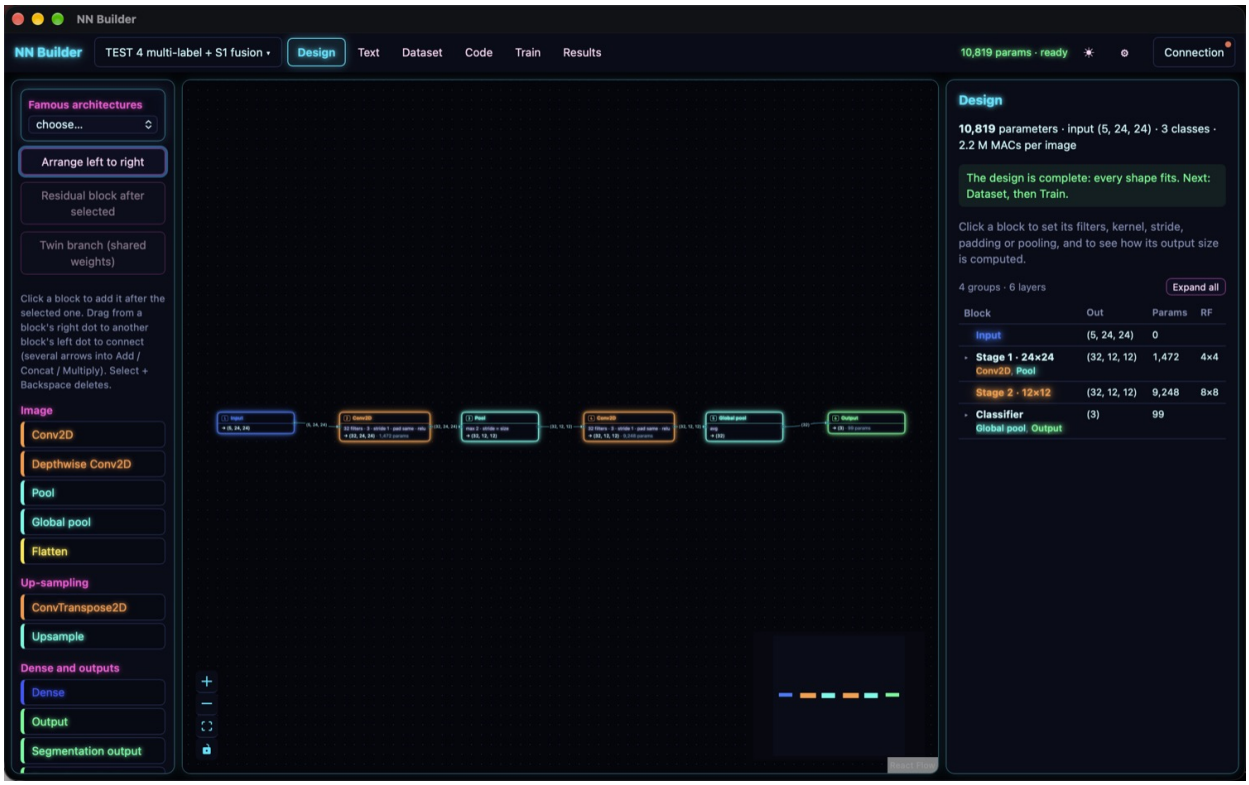


Figure 44: TEST 4 on the Design screen: five bands (three of Sentinel-2, two of Sentinel-1) enter one Input.

With early fusion the first convolution simply sees 5 channels ($3 \cdot 3 \cdot 5 \cdot 32 + 32 = 1\,472$). The Output has one unit per class ($32 \cdot 3 + 3 = 99$) and a sigmoid per unit instead of a softmax, so each class is a separate yes/no; the threshold per class can be tuned on the validation data.

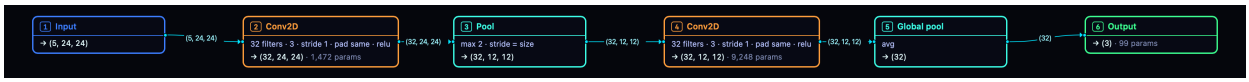


Figure 45: TEST 4 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

Id	Block	Settings	Output	Parameters
t1	Input	all bands	$5 \times 24 \times 24$	0
t2	Conv2D	32 filters, 3×3, same, relu	$32 \times 24 \times 24$	1 472
t3	Pool	max 2×2	$32 \times 12 \times 12$	0
t4	Conv2D	32 filters, 3×3, same, relu	$32 \times 12 \times 12$	9 248
t5	Global pool	avg	32	0
t6	Output	main output	3	99
	Total			**10 819**

17.5 TEST 5: two sensors at their own sizes

TEST 5 keeps each sensor at its own resolution (Sentinel-2 at 32×32, Sentinel-1 at 16×16), so nothing is invented by resizing, and joins them inside the network (mid fusion).

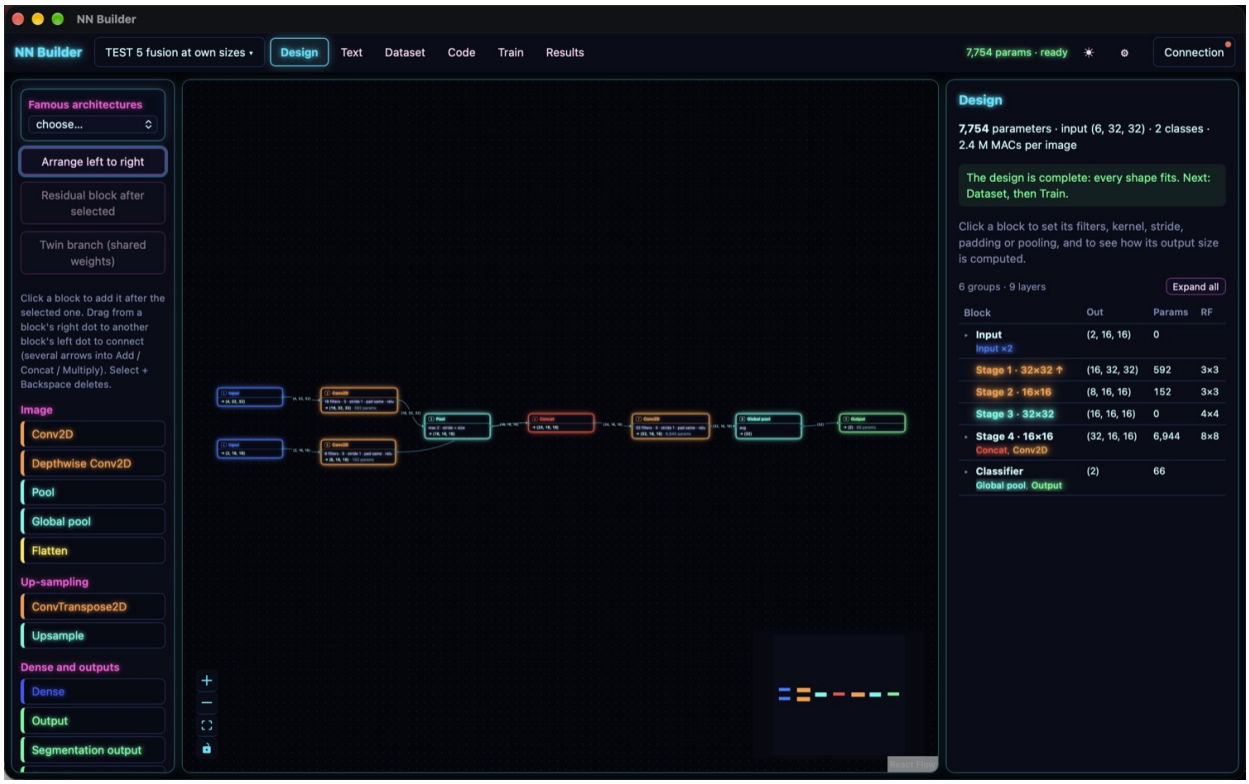


Figure 46: TEST 5 on the Design screen: two Inputs (band groups 1 and 2) with their own first layers, joined by Concat.



Figure 47: TEST 5 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

pooled to 16×16 so that it meets the radar branch (2 bands, $3 \cdot 3 \cdot 2 \cdot 8 + 8 = 152$) at the same size; Concat stacks the 16 and 8 channels into 24, and a joint convolution ($3 \cdot 3 \cdot 24 \cdot 32 + 32 = 6\,944$) mixes the sensors. Concat itself has no parameters.

Id	Block	Settings	Output	Parameters
t1	Input	band group 1	$4 \times 32 \times 32$	0
t4	Input	band group 2	$2 \times 16 \times 16$	0
t2	Conv2D	16 filters, 3×3, same, relu	$16 \times 32 \times 32$	592
t5	Conv2D	8 filters, 3×3, same, relu	$8 \times 16 \times 16$	152
t3	Pool	max 2×2	$16 \times 16 \times 16$	0
t6	Concat		$24 \times 16 \times 16$	0
t7	Conv2D	32 filters, 3×3, same, relu	$32 \times 16 \times 16$	6 944
t8	Global pool	avg	32	0
t9	Output	main output	2	66
	Total			**7 754**

17.6 TEST 6: a Siamese network for change

TEST 6 decides whether something changed between two images of the same place (before and after), with a custom loss written in Python.

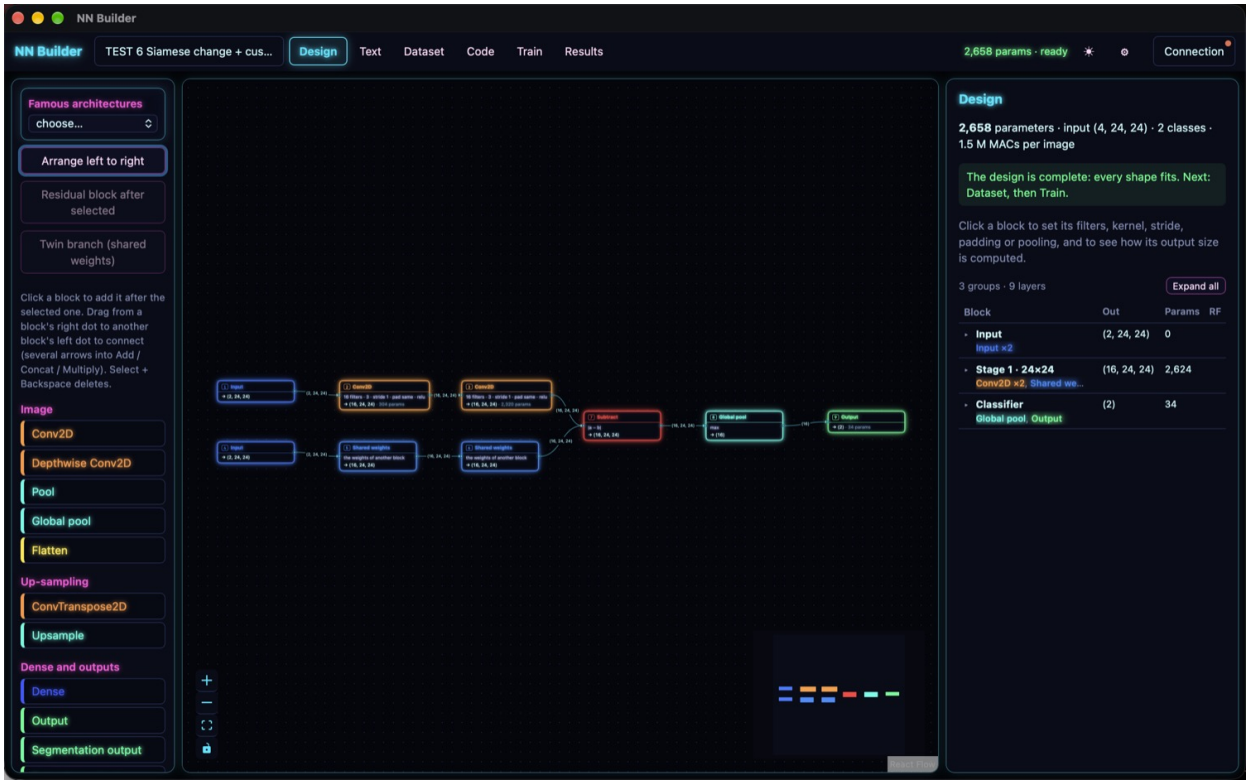


Figure 48: TEST 6 on the Design screen: two Inputs (before and after) through the same two convolutions, then the absolute difference.

Both images must be judged the same way, so the second branch does not have its own convolutions: its two blocks are *Shared weights* copies of the first branch's (**Twin branch** on the canvas). That is why the design has four convolution blocks but only two sets of weights ($304 + 2\,320$). Subtract with absolute value gives $|a - b|$ per pixel and channel, a global max pool keeps the strongest difference, and the Output says change / no change ($16 \cdot 2 + 2 = 34$). In total 2 658 parameters, the same number PyTorch counts for the generated code.

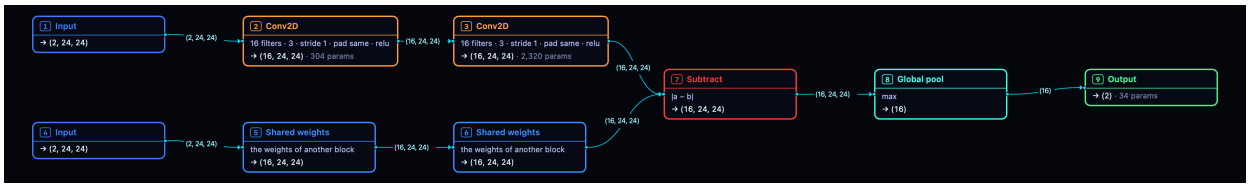


Figure 49: TEST 6 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

Id	Block	Settings	Output	Parameters
t1	Input	band group 1	$2 \times 24 \times 24$	0
t4	Input	band group 2	$2 \times 24 \times 24$	0
t2	Conv2D	16 filters, 3×3, same, relu	$16 \times 24 \times 24$	304
t3	Conv2D	16 filters, 3×3, same, relu	$16 \times 24 \times 24$	2 320
t5	Shared weights	weights of block t2	$16 \times 24 \times 24$	0
t6	Shared weights	weights of block t3	$16 \times 24 \times 24$	0
t7	Subtract	$\lvert a - b \rvert$	$16 \times 24 \times 24$	0
t8	Global pool	max	16	0
t9	Output	main output	2	34
	Total			**2 658**

17.7 TEST 7: a time series as a sequence

TEST 7 classifies a stack of six dates (2 bands, 16×16 pixels each) by how the image changes over time.

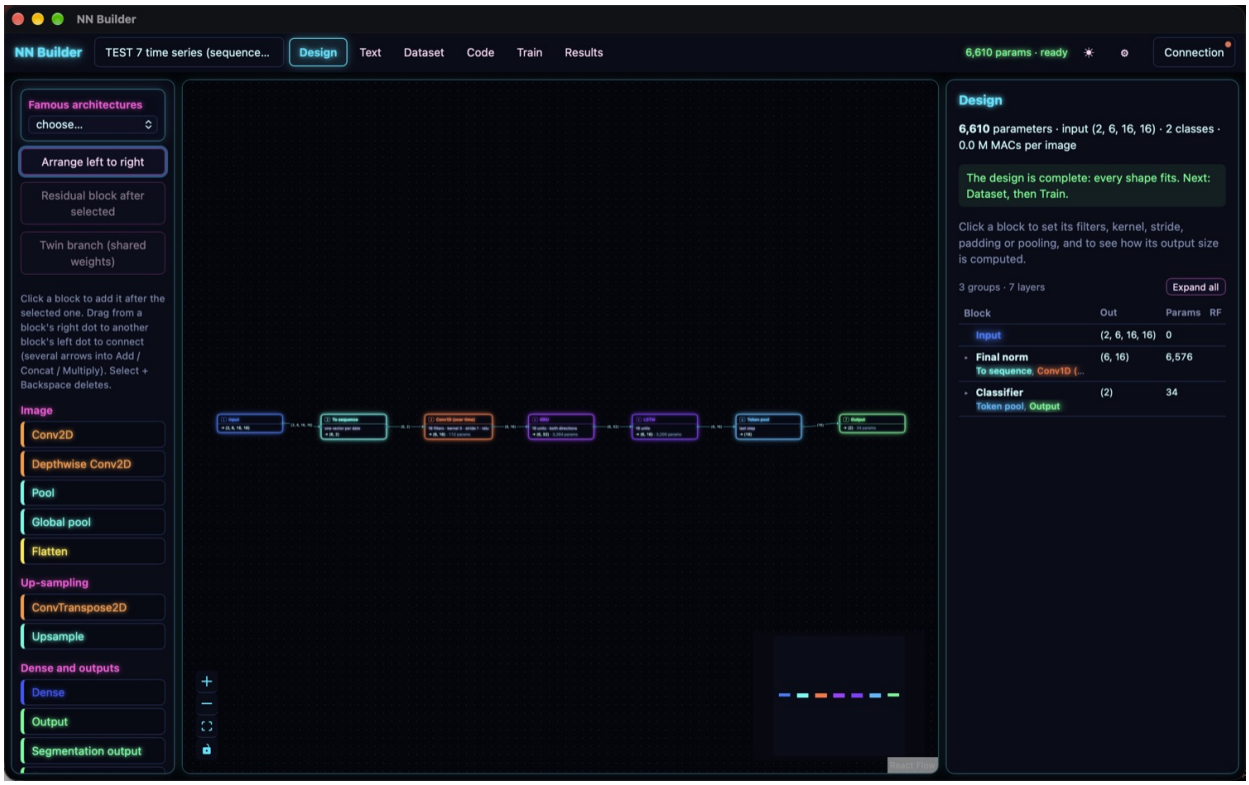


Figure 50: TEST 7 on the Design screen: To sequence, then a 1-D convolution, a bidirectional GRU and an LSTM over the six dates.



Figure 51: TEST 7 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as**

A Conv1D looks at three neighbouring dates at a time ($3 \cdot 2 \cdot 16 + 16 = 112$). A bidirectional GRU (16 units each way: $2 \cdot 3 \cdot (16 \cdot 16 + 16 \cdot 16 + 2 \cdot 16) = 3\,264$) reads the sequence forwards and backwards, an LSTM of 16 units on its 32 numbers ($4 \cdot (32 \cdot 16 + 16 \cdot 16 + 2 \cdot 16) = 3\,200$) reads it again, and *Token pool: last* keeps the state after the last date.

Id	Block	Settings	Output	Parameters
t1	Input	all bands	$2 \times 6 \times 16 \times 16$	0
t2	To sequence		6×2	0
t3	Conv1D	16 filters, kernel 3, relu	6×16	112
t4	GRU	16 units, both directions	6×32	3 264
t5	LSTM	16 units	6×16	3 200
t6	Token pool	last	16	0
t7	Output	main output	2	34
	Total			**6 610**

17.8 TEST 8: a crop map from a time series

TEST 8 labels every pixel (segmentation) from a stack of six dates: the kind of network used for crop-type maps.

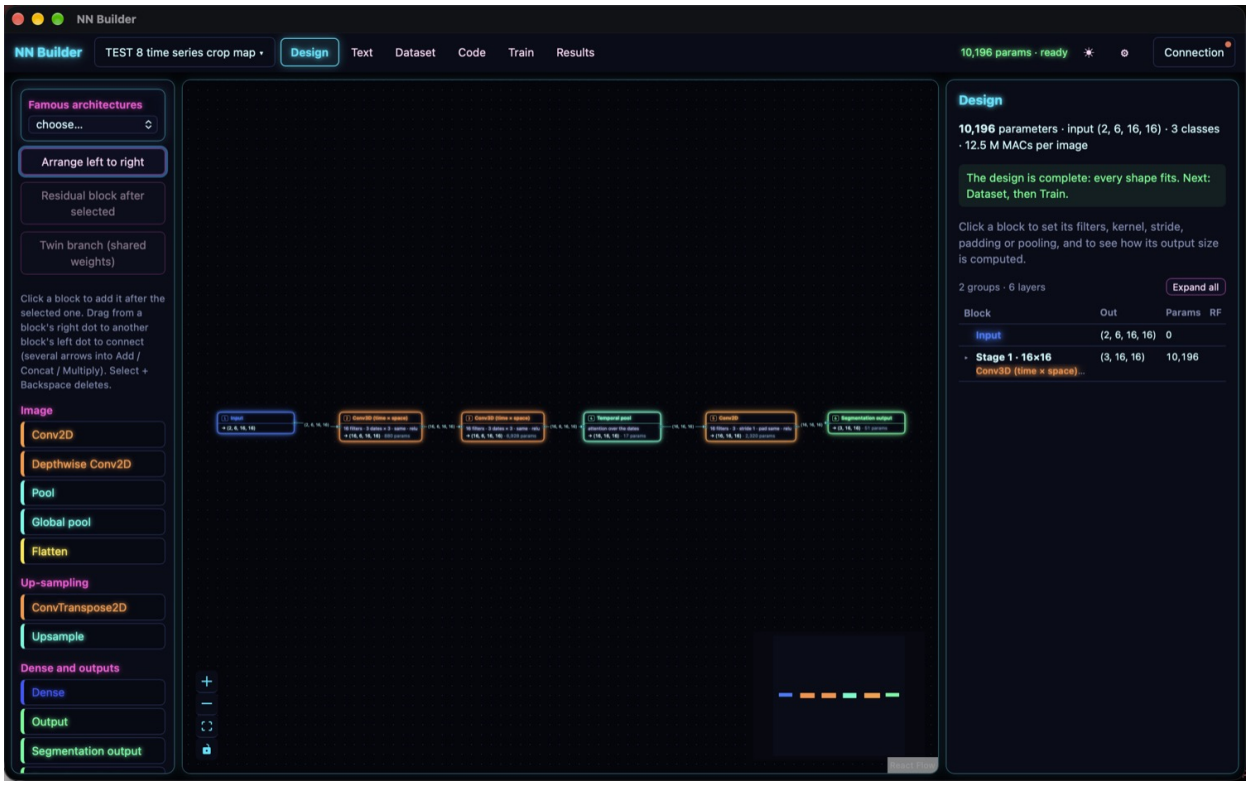


Figure 52: TEST 8 on the Design screen: two Conv3D layers over dates and pixels, temporal attention, a Conv2D and the Segmentation output.

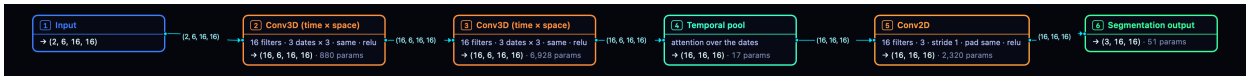


Figure 53: TEST 8 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

A Conv3D kernel spans 3 dates and 3×3 pixels ($3{\cdot}3{\cdot}3{\cdot}2{\cdot}16{+}16 = 880$, then $3{\cdot}3{\cdot}3{\cdot}16{\cdot}16{+}16 = 6\,928$), so it sees how each neighbourhood changes. *Temporal pool: attention* then weighs the six dates

per pixel and sums them into one image (17 weights: one score per channel plus a bias). A 2-D convolution refines it, and the Segmentation output gives 3 class scores per pixel (a 1×1 convolution, $16 \cdot 3 + 3 = 51$).

Id	Block	Settings	Output	Parameters
t1	Input	all bands	$2 \times 6 \times 16 \times 16$	0
t2	Conv3D	16 filters, 3×3 over 3 dates, relu	$16 \times 6 \times 16 \times 16$	880
t3	Conv3D	16 filters, 3×3 over 3 dates, relu	$16 \times 6 \times 16 \times 16$	6 928
t4	Temporal pool	attention	$16 \times 16 \times 16$	17
t5	Conv2D	16 filters, 3×3, same, relu	$16 \times 16 \times 16$	2 320
t6	Segmentation output		$3 \times 16 \times 16$	51
	Total			**10 196**

17.9 TEST 9: object detection

TEST 9 finds objects and draws boxes (two classes, squares and discs, in generated images with YOLO-format box labels).

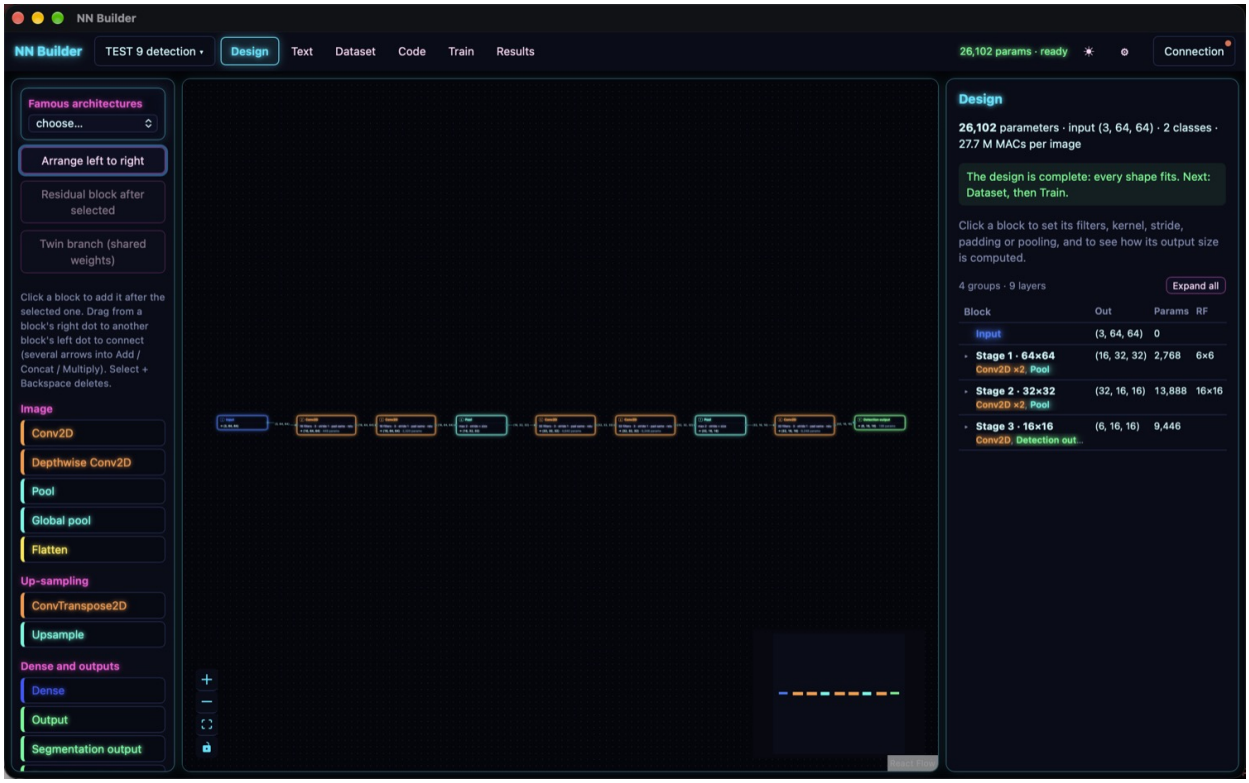


Figure 54: TEST 9 on the Design screen: four convolutions with two poolings, then the Detection output on a quarter-size map.



Figure 55: TEST 9 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

The backbone halves the 64×64 image twice (16 → 32 channels, 448 to 9 248 parameters per convolution). The Detection output is a 1×1 convolution that gives, for every cell of the 16×16 map, one score per class (*is an object centred here?*) and four numbers (box width, height and the centre's offset): $32 \cdot (2+4) + (2+4) = 198$. Each cell is 4×4 pixels of the input; the strongest peaks become the boxes.

Id	Block	Settings	Output	Parameters
t1	Input	all bands	$3 \times 64 \times 64$	0
t2	Conv2D	16 filters, 3×3, same, relu	$16 \times 64 \times 64$	448
t3	Conv2D	16 filters, 3×3, same, relu	$16 \times 64 \times 64$	2 320
t4	Pool	max 2×2	$16 \times 32 \times 32$	0
t5	Conv2D	32 filters, 3×3, same, relu	$32 \times 32 \times 32$	4 640
t6	Conv2D	32 filters, 3×3, same, relu	$32 \times 32 \times 32$	9 248
t7	Pool	max 2×2	$32 \times 16 \times 16$	0
t8	Conv2D	32 filters, 3×3, same, relu	$32 \times 16 \times 16$	9 248
t9	Detection output		$6 \times 16 \times 16$	198
	Total			**26 102**

17.10 TEST 10: unbalanced classes

TEST 10 trains on 200 images of one class and 20 of the other, where a careless network learns to always answer the common class.

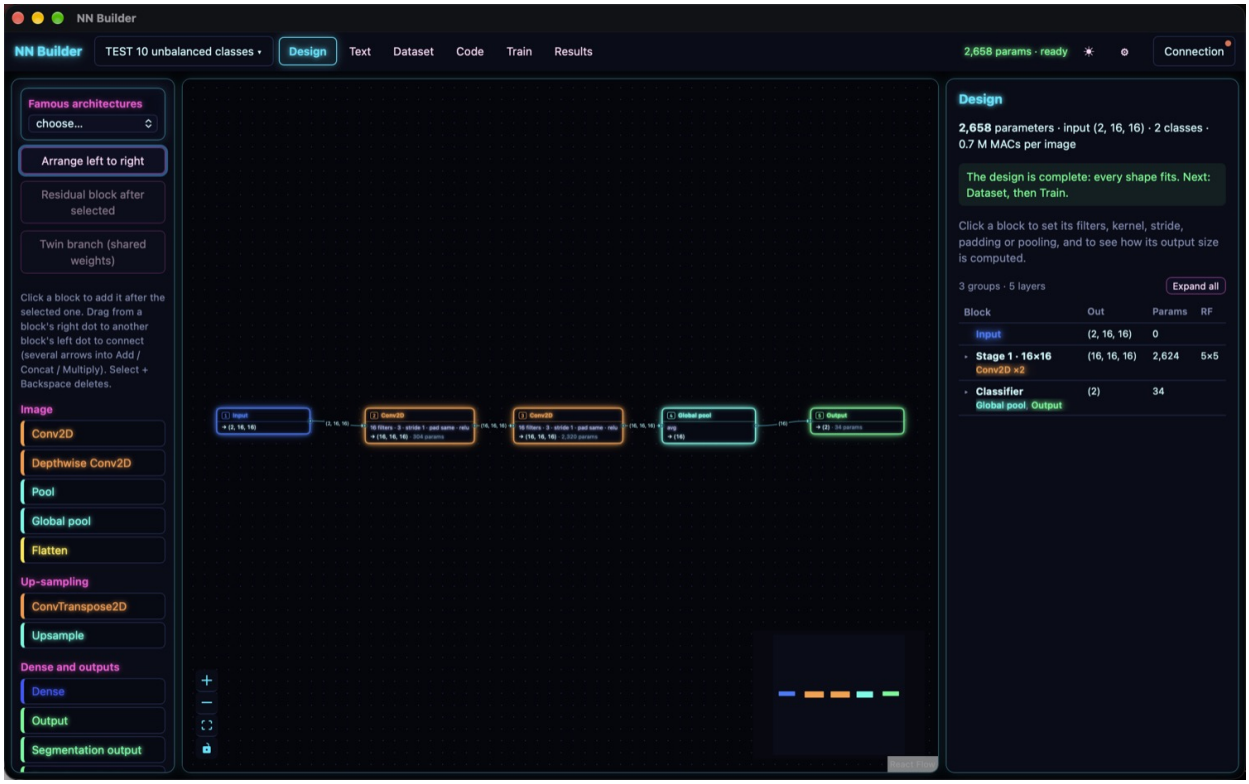


Figure 56: TEST 10 on the Design screen: a two-convolution classifier.

The network is plain ($304 + 2\,320 + 34 = 2\,658$ parameters); the test is in the Train tab: *inverse* class weights (0.55 for the common class, 5.5 for the rare one, so each class counts as much in total) and focal loss. Read folder in the Dataset tab showed the 10:1 imbalance and suggested the weights.



Figure 57: TEST 10 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

Id	Block	Settings	Output	Parameters
t1	Input	all bands	$2 \times 16 \times 16$	0
t2	Conv2D	16 filters, 3×3, same, relu	$16 \times 16 \times 16$	304
t3	Conv2D	16 filters, 3×3, same, relu	$16 \times 16 \times 16$	2 320
t4	Global pool	avg	16	0
t5	Output	main output	2	34
	Total			**2 658**

17.11 TEST 11: ViT-B/16 fine-tuned with LoRA

TEST 11 and TEST 12 start from the same ViT-B/16 with ImageNet weights and fine-tune it on EuroSAT RGB (1 epoch, 4 000 images); TEST 11 adapts it with LoRA, TEST 12 trains everything.

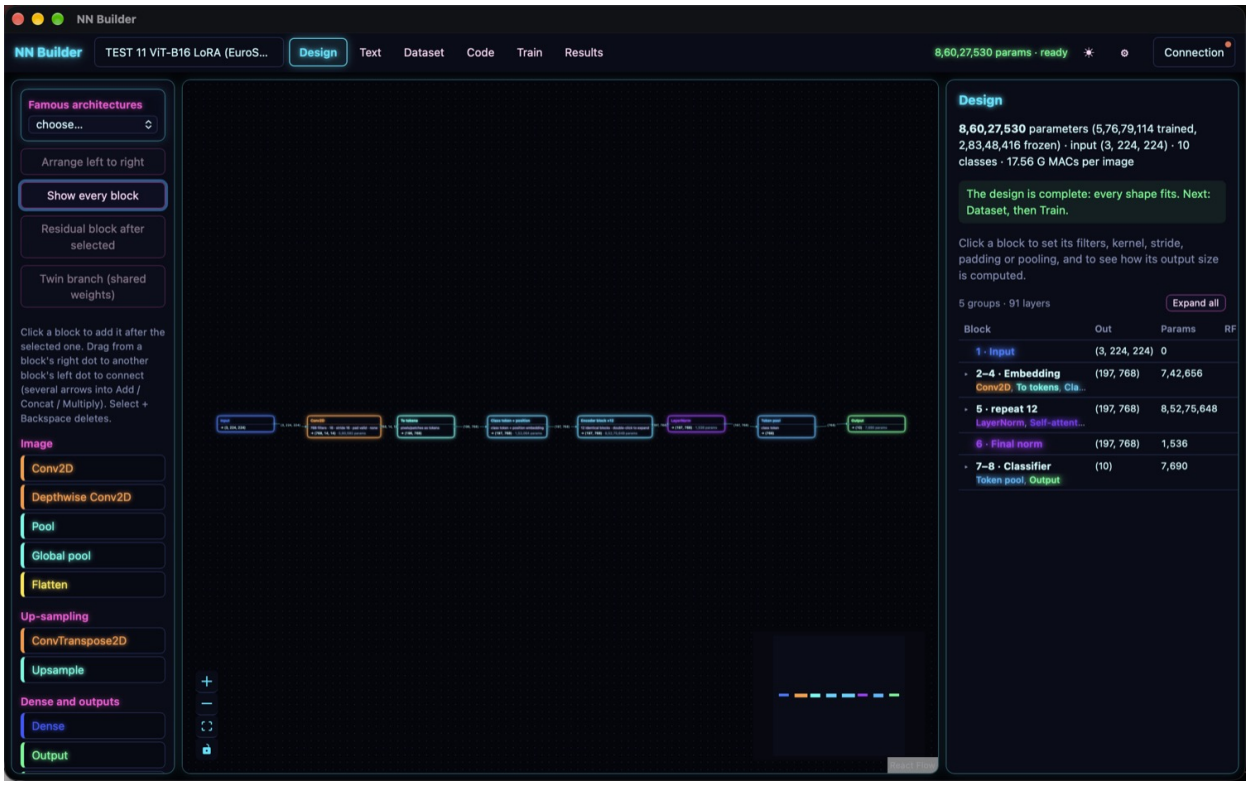


Figure 58: TEST 11 on the Design screen: the ViT-B/16 demo with its 12 encoder blocks collapsed (**Collapse repeated blocks**).



Figure 59: TEST 11 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size, with the repeated encoder blocks collapsed; long designs are split into rows here to fit the page.

ViT-B/16 cuts the 224×224 image into 196 patches with a 16×16 convolution ($16 \cdot 16 \cdot 3 \cdot 768 + 768 = 590\,592$), adds a class token and 197 learned positions ($768 + 197 \cdot 768 = 152\,064$) and runs 12 encoder blocks. Each block has two LayerNorms, a self-attention ($4 \cdot 768^2 + 4 \cdot 768 = 2\,362\,368$)

LoRA rank 4 on q, k and v: $3 \cdot 4 \cdot (768 + 768) = 18\,432$ new weights per block, 221 184 in all, while the 28 348 416 attention weights are frozen. With **Train: backbone** the run also keeps every other ImageNet weight fixed, so only the adapters and the new 10-class Output ($768 \cdot 10 + 10 = 7\,690$) learn: 228 874 weights, 0.27 % of the network, for 93.6 % after one epoch.

Block kind	Blocks	Parameters	of them frozen
Input	1	0	–
Conv2D	1	590 592	–
To tokens	1	0	–
Class token + positions	1	152 064	–
LayerNorm	25	38 400	–
Self-attention	12	28 569 600	28 348 416
Add	24	0	–
Dense	24	56 669 184	–
Token pool	1	0	–
Output	1	7 690	–
Total	91	**86 027 530**	28 348 416

17.12 TEST 12: ViT-B/16, full fine-tuning

TEST 12 is TEST 11 without LoRA: all 85 806 346 weights are trained.

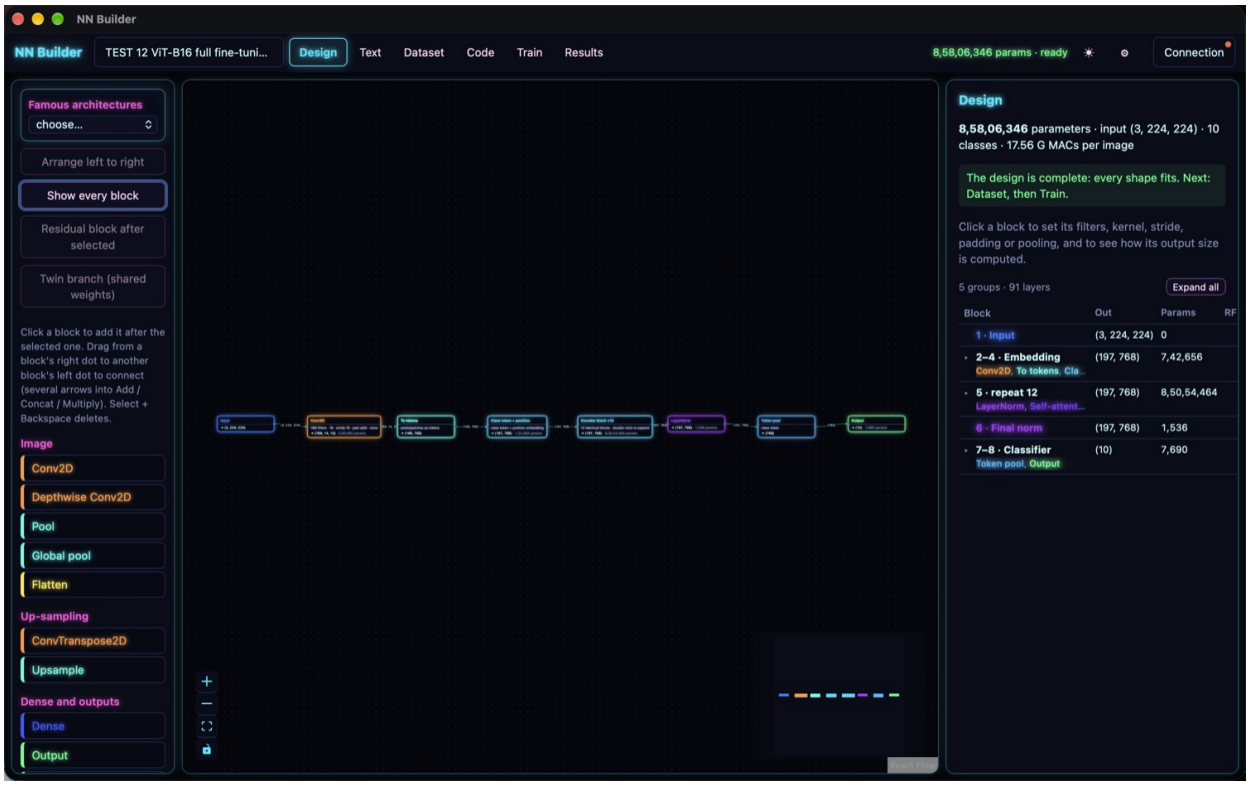


Figure 60: TEST 12 on the Design screen: the same ViT-B/16, collapsed.

The difference to TEST 11 is exactly the LoRA adapters ($86\,027\,530 - 85\,806\,346 = 221\,184$). Full fine-tuning reached 96.8 % against LoRA's 93.6 % in the same single epoch and about the same time (100 s per epoch on a T4); LoRA trains 375 times fewer weights and its checkpoints are correspondingly small. The MLPs (56.7 M) hold two thirds of the parameters, the attention layers one third.



Figure 61: TEST 12 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size, with the repeated encoder blocks collapsed; long designs are split into rows here to fit the page.

Block kind	Blocks	Parameters	of them frozen
Input	1	0	–
Conv2D	1	590 592	–
To tokens	1	0	–
Class token + positions	1	152 064	–
LayerNorm	25	38 400	–
Self-attention	12	28 348 416	–
Add	24	0	–
Dense	24	56 669 184	–
Token pool	1	0	–
Output	1	7 690	–
Total	91	**85 806 346**	–

17.13 TEST 13: every block of the language

TEST 13 trains `doc/examples/everything.nnb`, the design text that uses every block kind and language feature, on CIFAR-10 with the recommended recipe, three seeds, **save memory** and the TorchScript / ONNX export.

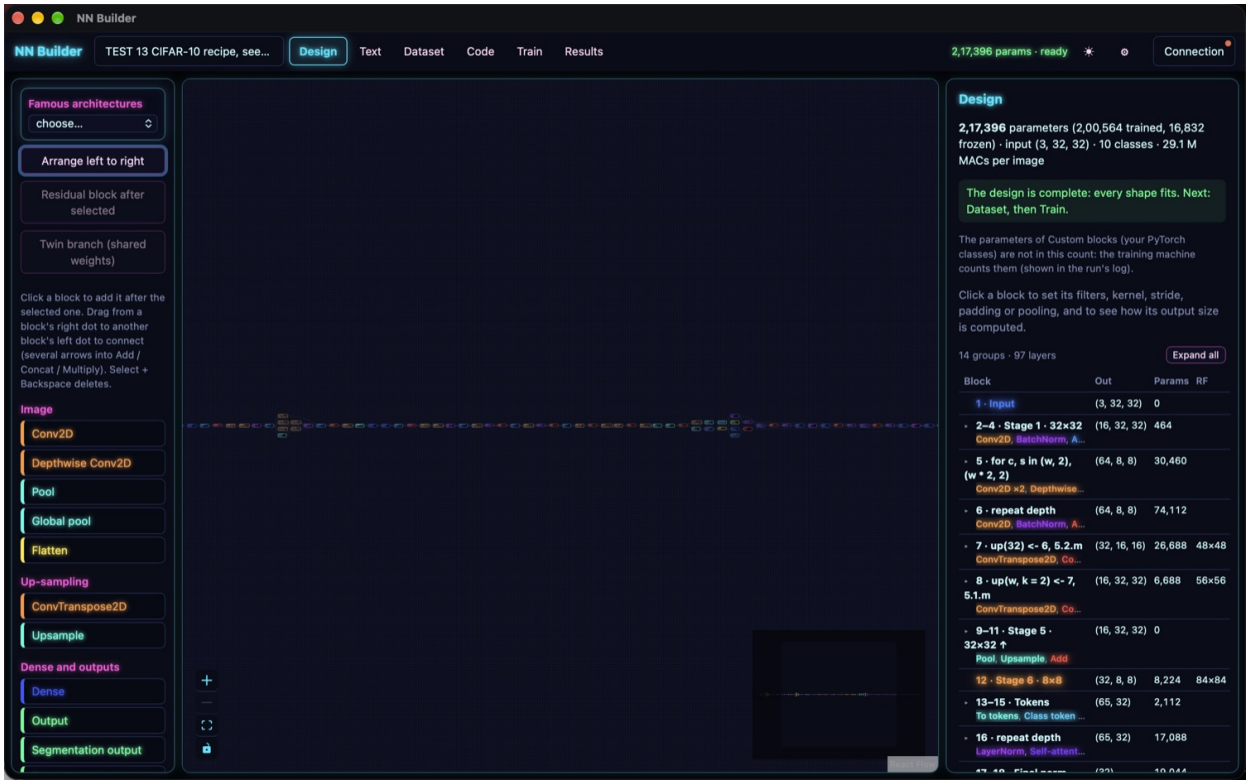


Figure 62: TEST 13 on the Design screen: the 97 blocks of `everything.nnb` (the canvas shows it from far; the layer table groups it).

It is a test of breadth, not a good classifier: Inception-like branches, depthwise convolutions, GroupNorm, a squeeze-and-excitation gate (Multiply), a U-Net style up-sampling path, a Custom block, a frozen convolution, shared weights with Subtract, attention with cross-attention, RMSNorm, SwiGLU, a mixture of experts, a LoRA Dense and an early exit. The table groups its

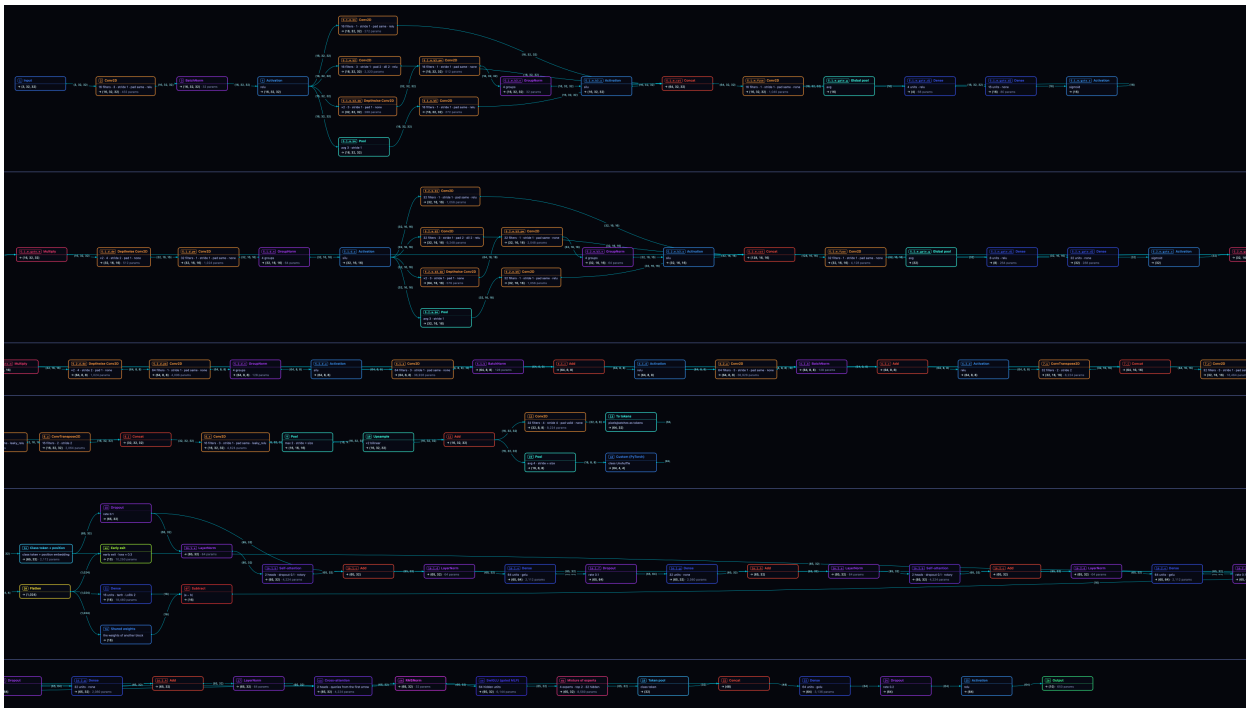


Figure 63: TEST 13 as a picture: **Save as PNG** on the Design screen (or **File → Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

Block kind	Blocks	Parameters	of them frozen
Input	1	0	–
Conv2D	18	132 672	432
BatchNorm	3	288	–
Activation	10	0	–
Depthwise Conv2D	4	2 400	–
Pool	4	0	–
groupnorm	4	288	–
Concat	5	0	–
Global pool	2	0	–
Dense	10	30 700	16 400
multiply	2	0	–
Add	7	0	–
convtranspose2d	2	10 288	–
upsample	1	0	–
To tokens	1	0	–
Custom (PyTorch)	1	0	–
Class token + positions	1	2 112	–
Flatten	1	0	–
Dropout	4	0	–
Early exit	1	10 250	–
LayerNorm	5	320	–
Shared weights	1	0	–
Self-attention	2	8 448	–
Subtract	1	0	–
cross_attention	1	4 224	–
RMSNorm	1	32	–
SwiGLU	1	6 144	–
Mixture of experts	1	8 580	–
Token pool	1	0	–
Output	1	650	–

Block kind	Blocks	Parameters	of them frozen
Total	97	**217 396**	16 832

17.14 TEST 14: a time series written as text

TEST 14 is `doc/examples/timeseries.nnb` pasted into the Text tab, with one setting changed on the canvas (the LSTM from 16 to 24 units) to check that only that line of the text changes.

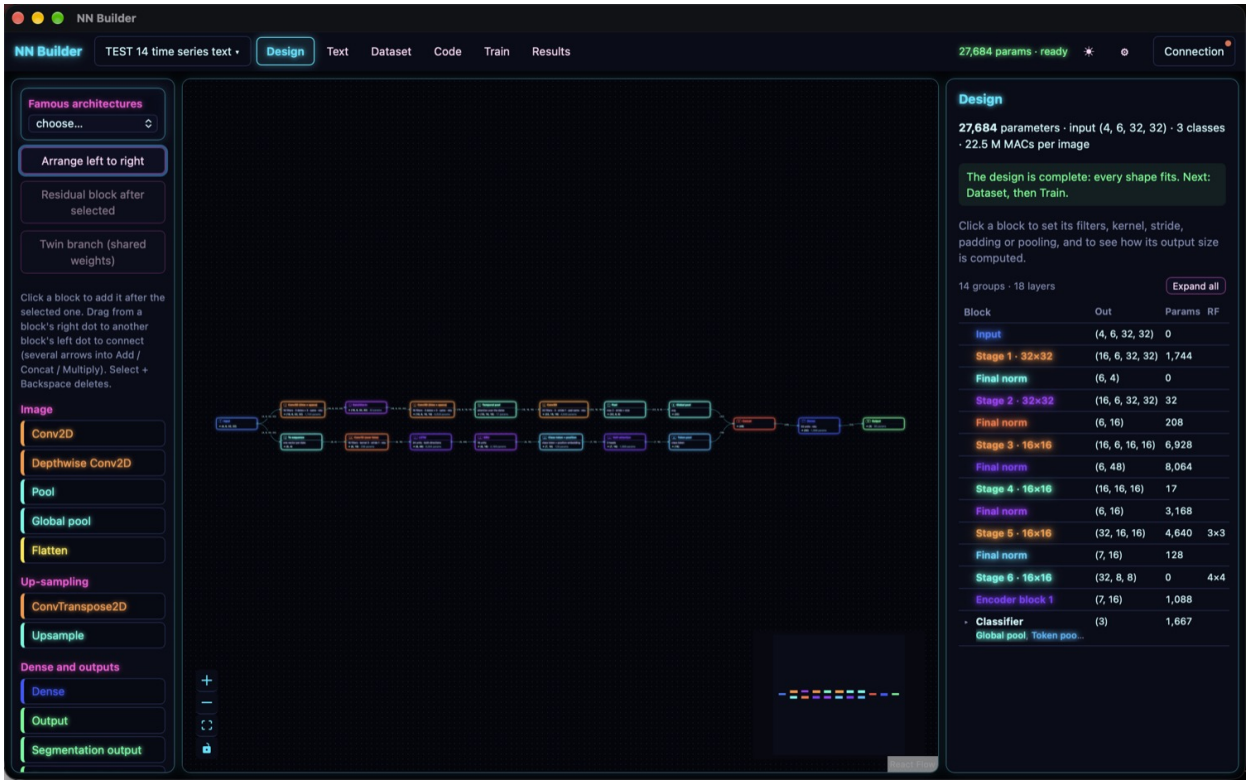


Figure 64: TEST 14 on the Design screen: two paths from the same input: Conv3D + temporal attention (image path) and a sequence path with Conv1D, LSTM, GRU and attention.

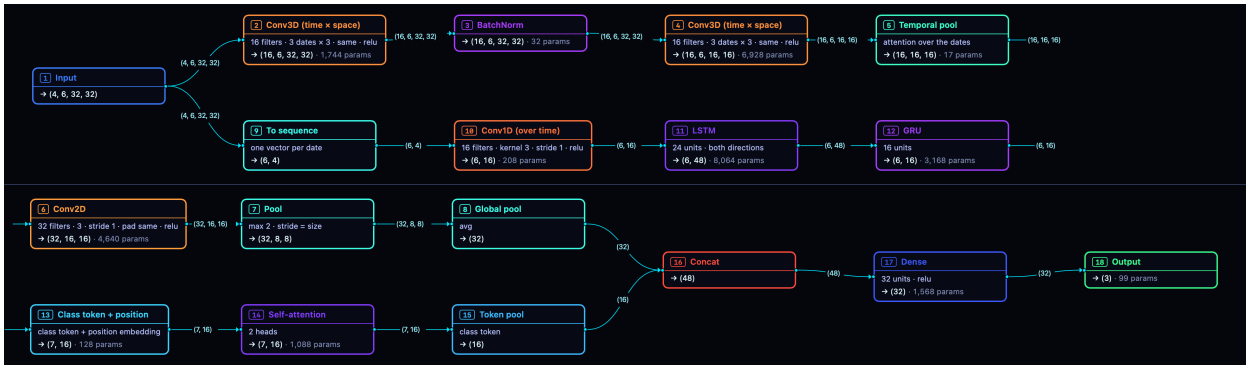


Figure 65: TEST 14 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

The input is 4 bands × 6 dates × 32×32 pixels. The *image path* (U-TAE style) runs two Conv3D layers (1 744 and 6 928), weighs the dates per pixel with temporal attention, and ends in a 2-D convolution (4 640) and a global pool. The *sequence path* (TSViT style) averages each date into a 4-number vector, then Conv1D ($3 \cdot 4 \cdot 16 + 16 = 208$), a bidirectional LSTM of 24 units

$(2\cdot4\cdot(16\cdot24+24^2+2\cdot24) = 8\,064$, the largest block after the edit), a GRU ($3\cdot(48\cdot16+16^2+2\cdot16) = 3\,168$) and a small transformer step (class token and 7 learned positions, 128; attention, 1 088). Concat joins the 32 + 16 numbers for the Dense and the Output.

Id	Block	Settings	Output	Parameters
t1	Input	all bands	$4 \times 6 \times 32 \times 32$	0
t2_2	Conv3D	16 filters, 3×3 over 3 dates, relu	$16 \times 6 \times 32 \times 32$	1 744
t9	To sequence		6×4	0
t3_2	BatchNorm		$16 \times 6 \times 32 \times 32$	32
t10	Conv1D	16 filters, kernel 3, relu	6×16	208
t4_2	Conv3D	16 filters, 3×3 over 3 dates, stride 2, relu	$16 \times 6 \times 16 \times 16$	6 928
t11	LSTM	24 units, both directions	6×48	8 064
t5	Temporal pool	attention	$16 \times 16 \times 16$	17
t12	GRU	16 units	6×16	3 168
t6	Conv2D	32 filters, 3×3, same, relu	$32 \times 16 \times 16$	4 640
t13	Class token + positions	positions learned	7×16	128
t7	Pool	max 2×2	$32 \times 8 \times 8$	0
t14	Self-attention	2 heads	7×16	1 088
t8	Global pool	avg	32	0
t15	Token pool	cls	16	0
t16	Concat		48	0
t17	Dense	32 units, relu	32	1 568
t18	Output	main output	3	99
	Total			**27 684**

17.15 TEST 15: segmentation of a large scene

TEST 15 trains on one large generated scene with a label image on the same grid, cut into 16×16 tiles; the test tiles are the eastern strip, with a column of tiles left out in between so that training and test pixels never touch.

A minimal fully convolutional network: two 3×3 convolutions (304 and 2 320) keep the full resolution, and the Segmentation output (1×1, $16 \cdot 3 + 3 = 51$) gives three class scores per pixel. Being fully convolutional, the same 2 675 weights work on any tile size.

Id	Block	Settings	Output	Parameters
t1	Input	all bands	$2 \times 16 \times 16$	0
t2	Conv2D	16 filters, 3×3, same, relu	$16 \times 16 \times 16$	304
t3	Conv2D	16 filters, 3×3, same, relu	$16 \times 16 \times 16$	2 320
t4	Segmentation output		$3 \times 16 \times 16$	51
	Total			**2 675**

17.16 TEST 16: the Small CNN on AWS

TEST 16 is the Small CNN demo, used to test training on Amazon EC2: start and stop of the instance, Prepare, fetching results from a stopped instance, the app closed during a run, and Stop.

Three stages of two 3×3 convolutions (without biases, as BatchNorm follows) at 32, 64 and 128 channels; each stage ends in max pooling and dropout. The convolutions grow from 864 to 147 456 weights, but the largest block is the Dense layer after Flatten: $2\,048 \cdot 128 + 128 = 262\,272$, almost half of the 551 018. That is typical of CNNs that flatten instead of pooling globally.

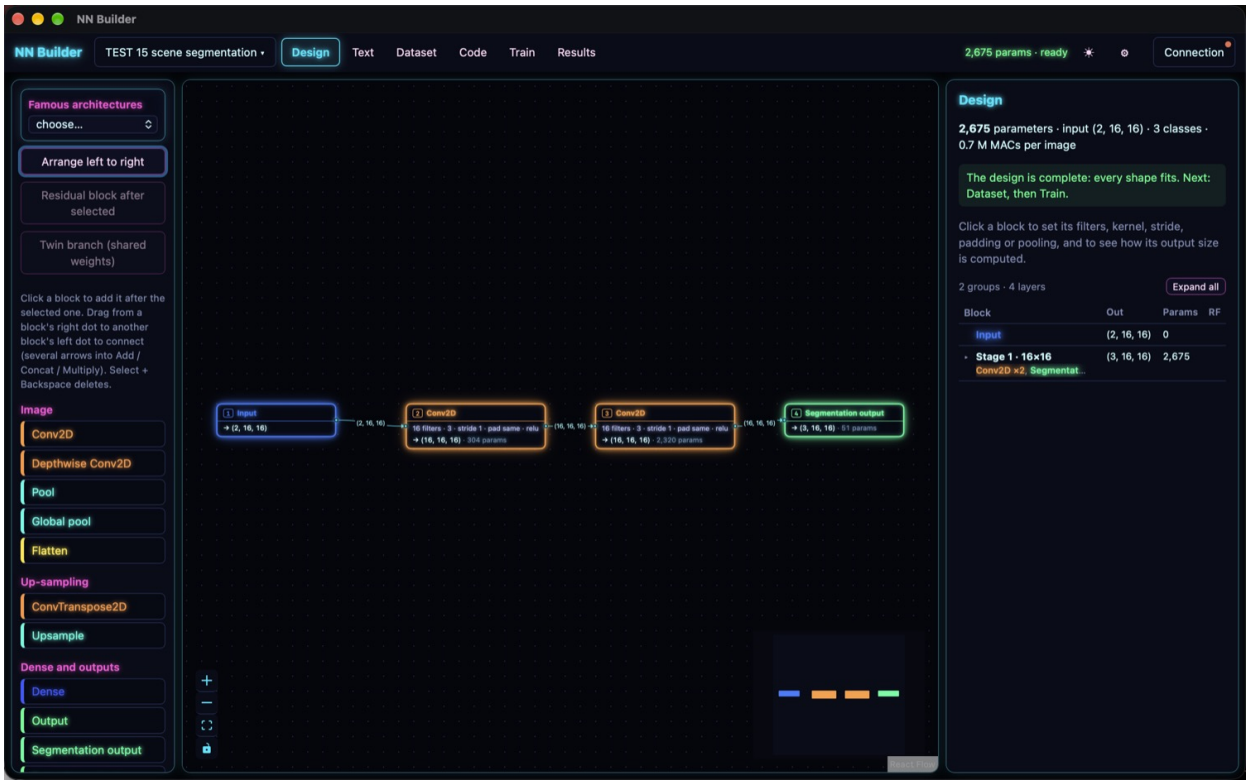


Figure 66: TEST 15 on the Design screen: two convolutions and the Segmentation output.



Figure 67: TEST 15 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

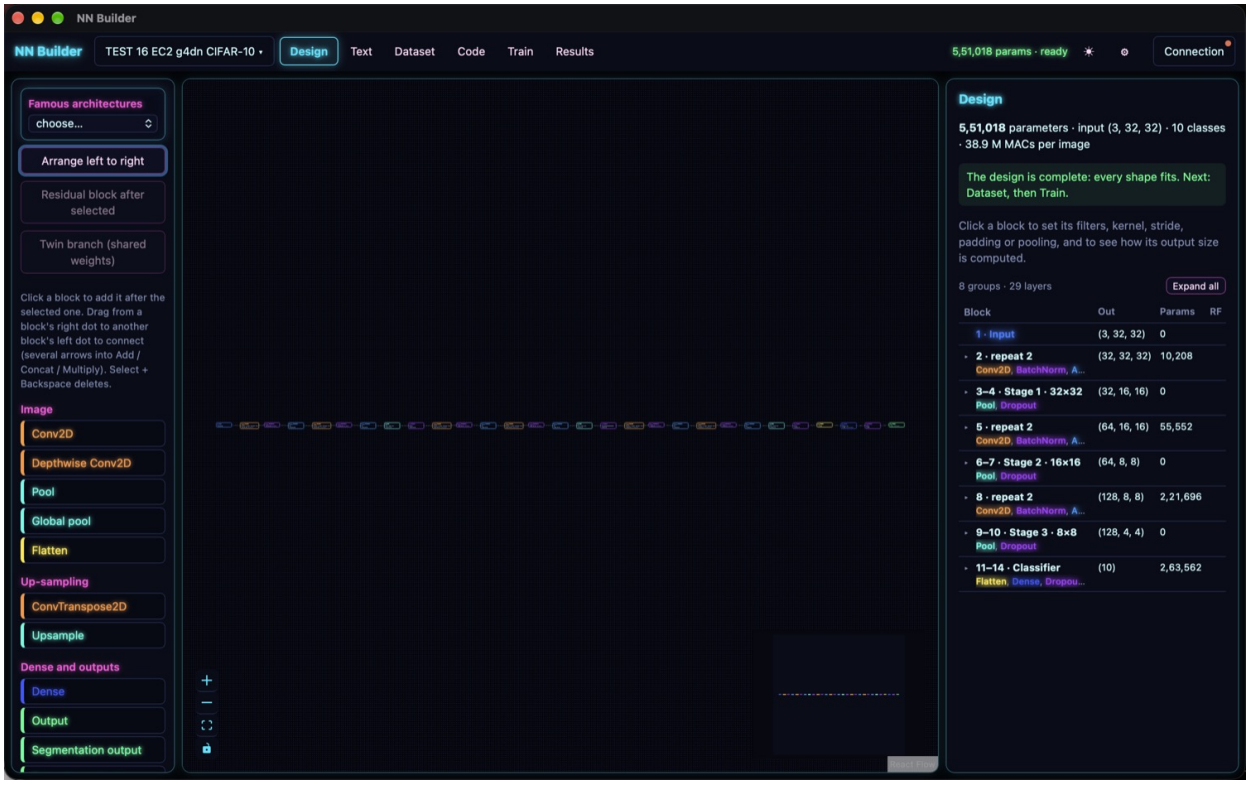


Figure 68: TEST 16 on the Design screen: the Small CNN (CIFAR-10) demo: three VGG-style stages and a Dense classifier.

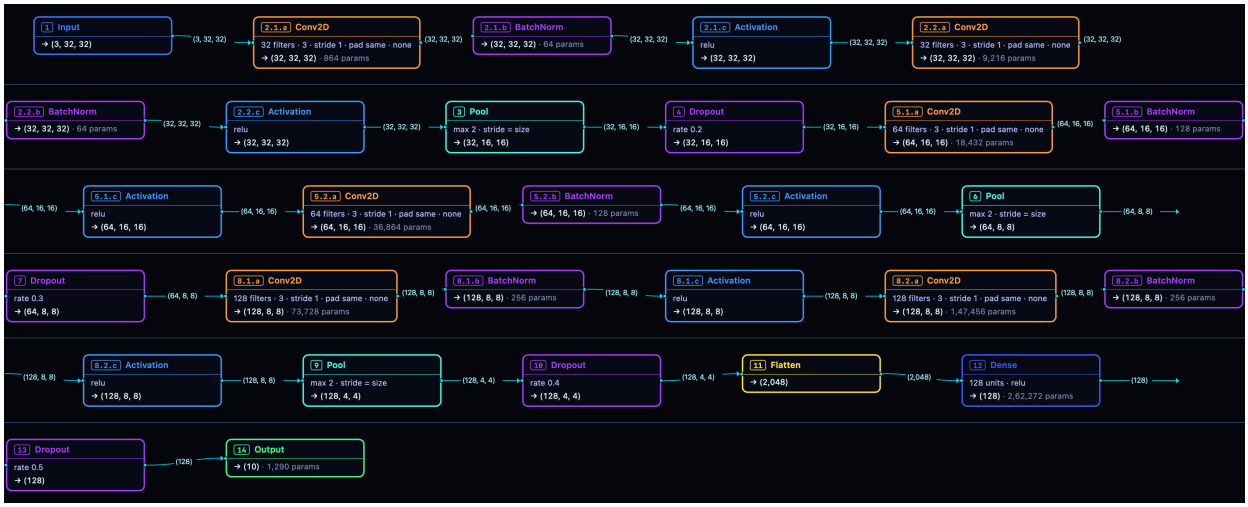


Figure 69: TEST 16 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

Id	Block	Settings	Output	Parameters
n1	Input	all bands	3 × 32 × 32	0
n2	Conv2D	32 filters, 3 × 3, same	32 × 32 × 32	864
n3	BatchNorm		32 × 32 × 32	64
n4	Activation	relu	32 × 32 × 32	0
n5	Conv2D	32 filters, 3 × 3, same	32 × 32 × 32	9 216
n6	BatchNorm		32 × 32 × 32	64
n7	Activation	relu	32 × 32 × 32	0
n8	Pool	max 2×2	32 × 16 × 16	0
n9	Dropout	rate 0.2	32 × 16 × 16	0
n10	Conv2D	64 filters, 3 × 3, same	64 × 16 × 16	18 432
n11	BatchNorm		64 × 16 × 16	128
n12	Activation	relu	64 × 16 × 16	0
n13	Conv2D	64 filters, 3 × 3, same	64 × 16 × 16	36 864
n14	BatchNorm		64 × 16 × 16	128
n15	Activation	relu	64 × 16 × 16	0
n16	Pool	max 2×2	64 × 8 × 8	0
n17	Dropout	rate 0.3	64 × 8 × 8	0
n18	Conv2D	128 filters, 3 × 3, same	128 × 8 × 8	73 728
n19	BatchNorm		128 × 8 × 8	256
n20	Activation	relu	128 × 8 × 8	0
n21	Conv2D	128 filters, 3 × 3, same	128 × 8 × 8	147 456
n22	BatchNorm		128 × 8 × 8	256
n23	Activation	relu	128 × 8 × 8	0
n24	Pool	max 2×2	128 × 4 × 4	0
n25	Dropout	rate 0.4	128 × 4 × 4	0
n26	Flatten		2048	0
n27	Dense	128 units, relu	128	262 272
n28	Dropout	rate 0.5	128	0
n29	Output	main output	10	1 290
	Total			**551 018**

17.17 TEST 17: real Earth Engine exports, S2 + S1

TEST 17 reads 40 real Sentinel-1 / Sentinel-2 image pairs exported from Google Earth Engine (S2 Level-1C, 13 bands; S1 VV and VH in dB; LZW-compressed GeoTIFFs) and checks them on Kaggle.

The two groups are stacked into 15 channels (13 + 2), so the first convolution has $3 \cdot 3 \cdot 15 \cdot 32 + 32 = 4\,352$ weights. The data check read every pair, confirmed the radar was already in decibels (VV mean −13.3 dB, VH −21.6 dB) and passed one batch of 2×15×64×64 through the network. Testing it found four bugs (LZW files on Kaggle, the band names of the second group, a 13-band header and a full path as the second group's folder), all fixed.

Id	Block	Settings	Output	Parameters
t1	Input	all bands	15 × 64 × 64	0
t2	Conv2D	32 filters, 3 × 3, same, relu	32 × 64 × 64	4 352
t3	BatchNorm		32 × 64 × 64	64
t4	Pool	max 2×2	32 × 32 × 32	0
t5	Conv2D	64 filters, 3 × 3, same, relu	64 × 32 × 32	18 496
t6	Global pool	avg	64	0
t7	Output	main output	10	650
	Total			**23 562**

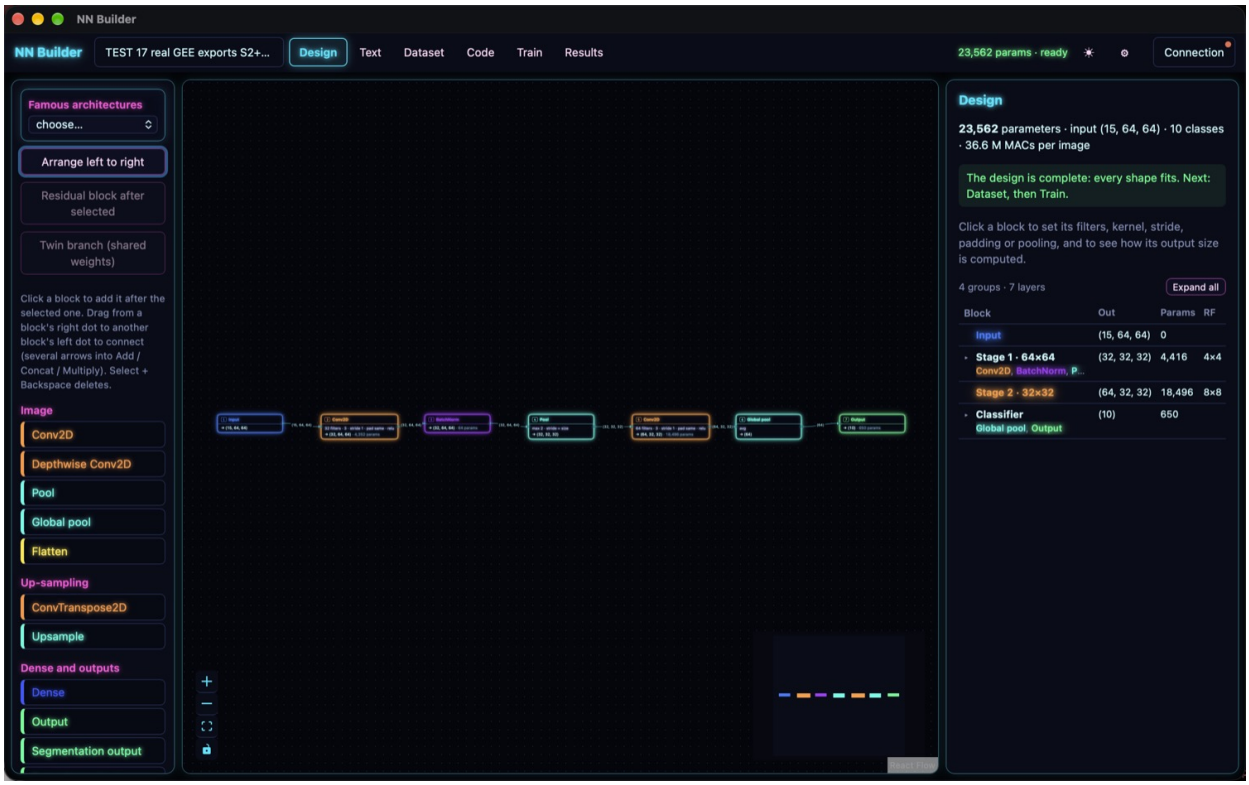


Figure 70: TEST 17 on the Design screen: an early-fusion CNN on 15 channels.

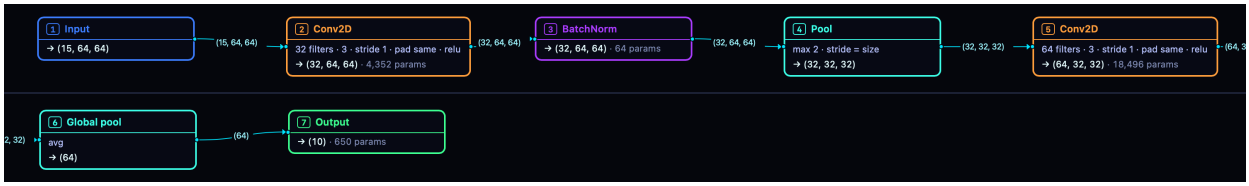


Figure 71: TEST 17 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

17.18 TEST 19: the app's Earth Engine export

TEST 19 uses data made by the application's own Earth Engine script: Sentinel-2 (B2, B3, B4, B8, 2024 median) over a 5.5 km tile with ESA WorldCover as the label, fetched by the Kaggle notebook straight from a Google Drive link.

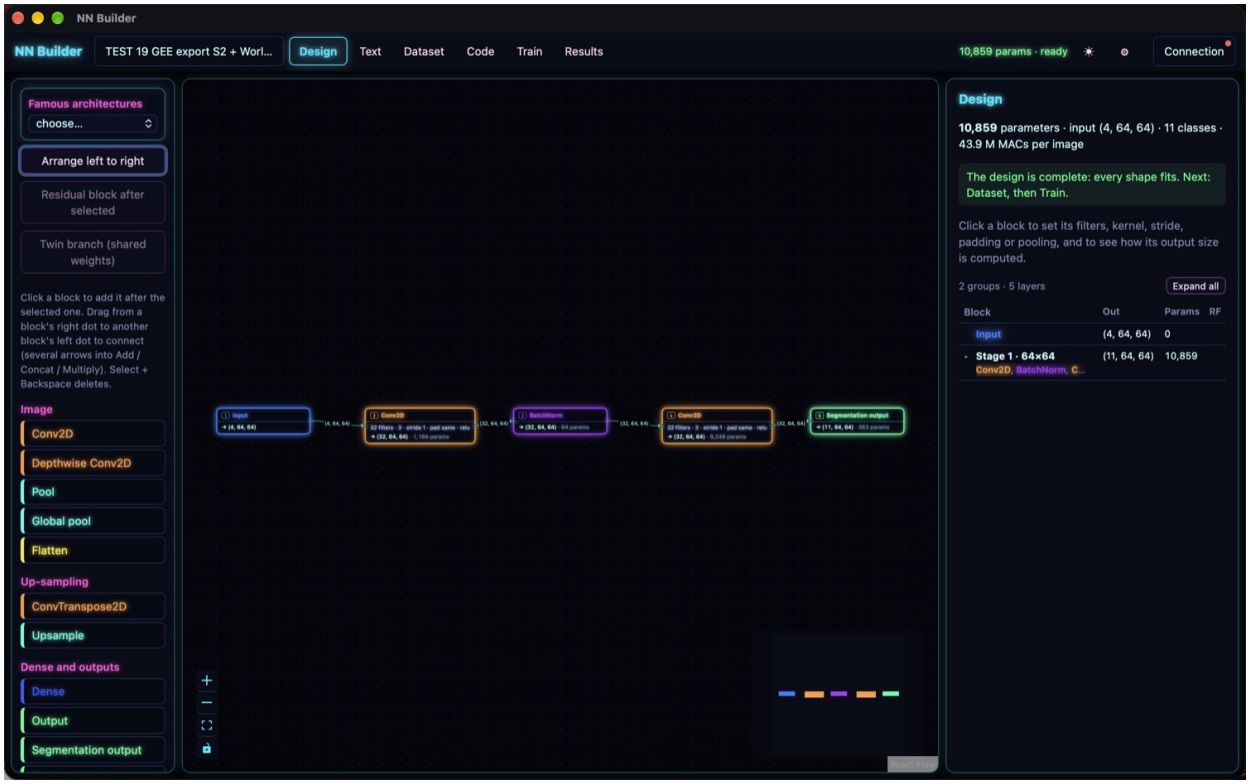


Figure 72: TEST 19 on the Design screen: two convolutions and an 11-class Segmentation output.



Figure 73: TEST 19 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

The network is small on purpose (1 184 + 64 + 9 248 + 363 = 10 859 parameters): the test is the data path. The Segmentation output has one score per WorldCover class ($32 \cdot 11 + 11 = 363$). After 10 epochs on 48 tiles, built-up land (IoU 0.71) and tree cover (0.67) are learnt; the mean IoU over all 11 classes is only 0.148 because four classes do not occur in the tile and count as 0.

Id	Block	Settings	Output	Parameters
t1	Input	all bands	$4 \times 64 \times 64$	0
t2	Conv2D	32 filters, 3×3, same, relu	$32 \times 64 \times 64$	1 184
t3	BatchNorm		$32 \times 64 \times 64$	64
t4	Conv2D	32 filters, 3×3, same, relu	$32 \times 64 \times 64$	9 248
t5	Segmentation output		$11 \times 64 \times 64$	363
	Total			**10 859**

17.19 TEST 24: every classifier block once

TEST 24 is one small CIFAR-10 network that uses each of the 36 blocks a classifier can use exactly once, with every block's settings chosen by hand. It was written in the Text tab, pasted into the application window, and checked there: the Design screen reports *The design is complete: every shape fits.* TEST 13 does the same for the text language, with many blocks repeated; here each kind appears once, so the design can be read block by block. It shows the blocks working together; it is not meant to reach a high accuracy.

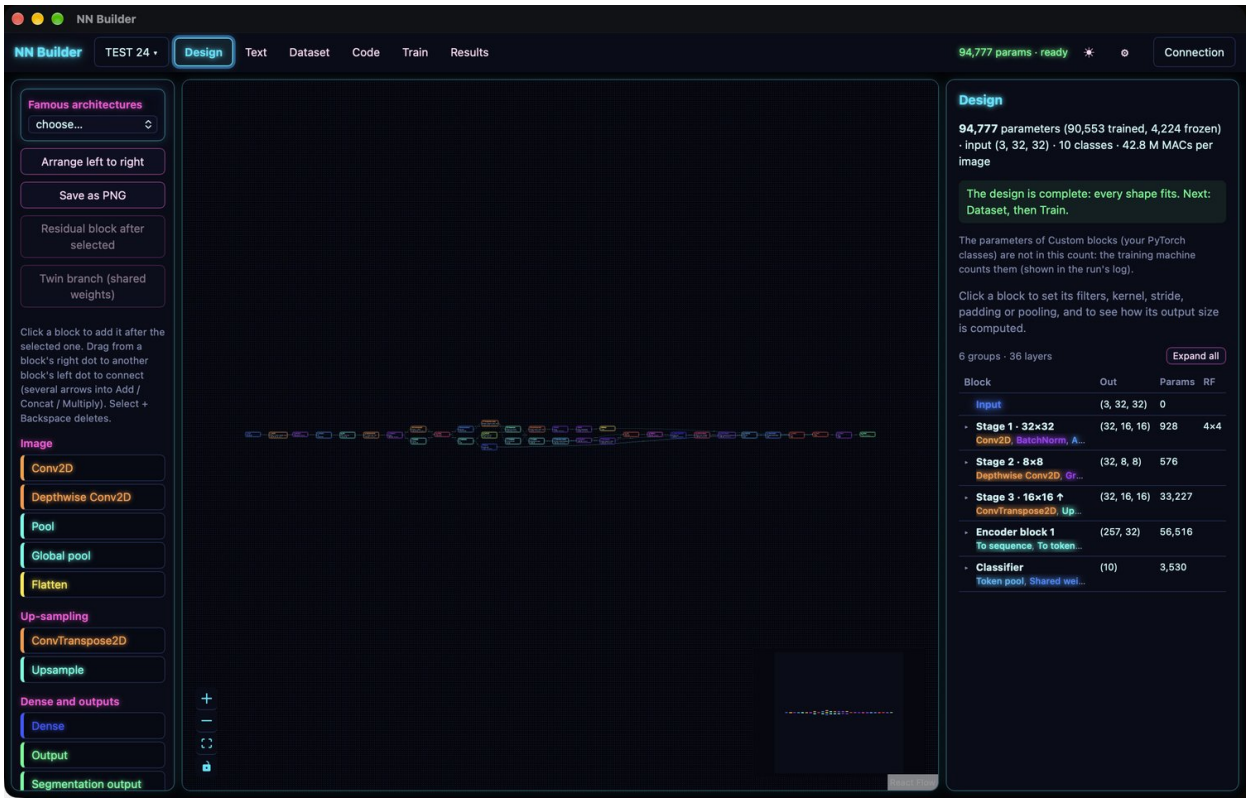


Figure 74: TEST 24 on the Design screen: 94 777 parameters (90 553 trained, 4 224 frozen), input $3 \times 32 \times 32$, 10 classes, 42.8 M MACs per image.

The four parts.

- *CNN stem* (t1–t10). Conv2D, BatchNorm and ReLU, then a max pool and a depthwise convolution take the image to $32 \times 8 \times 8$; a 4×4 kernel with padding 1 halves 16 to exactly 8 (with 3×3 the application warns that the last row and column are dropped). GroupNorm (8 groups of 4 channels) follows. Two kinds of up-sampling bring the map back to 16×16: a learned ConvTranspose2D and a fixed bilinear Upsample, multiplied pixel by pixel.
- *Time path* (t11–t18). A Custom block, the class `Views` in the design's own PyTorch code, stacks the map as it is, mirrored left–right, upside down and both, as four "dates". Its line `# out: c, 4, h, w` declares the shape $32 \times 4 \times 16 \times 16$. Conv3D looks at three neighbouring views at once ($3 \cdot 3 \cdot 3 \cdot 32 \cdot 32 + 32 = 27\,680$). Temporal pool (attention, $32 + 1$ weights) merges the views into one map for the transformer; To sequence turns them into a 4-step sequence for Conv1D, LSTM ($4(32 \cdot 32 + 32^2 + 2 \cdot 32) = 8\,448$) and a two-way GRU ($2 \cdot 3(32 \cdot 32 + 32^2 + 2 \cdot 32) = 12\,672$), flattened to 256 numbers.
- *Transformer path* (t19–t28). The 16×16 map becomes 256 tokens of 32 features; a class token and learned positions make 257 ($32 + 257 \cdot 32 = 8\,256$). Self-attention has LoRA rank 4: its $4 \cdot 32^2 + 4 \cdot 32 = 4\,224$ weights are frozen and only the $4\,992 - 4\,224 = 768$ weights of the rank-4 adapters train. Then come a residual Add, RMSNorm, a SwiGLU MLP ($3 \cdot 32 \cdot 64 = 6\,144$),

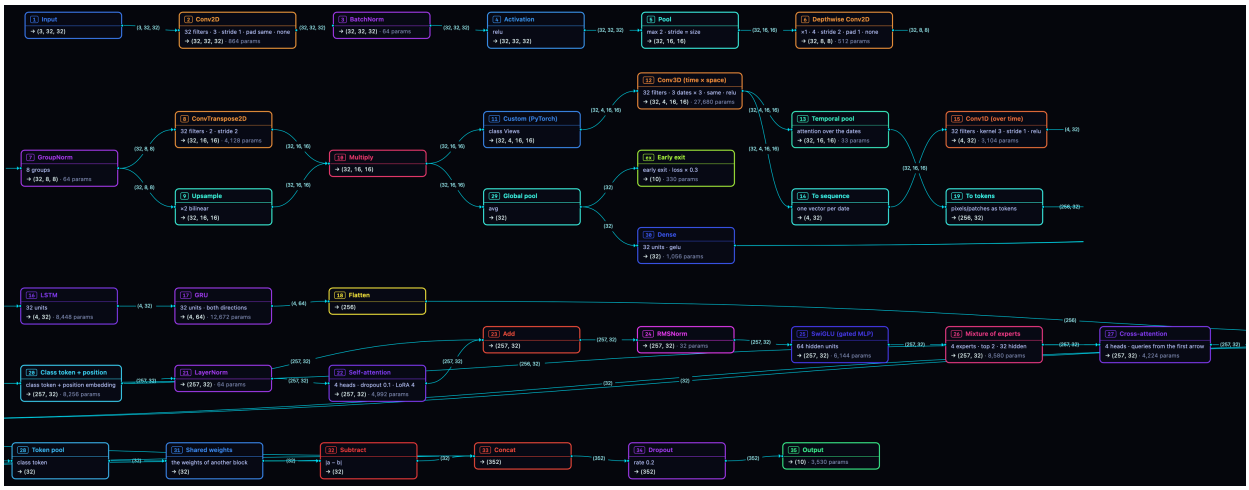


Figure 75: TEST 24 as a picture: **Save as PNG** on the Design screen (or **File** → **Export Design as PNG**) draws the whole design at full size; long designs are split into rows here to fit the page.

a mixture of 4 experts with 2 per token, and cross-attention in which the processed tokens (queries) look at the raw patch tokens t19 (keys and values). Token pool keeps the class token.

- *Head* (t29–t35). Global pool gives the CNN path's vector, which also feeds an early exit (share of the loss 0.3). Dense t30 maps it to 32 numbers, and the Shared-weights block t31 applies the same Dense to the transformer vector (the Siamese idea: both in one space). Subtract gives $|a - b|$; Concat joins $32 + 32 + 32 + 256 = 352$ numbers; Dropout 0.2 and the Output ($352 \cdot 10 + 10 = 3\,530$) end the network.

Two blocks of the palette are not used: *Segmentation output* and *Detection output* need a dataset with masks or boxes, and a network has one kind of output (TEST 15 and TEST 9 use them).

Training settings. CIFAR-10 photos are objects that may face either way, so the run uses a horizontal flip and a random crop with 4 pixels of padding. The recipe is **Use recommended** (AdamW, weight decay 0.05, cosine decay, label smoothing 0.1, gradient clipping at 1) with the warm-up cut to 2 of 20 epochs, batch 128, learning rate 0.001, and a 10 % validation part with early stopping after 5 epochs without improvement. The application estimates 0.75 GB of GPU memory at batch 128.

Id	Block	Settings	Output	Parameters
t1	Input	CIFAR-10	$3 \times 32 \times 32$	0
t2	Conv2D	32 filters, 3×3, same, none, no bias	$32 \times 32 \times 32$	864
t3	BatchNorm		$32 \times 32 \times 32$	64
t4	Activation	relu	$32 \times 32 \times 32$	0
t5	Pool	max 2×2	$32 \times 16 \times 16$	0
t6	Depthwise Conv2D	$\times 1$, 4×4, stride 2, pad 1, none, no bias	$32 \times 8 \times 8$	512
t7	GroupNorm	8 groups	$32 \times 8 \times 8$	64
t8	ConvTranspose2D	32 filters, 2×2, stride 2, none	$32 \times 16 \times 16$	4 128
t9	Upsample	$\times 2$ bilinear, from t7	$32 \times 16 \times 16$	0
t10	Multiply	t8 $\times$ t9	$32 \times 16 \times 16$	0
t11	Custom (PyTorch)	class `Views`, `# out: c, 4, h, w`	$32 \times 4 \times 16 \times 16$	0
t12	Conv3D	32 filters, $3 \times 3 \times 3$ dates, same, relu	$32 \times 4 \times 16 \times 16$	27 680
t13	Temporal pool	attention	$32 \times 16 \times 16$	33
t14	To sequence	from t12	4×32	0
t15	Conv1D	32 filters, kernel 3, same, relu	4×32	3 104
t16	LSTM	32 units	4×32	8 448

Id	Block	Settings	Output	Parameters
t17	GRU	32 units, both directions	4×64	12 672
t18	Flatten		256	0
t19	To tokens	patches, from t13	256×32	0
t20	Class token + position	class token, learned positions	257×32	8 256
t21	LayerNorm		257×32	64
t22	Self-attention	4 heads, dropout 0.1, LoRA rank 4	257×32	4 992
t23	Add	t20 + t22	257×32	0
t24	RMSNorm		257×32	32
t25	SwiGLU	64 hidden units	257×32	6 144
t26	Mixture of experts	4 experts, top 2, 32 hidden	257×32	8 580
t27	Cross-attention	4 heads; queries t26, keys/values t19	257×32	4 224
t28	Token pool	class token	32	0
t29	Global pool	avg, from t10	32	0
tex	Early exit	share of the loss 0.3	10	330
t30	Dense	32 units, gelu	32	1 056
t31	Shared weights	weights of t30, on t28	32	0
t32	Subtract	$\lvert t30 - t31 \rvert$	32	0
t33	Concat	t30, t31, t32, t18	352	0
t34	Dropout	rate 0.2	352	0
t35	Output	main output	10	3 530
	Total	(4 224 of them frozen)		**94 777**

18 Troubleshooting

"design has problems". Open Design; the red block says why (Section 5.1). Blocks after it are greyed out until it is fixed.

"dataset incomplete" or no classes. The class list is empty (Hugging Face: **Read info** and a ClassLabel column; Kaggle: type the class folder names; regression: the target columns; detection: the object classes). Text datasets cannot be used.

The one-batch check stops the run. A layer's real shape differed from the application's (for example images of an unexpected size in a custom dataset). The message names the layer; check the image size and channels in the Dataset tab.

"CUDA out of memory". With **Split a batch that does not fit** on (the default) the run handles this itself and continues more slowly. If even one image at a time does not fit, the run stops with a message: switch on **Save memory**, keep mixed precision on, or use a smaller input size, network or a larger GPU (Section 7).

Kaggle run fails to start. The account must be phone-verified for GPU and internet access; the token must be current.

SSH machine does not answer. Password machines: press **Connect** again (the login lasts a few hours); key machines: **Test**. A running training is not affected.

EC2: "no public address" / ssh never answers. Give the instance a public IP (or an Elastic IP), allow port 22 in its security group, and use its key file. A new account may have a vCPU quota of 0 for G instances: request an increase.

EC2 spot: the run says "interrupted". AWS took the capacity back; with a persistent request and behaviour *stop* it resumes by itself. To continue now, use **Continue on another machine** (needs the checkpoint copy on the Mac).

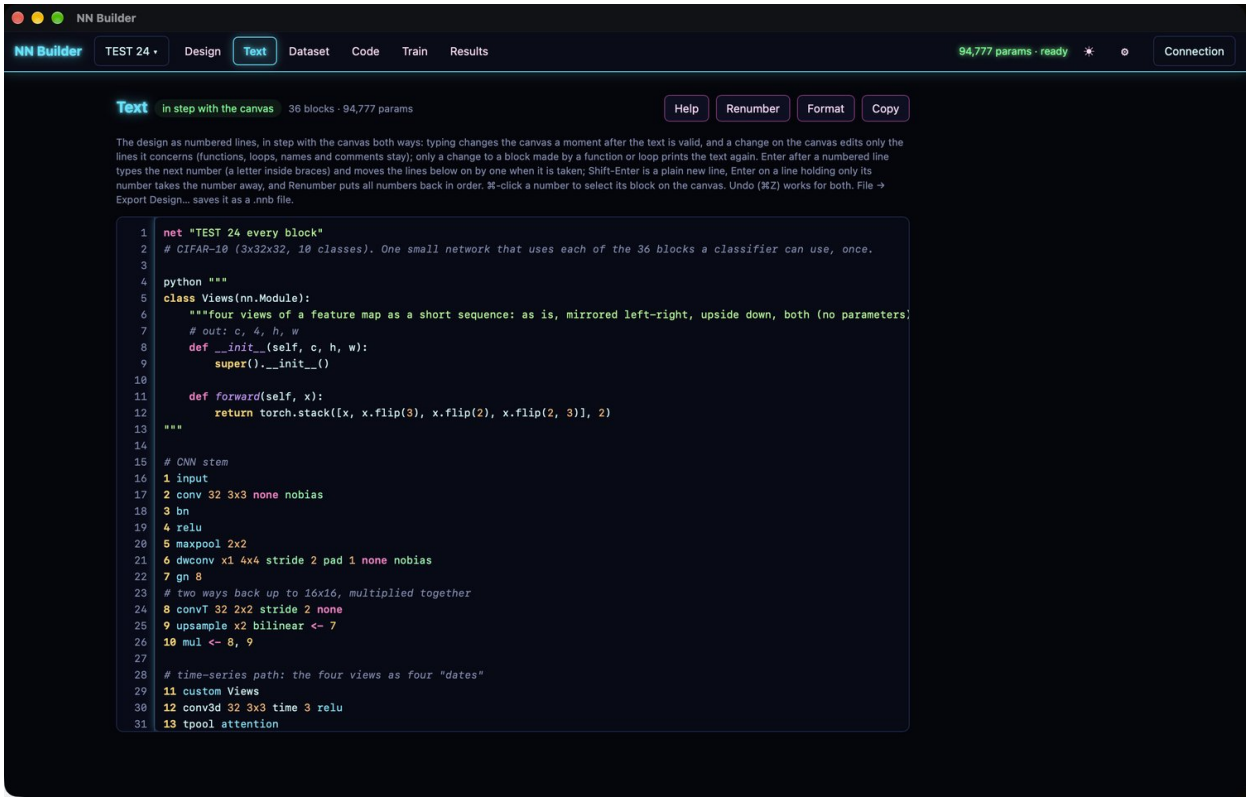


Figure 76: TEST 24 on the Text tab: the design's PyTorch code (the `Views` class) and the CNN stem. Each line is a block; `<-` names other inputs.

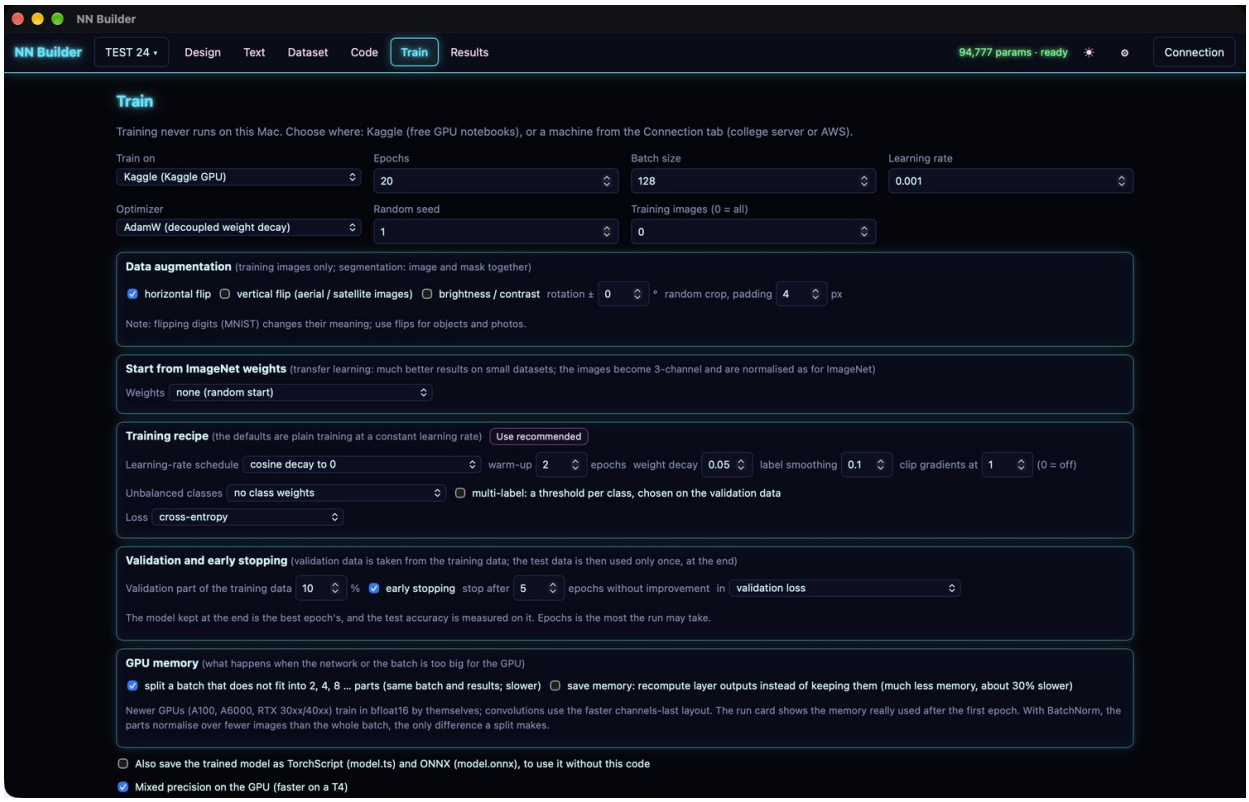


Figure 77: TEST 24 on the Train tab: augmentation, the recommended recipe with a 2-epoch warm-up, and the validation part.

A demo disappeared. Unsaved demos are discarded when another demo or project is opened or the application quits; save them with `Cmd+S`.

"CHECK: shapes differ ... (the class's # out: line declares the expected shape)". A Custom block's PyTorch class returns another shape than its `# out:` line says: correct the line or the class (Section 8.1).

"the image must be a whole number of map cells". The Detection output's map must divide the image evenly (e.g. 64 and 16): change the pools before it or the image size.

"MEMORY: a GPU ran out of memory" with several GPUs. With several GPUs a batch is not split automatically: a smaller batch, **save memory**, or one GPU (Section 8.4).

macOS asks for the Keychain after an update. An unsigned new build of the application is a new program for macOS: **Always Allow** once. A self-signed code-signing certificate (FUTURE_PLAN.md) would end this.

Training is unstable (loss jumps or becomes NaN). Lower the learning rate (by 10×), keep mixed precision on (the script scales the loss), add BatchNorm / LayerNorm, check that the labels match the class list.

High training accuracy, low validation accuracy. Overfitting: augmentation, dropout, fewer parameters (Global pool instead of Flatten + large Dense), more data, fewer epochs.

19 Privacy and security

- Tokens and keys live only in the system keychain; they are passed to one call at a time, and written on a remote machine only when a run needs them, as files readable by their owner.
- The design computer never downloads datasets or runs training; it reads dataset descriptions only.
- Unsaved demos are kept in the application's own folder and deleted when no longer needed; projects live where you choose.
- Updates come only from the project's GitHub releases and are checked against the published SHA-256 checksum.

A Glossary

Activation function a non-linear function applied element-wise (ReLU, GELU, ...).

Batch the images processed together in one training step.

Channel one "layer" of an image or feature map (3 for RGB; a convolution with 64 filters produces 64).

Checkpoint a saved training state (weights, optimiser, position) from which a run can resume.

Epoch one pass over the whole training set.

Feature map the output of a convolution: one image per filter.

Kernel / filter the small window of weights a convolution slides over the image.

mIoU mean over the classes of (overlap / union) of predicted and true masks.

Padding pixels added around the image so the window can reach the borders.

Parameter a trainable number (weight or bias).

Receptive field the input region one output value depends on.

Spot instance spare cloud capacity at a lower price that the provider can take back.

Stride how far the window moves at each step.

Token one element of a sequence a transformer works on (here: an image patch or row).

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